

SERVICE MANUAL

for

CATERPILLAR[®]

D6 TRACTOR

SERIAL NUMBERS 37A1-UP AND 44A1-UP

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Caterpillar Tractor Co., General Offices, Peoria, Illinois. • Caterpillar Americas Co., Peoria, Illinois. • Caterpillar Overseas S.A., Geneva. • Caterpillar of Australia Pty. Ltd., Melbourne. • Caterpillar Brasil S.A., São Paulo. • Caterpillar Tractor Co. Ltd., Glasgow. • Caterpillar of Canada Ltd., Toronto. • Caterpillar France S.A., Grenoble. • Caterpillar (Africa) (Pty.) Ltd., Johannesburg. • Caterpillar Mexicana S.A. de C.V., Monterrey.

General Instructions

These general instructions will be extremely helpful in following the detailed instructions in the manual. They should be read and kept in mind while assembling or disassembling the machine.

SYMBOLS

Throughout this Service Manual, symbols are used on the illustrations to point out parts which are to be removed or disconnected. These removals or disconnections are indicated by the following code:

 Remove  Disconnect

The part to be removed or disconnected is pointed out by a number in the symbol and this number corresponds to the part description listed in the caption copy beneath the picture. The disconnect arrow points as close as possible to point of disconnection.

SAFETY AND WORKMANSHIP SUGGESTIONS

There are certain practices which should be followed in the interest of safety and good workmanship when working around machinery.

Always show proper respect for weight. Do not attempt to lift heavy parts when a hoist should be used. Never leave heavy parts in an unstable position. When raising a portion of a machine, be sure the machine is blocked securely at front and rear. The weight should be supported by blocks rather than lifting equipment.

When using a hoist, follow the recommendations outlined in the manual. Fabricate and use the illustrated lifting devices in conjunction with the hoist to achieve the proper balance of the assemblies being lifted and to assure safe handling. (See special topic, GENERAL INSTRUCTIONS, DIESEL ENGINE REMOVAL).

Tools

All service tools should be kept in first-class condition. Use the proper tool for the job at hand. Special service tools are available for specific jobs and should be used when recommended. The use of these tools will save time and prevent damage to parts.

In the manual, puller arrangements are illustrated for separating tightly fitted parts. However, if the machine is being reconditioned in a shop, it might be easier and faster to use a press. When pulling a bearing or gear from a shaft, always use a centering spacer between the forcing screw and end of shaft.

Disassembly

If a part offers unexpected resistance to removal, check carefully to see that all nuts and bolts have been removed before using force. Possibly some other part is interfering and should be removed first. Parts which are fitted together with tapered splines are always very tight. If they are not tight when disassembled, inspect the tapered splines and discard the part if the splines are worn.

Identical parts should be kept in order or marked so they can be reinstalled in the same position from which they were removed.

When shims are used, be sure to remove all of them. Tie the shims together and identify them as to their location. Keep them clean and flat until they are reinstalled.

INSPECT FREQUENTLY AND CORRECT MINOR TROUBLES

A bearing changed in time will save a shaft. An oil leak, corrected, prevents loss of lubricant and an overheated bearing. A nut tightened in time will prevent the loss or breakage of an associated part.

MAINTAIN ADJUSTMENTS

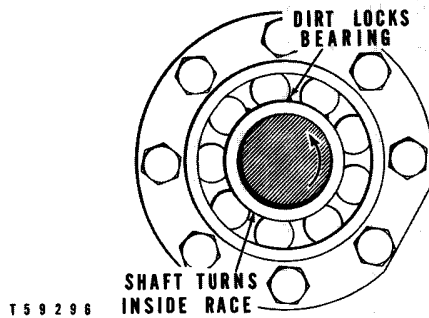
Operating adjustments have been kept to a minimum on Caterpillar-built machines but they are important and should be carefully maintained.

Follow the recommendations in the Operation and Maintenance Instructions.

KEEP DIRT OUT

The most important single item in preserving the long life of the machine is to keep dirt out of vital working parts. Caterpillar Tractor Co. has taken precautions to safeguard against dirt entering working parts. Enclosed compartments, seals and filters have been provided to keep the supply of air, and lubricants clean. It is important that the effectiveness of these safeguards be maintained. Filters should be replaced or cleaned regularly. Worn seals or broken gaskets should be replaced immediately. Be sure seal lips are facing in the correct direction.

Anti-friction bearings, properly lubricated with clean lubricant, will last indefinitely. Abrasives in the lubricant will cause rapid wear on the extremely hard races and balls or rollers. Dirt in an anti-friction bearing can cause the bearing to lock resulting in the shaft turning in the inner race or the outer race turning within the cage. Dirt and abrasives in lubricants will embed in sleeve bearings and act like fine sandpaper against the shaft causing extremely rapid wear.



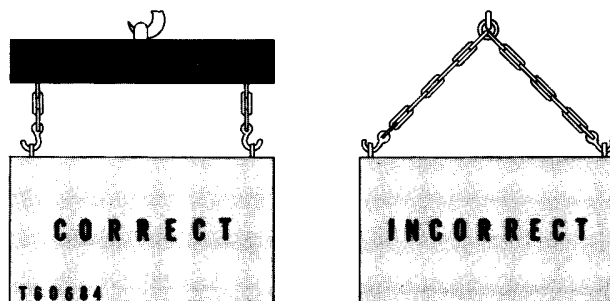
EFFECT OF DIRT IN BEARING

Lubricant must be changed at recommended intervals. Use clean containers. Before removing the filler cap, brush away the dirt.

Whenever hydraulic, fuel, lubricating oil, air or brake lines are to be disconnected, clean the point of disconnection as well as the adjacent area. As soon as a disconnection is made, cap, plug or tape the line or opening to prevent entry of foreign material. The same recommendations for cleaning and covering apply when access covers or inspection plates are removed.

DIESEL ENGINE REMOVAL

Unless otherwise specified, all engine removals should be accomplished using an adjustable lifting beam. For maximum efficiency of the lifting apparatus, all supporting members (chains and cables) should be parallel to each other and as near perpendicular as possible to the top of the cylinder head. See the illustration.



LIFTING ENGINE

In cases where it is necessary to remove the engine on an angle, it must be remembered that the lifting capacity of the lifting eyes on the engine diminishes as the angle between the supporting members and cylinder head becomes less than 90°. Lifting eyes and brackets should never be bent and should only have stress in tension. A length of pipe and a washer can be utilized, as shown, to help relieve these stresses on lifting eyes.



LIFTING EYE

CLEANING AND INSPECTION

Clean all parts thoroughly after they are removed and inspect them. Be sure all lubricant passages and oil holes are open. Badly worn or damaged parts should not be reinstalled on the machine. Cover all parts to keep them clean until they are installed and leave the new parts in the container.

Anti-friction bearings should receive special handling. As soon as a bearing is removed, cover it to keep dirt and abrasives out. Wash bearings in non-flammable cleaning solution and allow bearings to drain dry. Do not blow dry with air pressure or spin the bearings. Inspect the races and balls or rollers. Discard the bearings if they are pitted, scored or burned. If the bearing is serviceable, coat it with light oil and wrap it in clean paper. Do not unwrap new bearings until time of installation.

Replace all damaged O-rings and gaskets. In some assemblies and systems, it is recommended that O-rings be replaced when components are being re-assembled.

Inspect parts carefully. Refer to the topic, SPECIFICATIONS, as a guide for parts replacement.

SPECIFICATIONS

The manual SPECIFICATIONS sections contain dimensions for new parts and a suggested guide for amount of wear that might normally be considered a reasonable figure to use in determining when to replace parts. These wear dimensions are shown after the description, "permissible clearance". The specifications are provided to help determine when new parts should be installed. This information should not be regarded as the only factor in determining when to replace parts. Past performance should be the governing factor for part replacement. Even

though parts are worn they may still be satisfactory. The remaining service life of these parts must be considered when deciding if parts should be replaced. If a machine is disassembled for reconditioning, it is well to replace parts not completely worn out if the remaining service life is deemed short.

ASSEMBLY

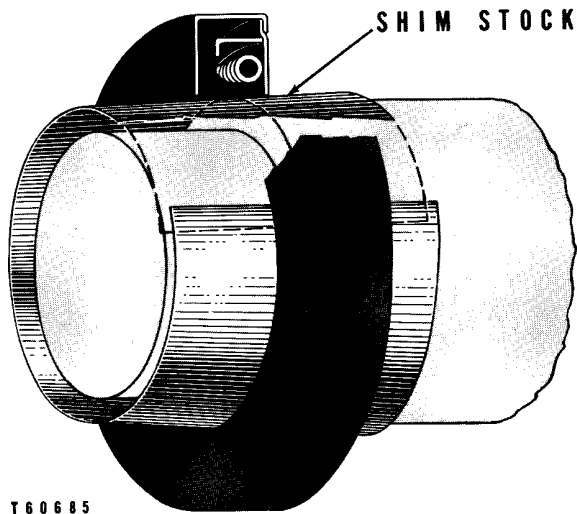
Clean the rust preventive compound from all machined surfaces of new parts before installing them. Be sure to install parts in the proper location and position.

Bearings which have a shrink fit should be heated in oil not to exceed 250° Fahrenheit, before attempting to install them. Other parts which require a shrink fit can be heated to a higher temperature in an electric oven or by other suitable means, but not to exceed a temperature which would remove hardness.

Parts which are heated to facilitate assembly must sometimes be allowed to cool and then pressed together again. Often the parts will separate as they cool and shrink.

When one part is pressed into another, use white lead or a suitable prepared compound to lubricate the mating surfaces. Tapered parts, however, should be assembled dry. Before assembling parts with tapered splines be sure the splines are clean, dry and free from burrs. Then press the parts together tightly.

When installing new bearings or reinstalling used bearings, lubricate the bearings liberally. Bearings that are to be preloaded must have a liberal film of oil over the entire assembly before the preloading procedure is carried out.



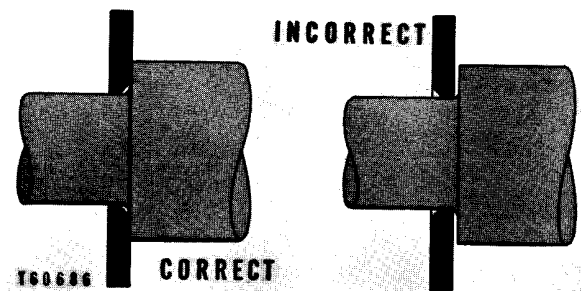
PROTECTING SEAL DURING INSTALLATION

Lubricant should be applied to the lip of all lip-type rubber seals before installation. This will prevent damage to the seal during the initial running until the oil being sealed has contacted the sealing face. Apply lubricant to an O-ring seal before installation to prevent rolling and twisting of the seal.

Silicone seals are subject to swelling when cleansed or immersed in kerosene or diesel fuel.

Packing-type seals should always be renewed if the contacting part is removed.

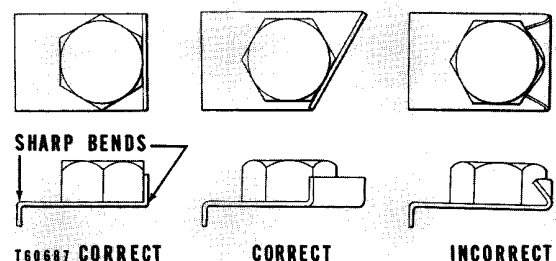
When installing a bearing, spacer or washer against a shoulder on a shaft, be sure the chamfered side is toward the shoulder. If the washer is turned in the wrong direction the radius may interfere and prevent washer from seating against the shoulder.



HOW TO INSTALL SPACER AGAINST SHOULDER ON SHAFT

Do not install sleeve bearings by driving them in with a hammer. Use a press if possible and be sure to apply the pressure directly in line with the bore. If a bearing must be driven in, use a bearing driver or a bar with a smooth flat end. If the sleeve bearing has an oil hole, be sure it is aligned with the oil hole in the part into which it is assembled.

Install gaskets where required and use new ones if necessary. Never use cork or felt gaskets or seals a second time. Be sure the holes in the gaskets correspond with the lubricant passages in the mating parts. If it is necessary to make gaskets, select stock of the proper type and thickness and be sure to cut sufficient holes in the right places. Blank gaskets can cause serious damage.



HOW TO INSTALL FLAT METAL LOCKS

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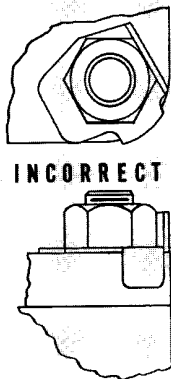
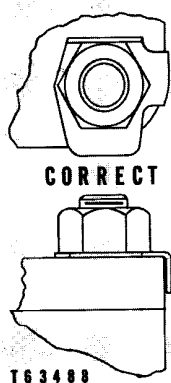
GENERAL INSTRUCTIONS

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Use bolts of the correct length. A bolt which is too long may "bottom" before the head is tight against the part it is to hold, and in addition the threads may be damaged when the bolt is removed.

If a bolt is too short, there may not be enough threads to hold the part securely.

Lockwashers, cotter pins or flat metal locks should be used to lock each nut and bolt. Flat metal locks must be installed properly to be effective. Bend one end of the lock sharply around the edge of the part. Bend the other end sharply against one flat surface of the nut or bolt head. Do not bend the lock against more than one side of the nut.



METHOD FOR LOCK POSITIONING AND BENDING

When assembling a machine, complete each step in turn. Do not leave one part partially assembled and start assembling some other part. Make all adjustments as recommended. Always check the job after it is completed to see that nothing has been overlooked.

PREPARATION FOR STARTING

After assembling a machine, lubricate it thoroughly. Fill the various compartments with the type and grade of lubricant recommended in the Operation and Maintenance Instructions.

Recheck the various adjustments by operating the machine before returning it to the job.

TORQUE VALUES

Unless reference is made in the copy to the topic, SPECIFICATIONS, the standard torque values listed in the chart should be used for standard heat-treated bolts and studs and steel nuts when assembling Caterpillar Equipment.

GENERAL USAGE

(Tightening Torques)

SIZE (INC and NF)	TORQUE (Pound Feet)	SIZE (INC and NF)	TORQUE (Pound Feet)
1/4"	6-12	3/4"	230-300
5/16"	13-23	7/8"	360-480
3/8"	27-37	1"	560-720
7/16"	40-60	1 1/8"	700-900
1/2"	65-85	1 1/4"	880-1120
9/16"	95-125	1 3/8"	1050-1350
5/8"	130-170	1 1/2"	1300-1700

HYDRAULIC VALVE BODIES

(Tightening Torques)

SIZE (INC and NF)	TORQUE (Pound Feet)
5/16"	13-17
3/8"	24-30
7/16"	39-47
1/2"	60-70
5/8"	118-142

DECIMAL EQUIVALENTS

The decimal dimensioning system provides a more accurate means of specifying dimensions and tolerances. The following table includes the standard conversions from fractions to decimals. Decimals are shown in the standard two, three and four place designation.

1/4.....	.02	.016	.0156	3/4.....	.52	.516	.5156
1/2.....	.03	.031	.0312	1.....	.53	.531	.5312
3/4.....	.05	.047	.0469	5/8.....	.55	.547	.5469
1.....	.06	.062	.0625	7/8.....	.56	.562	.5625
1 1/8.....	.08	.078	.0781	1 1/4.....	.58	.578	.5781
1 1/4.....	.09	.094	.0938	1 3/8.....	.59	.594	.5938
1 1/2.....	.10			1 1/2.....	.60		
1 3/4.....	.11	.109	.1094	1 3/4.....	.61	.609	.6094
2.....	.12	.125	.1250	2.....	.62	.625	.6250
2 1/8.....	.14	.141	.1406	2 1/4.....	.64	.641	.6406
2 1/4.....	.16	.156	.1562	2 3/4.....	.66	.656	.6562
2 3/4.....	.17	.172	.1719	3.....	.67	.672	.6719
3.....	.19	.188	.1875	3 1/8.....	.69	.688	.6875
3 1/8.....	.20	.203	.2031	3 1/4.....	.70	.703	.7031
3 1/4.....	.22	.219	.2188	3 3/4.....	.72	.719	.7188
3 3/4.....	.23	.234	.2344	4.....	.73	.734	.7344
4.....	.25	.250	.2500	4 1/8.....	.75	.750	.7500
4 1/8.....	.27	.266	.2656	4 1/4.....	.77	.766	.7656
4 1/4.....	.28	.281	.2812	4 3/4.....	.78	.781	.7812
4 3/4.....	.30	.297	.2969	5.....	.80	.797	.7969
5.....	.31	.312	.3125	5 1/8.....	.81	.812	.8125
5 1/8.....	.33	.328	.3281	5 1/4.....	.83	.828	.8281
5 1/4.....	.34	.344	.3438	5 3/4.....	.84	.844	.8438
5 3/4.....	.36	.359	.3594	6.....	.86	.859	.8594
6.....	.38	.375	.3750	6 1/8.....	.88	.875	.8750
6 1/8.....	.39	.391	.3906	6 1/4.....	.89	.891	.8906
6 1/4.....	.40			6 3/4.....	.90		
6 3/4.....	.41	.406	.4062	7.....	.91	.906	.9062
7.....	.42	.422	.4219	7 1/8.....	.92	.922	.9219
7 1/8.....	.44	.438	.4375	7 1/4.....	.94	.938	.9375
7 1/4.....	.45	.453	.4531	7 3/4.....	.95	.953	.9531
7 3/4.....	.47	.469	.4688	8.....	.97	.969	.9688
8.....	.48	.484	.4844	8 1/8.....	.98	.984	.9844
8 1/8.....	.50	.500	.5000	8 1/4.....	.98	.984	.9844
8 1/4.....				8 3/4.....	.99	.984	.9844
8 3/4.....				9.....	1.00	1.000	1.0000
9.....							

ENGINE

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ENGINE**ISSUED 9-62****SPECIFICATIONS**

See the topic, SPECIFICATIONS, in the GENERAL INSTRUCTIONS.

Number of cylinders 6
 Bore and stroke 4.5 x 5.5
 Firing order 1-5-3-6-2-4

Accessory Drive

Backlash between accessory drive
 gear and accessory drive idler
 gear002 - .011 in.
 Gear retaining nut torque 190 - 210 lb. ft.
 Idler gear:
 Backlash between accessory drive
 idler gear and camshaft gear002 - .011 in.
 End clearance002 - .009 in.
 Permissible end clearance018 in.
 Bearing bore 1.000 - 1.001 in.
 Bearing clearance001 - .003 in.
 Permissible bearing clearance007 in.

Camshaft

Backlash between camshaft gear
 and crankshaft gear001 - .009 in.
 Bearing journal diameter 2.3105 - 2.3115 in.
 Bearing clearance001 - .005 in.
 Permissible bearing clearance010 in.
 End clearance004 - .009 in.
 Permissible end clearance025 in.
 Retaining nut torque 165 - 185 lb. ft.

Connecting Rod

Bearing clearance0032 - .0061 in.
 Permissible bearing clearance012 in.
 Bolt nut torque 85 lb. ft.
 Center-to-center distance 8.999 - 9.001 in.
 Piston pin bearing bore 1.5014 - 1.5020 in.

Crankshaft

Main journal diameter 3.499 - 3.500 in.
 Permissible main journal wear008 in.
 Main bearing clearance (Earlier)0053 - .0082 in.
 Permissible bearing clearance (Earlier)013 in.
 Main bearing clearance (Later—
 with lead-tin overlay)0030 - .0059 in.
 Permissible bearing clearance (Later)010 in.
 End clearance0110 - .0180 in.
 Permissible end clearance035 in.
 Main bearing nut torque 155 lb. ft.
 Connecting rod journal diameter 2.999 - 3.000 in.
 Permissible journal wear007 in.
 Permissible out-of-round004 in.
 Pulley retaining bolt torque 210 - 250 lb. ft.

Cylinder Block

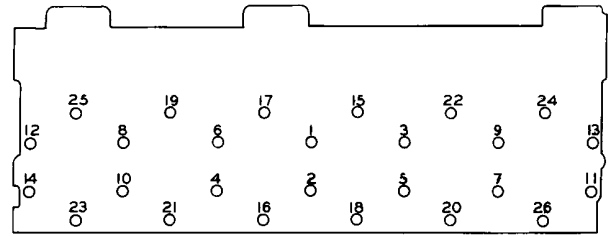
Main bearing original bore
 dimension 3.8155 - 3.8165 in.

Cylinder Head

Tighten bolts in sequence shown
 in illustration:
 Initial 85 lb. ft.
 Final 120 - 130 lb. ft.

Cylinder Liner (Use 5F7362 Adapter

Plate for Removal)
 Inside diameter 4.500 - 4.501 in.

**CYLINDER HEAD BOLT TIGHTENING SEQUENCE**

Permissible liner wear (increase in
 diameter at top of ring travel)015 in.
 Counterbore dimension in block400 - .402 in.
 Flange thickness403 - .405 in.

Engine Support

Bolt torque 175 - 225 lb. ft.

Fan Belt Tension

Slack at point midway
 between pulley 9/16 - 13/16 in.

Flywheel

Retaining bolt torque 118 - 142 lb. ft.

Flywheel Housing

Retaining bolt torque 60 - 70 lb. ft.

Fuel Injection Equipment

Fuel injection pump timing (before
 top center) 11°
 Fuel injection pump timing dimension
 setting (off engine using 2M5219
 Gauge) 4.2655 - 4.2695 in.

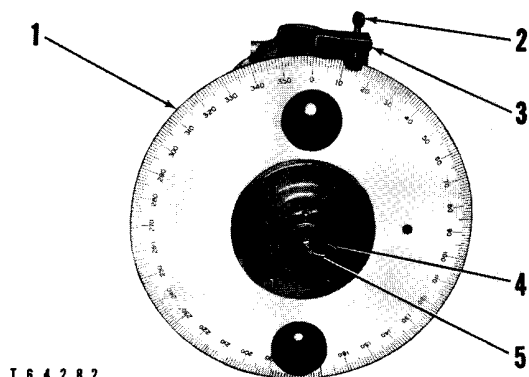
LIFTER SETTINGS IN DEGREES (OFF ENGINE)

LIFTER NO. (Numbered con- secutively front to rear)	TIMING PLATE DEGREES
1	180°
2	60°
3	300°
4	120°
5	240°
6	0°

Fuel injection pump timing dimension
 setting (on engine with pistons at top
 center using 2M5219 Gauge) 4.2264 - 4.2304 in.
 Fuel pump plunger length 2.5935 - 2.5937 in.
 Permissible wear (decrease in
 length of plunger)005 in.
 Injection pump retaining bushing
 torque 140 - 160 lb. ft.

ENGINE SPECIFICATIONS

ISSUED 9-62



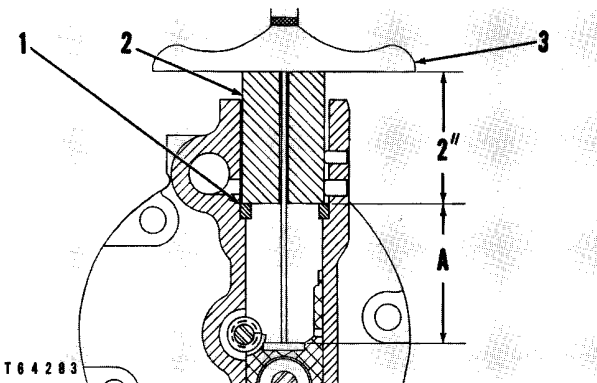
T 6 4 2 8 2

TIMING DIMENSION SETTING (OFF ENGINE)

1-1F8747 Plate Assembly.* 2-Lockscrew. 3-2M5220 Pointer Assembly. 4-Washer. 5-Bolt.

*On earlier 1F8747 Plate Assemblies the drive shaft must be replaced with a 2M5218 Shaft Assembly.

Injection nozzle orifice (use 5B2178
Drill for cleaning)025 in.
Nozzle retaining nut torque 100 - 110 lb. ft.



T 6 4 2 8 3

CHECKING TIMING DIMENSION SETTING

1-Spacer. 2-2M5219 Gauge. 3-4-5" depth micrometer.
A-Timing dimension setting.

If the reading is less than 4.2655 inches (off engine) replace the spacer (1) with a .004 inch thicker spacer.
If the reading is greater than 4.2695 inches (off engine) replace the spacer (1) with a .004 inch thinner spacer.

Fuel rack:

Bearing bore5007 - .5013 in.
Bearing clearance002 - .003 in.
Permissible clearance005 in.

Camshaft:

Bearing bore 1.8750 - 1.8760 in.
Bearing clearance0015 - .0035 in.
Permissible clearance010 in.

Fuel Transfer Pump

Clearance between gears and covers, total0010 - .0022 in.
Permissible clearance0035 in.
Bearing bore4950 - .4956 in.
Shaft diameter4936 - .4938 in.
Bearing clearance0012 - .0019 in.
Permissible bearing clearance003 in.

Plunger spring:

Free length 1 41/64 in.
Pounds force 2.16 - 2.54
When compressed to 1 1/32 in.
Drive gear retaining nut torque 10 lb. ft.
Retainer Flush within .002 in.

Generator

Generator	12V-24A DR1105129	24V-15A DR1105213
Brush spring tension	16 oz.	16 oz.
Rotation (viewing driving end)	clockwise	clockwise
Circuit	A	A
Field current (80°F)		
Volts	12	24
Amps	1.54-1.67	.94-1.02
Output (cold)		
Volts	14.0	28.5
Amps	24	15
Approximate RPM	1760	1800

Generator Belt Tension

Slack at point midway between pulleys 9/16 - 13/16 in.

Generator Regulator

Generator Regulator	12V-24A DR1119625	24V-15A DR1119626
Circuit	A	A
Polarity	Neg. ground	Neg. ground
Cut-out relay:		
Armature-yoke air gap (in.)	.020	.017
Point opening (in.)	.020	.032
Closing voltage range	11.8-13.0	22.8-25.2
Voltage regulator:		
Armature-yoke air gap (in.)	.067	.067
Point opening (in.)	.016	.016
Voltage setting range	14.1-14.9	27.4-29.4
Current regulator:		
Armature-yoke air gap (in.)	.075	.075
Current setting range (amps)	22-26	14-16

Governor

Backlash between drive gear and driven gear002 - .006 in.

Oil Pump

Clearance between the gears and covers, total002 - .004 in.
Shaft diameter7404 - .7410 in.
Bearing bore7420 - .7430 in.
Bearing clearance0010 - .0026 in.
Permissible bearing clearance005 in.
Idle gear:
Backlash between idler gear and crankshaft gear001 - .006 in.
Bearing bore 1.2505 - 1.2515 in.
Bearing clearance0020 - .0036 in.
Permissible clearance008 in.
Drive gear:
Backlash between drive gear and idler gear001 - .007 in.
Retaining nut torque 60 lb. ft.

Piston Pins

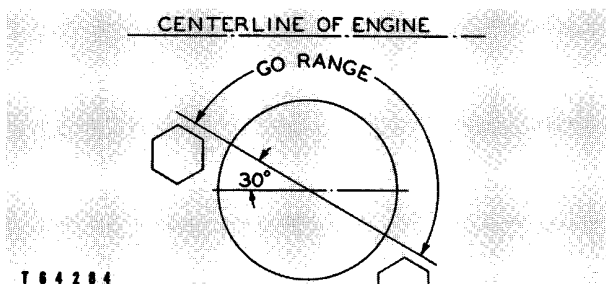
Clearance in rod bearing	.0009 - .0019 in.
Permissible clearance in rod	.006 in.
Permissible clearance in piston	.006 in.

Piston Rings

Piston ring side clearance:	
Top ring	.0030 - .0041 in.
Intermediate ring	.0025 - .0036 in.
Oil control ring	.0015 - .0035 in.
Permissible clearance (top compression ring)	.010 in.
Ring gap:	
Top ring	.017 - .023 in.
Intermediate ring	.018 - .024 in.
Oil control ring	.013 - .023 in.

Precombustion Chamber

Torque	140 - 160 lb. ft.
Glow plug torque	8 - 10 lb. ft.

**PRECOMBUSTION CHAMBER POSITIONING DIAGRAM**

The precombustion chamber gasket is supplied in two different thicknesses. With the proper thickness gasket and the recommended torque to tighten the chambers into the head, the glow plugs and electrical connections will be positioned in the "go" range. The "go" range is the position in which the glow plugs will be clear of fuel lines and other points of interference, when installed in the precombustion chamber.

The diagram can be used to position the precombustion chambers if new gaskets are installed.

Rocker Arm

Bearing bore	.7260 - .7266 in.
Clearance between shaft and bearing	.0010 - .0026 in.
Permissible clearance	.008 in.
Retaining bolt torque	120 lb. ft.

Starting Motor (24V — Delco-Remy 1113818)

Brush spring tension (minimum)	80 oz.
Rotation (viewing drive end)	Clockwise
No load test:	
Volts	23.0
Minimum Amps	60
Maximum Amps	90
Minimum RPM	7000
Maximum RPM	10,700

Lock test:

Approximate volts	3.5
Amps	500
Minimum torque	26 lb. ft.

Timing Gear Housing

Retaining bolt torque	27 - 37 lb. ft.
-----------------------	-----------------

Turbocharger (Schwitzer 4D used on normalized engines)

Torque on turbine housing and compressor cover band clamp bolts	10 lb. ft.
Torque on compressor back plate to bearing housing bolts	17 lb. ft.
Torque on impeller nut (cold)	30 lb. ft.
Torque on thrust bearing retaining bolts	5 lb. ft.
Clearance between impeller and compressor back plate	.020 - .022 in.
End clearance	.004 - .006 in.
Permissible end clearance	.008 in.
Bearing diameter (ID)	.6880 - .6883 in.
Bearing diameter (OD)	1.0585 - 1.0590 in.
Permissible bearing clearance (satisfactory if compressor wheel and/or turbine wheel have not rubbed housing or housings)	
Journal diameter	.6862 - .6865 in.
Housing bore diameter	1.0625 - 1.0630 in.
Thickness of thrust bearing	.025 - .026 in.
Thickness of thrust rings	.099 - .100 in.
Width of groove in sleeve	.128 - .130 in.
Side clearance in groove of sleeve with two piston rings installed	.002 - .006 in.
Maximum piston ring width	.062 in.
Maximum piston ring groove width (turbine)	.068 in.

Valves

Exhaust:	
Clearance (hot)	.025 in.
Stem clearance in bushing	.003 - .005 in.
Permissible clearance with new valve	.009 in.
Inlet:	
Clearance (hot)	.015 in.
Stem clearance in bushing	.003 - .005 in.
Permissible clearance with new valve	.009 in.

Valve Lifters

Lifter diameter	1.1225 - 1.1235 in.
Bore diameter	1.125 - 1.126 in.
Permissible clearance	.012 in.

Valve Spring

Pounds force	53.25 - 58.75
When compressed to	2 5/16 in.

Valve Timing

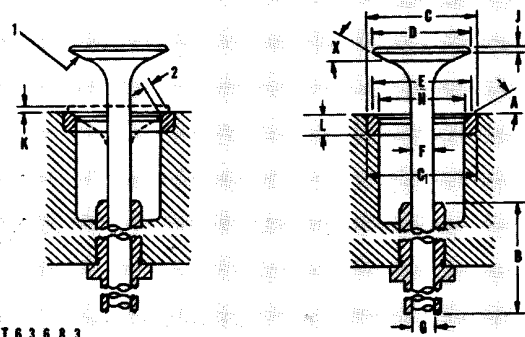
With valve clearances set correctly hot, dial indicator mounted above valve stem, reading taken with valve .075 in. off its seat.

ENGINE SPECIFICATIONS

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Exhaust opens (before bottom center)	20° 58'
Exhaust closes (before top center)	2° 19'
Inlet opens (after top center)	0° 15'
Inlet closes (after bottom center)	8° 31'

Valve and Valve Seat Specifications



VALVE AND VALVE SEAT SPECIFICATIONS

1-Valve face. 2-Valve seat. A-Valve seat angle. B-Valve guide length. C-Valve seat insert diameter. C₁-Bore for valve seat insert. D-Valve head diameter. E-Outside diameter of valve seat face. F-Valve stem diameter. G-Valve guide bore. H-Inside diameter of valve seat face. J-Valve lip thickness. K-Measurement from top of valve to face of head with valve seated. L-Depth of bore for valve seat inserts. X-Valve face angle.

	INLET	EXHAUST
A	30°	30°
B	3.36"	3.36"
C	2.0030"- 2.0040"	1.8780"- 1.8790"
C ₁	2.0000"- 2.0010"	1.8750"- 1.8760"
D	1.890"- 1.900"	1.766"- 1.776"
E	1.849"- 1.859"	1.725"- 1.735"
F	.3715"- .3725"	.3715"- .3725"
G	.3755"- .3765"	.3755"- .3765"
H	1 3/4"	1 5/8"
J	3/64" MIN.	3/64" MIN.
K	.128"	.117"
L	.448"-.450"	.448"-.450"
X	29 1/4°	29 1/4°

Maximum permissible valve seat width (2)095 in.

Water Pump

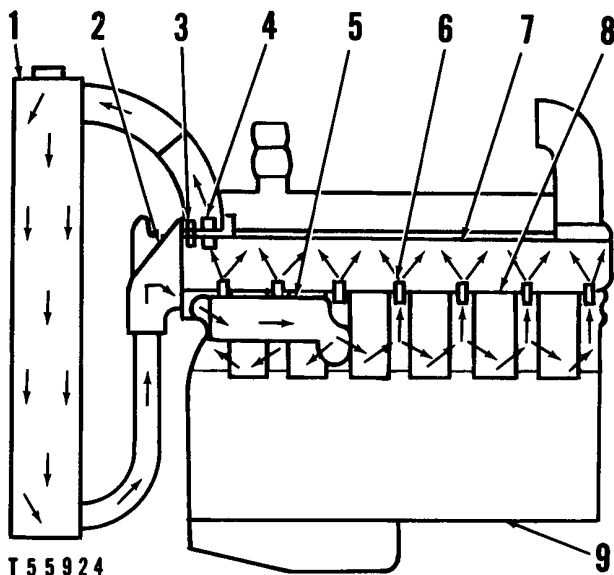
Impeller retaining nut torque 30 lb. ft.
Pulley retaining nut torque 118 - 142 lb. ft.

Water Temperature Regulator

Opening temperature for testing
(at atmospheric pressure) 175° - 186°

Cooling System

A centrifugal-type pump is mounted on the front face of the cylinder block. The pump and fan pulley are mounted on the same shaft and are belt driven from the crankshaft pulley.



COOLING SYSTEM

1-Radiator core. 2-Water pump. 3-By-pass. 4-Regulator.
5-Oil cooler. 6-Water director. 7-Cylinder head. 8-Cylinder liner. 9-Cylinder block.

Coolant flow from the outlet side of the centrifugal pump (2) is delivered directly to the oil cooler (5). After passing through the oil cooler, the coolant is directed to the cylinder block (9). Here the coolant flows around the cylinder liners (8) and through water directors (6) into the cylinder head (7). The water directors direct coolant to the area around the exhaust valves and precombustion chambers.

From the areas around the valves and precombustion chambers, the coolant flows through a cored passage in the head to the water temperature regulator (4). After the coolant becomes sufficiently warm, the regulator opens to permit flow into the radiator core (1). Coolant flows down through the radiator core during which time it is cooled by air flowing around the core. Then the coolant returns to the suction side of the pump. Thus, the cycle is completed with the regulator in the open position.

A limited amount of coolant continuously flows from the forward end of the cylinder block to the suction side of the water pump through the by-pass (3) to assure circulation when the regulator is closed and improve regulator response.

When the engine is equipped with a gasoline starting engine, some coolant is directed from the cylinder block to the starting engine water jacket and returns to the diesel engine cylinder head.

OVERHEATING

If difficulty is experienced with the engine overheating, check the following possible causes:

1. Coolant Level.

Insufficient coolant in the cooling system. If the coolant level has been allowed to fall so low that the coolant is no longer properly circulating the engine should be stopped immediately and allowed to cool before adding coolant. If there is still good circulation, add make-up coolant slowly while the engine is running. These precautions will minimize the possibility of cracking the cylinder head.

2. Fan.

Loose fan belt.

3. Radiator Clogging — (External)

Accumulation of leaves, mud and debris around the radiator core. In some cases, these accumulations can be removed by flushing with water or compressed air from the back of the radiator. It may become necessary, eventually, to remove the radiator to clean the core effectively.

4. Water Temperature Regulator.

Failure of water temperature regulator to open. Check the regulator for opening temperature as described in the topic, TESTING TEMPERATURE REGULATOR. Observe the amount of scale deposited on the regulator. Too much scale will obstruct operation. The engine should not be operated with the regulator removed.

5. Water Pump.

A badly corroded or worn water pump impeller, or loose drive gear.

6. Internal Clogging.

Excessive scale or sediment deposits in the heat exchanger, cylinder head or block. Such deposits can cause serious damage to the engine, by retarding the transfer of heat from the head and cylinders to the coolant. In such cases, the water temperature may not be above normal. However, loose scale and sediment may deposit in water passages to such an extent that circulation will be retarded, in which case the water temperature may go above normal.

ENGINE COOLING SYSTEM

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To check for lime and scale in the cooling system, remove one of the precombustion chambers and inspect the surface which comes in contact with the coolant. To remove hard scale, follow the directions in accordance with the Operation and Maintenance Instructions.

7. Continuous Overload.

Operating a machine at full governed speed under a continuous overload, which lugs the engine speed down below its rated speed, may also cause overheating. As a correction, the load should be reduced to allow the engine to operate at rated speed.

8. Altitude.

The altitude at which the machine is operating should be considered when boiling is encountered. The horsepower of the engine is decreased as the altitude increases. Also, the boiling point of water is lower at higher altitudes.

9. Water Temperature Indicator.

It may be that the indicator is not registering correctly. If the indicator is suspected of giving a false reading, install a new one and check the reading.

10. Overheating can also be caused by combustion gases escaping by the precombustion chamber gaskets or cylinder head gaskets and water ferrules and seals. The gases can accumulate in the cylinder head and block or restrict the flow of coolant through the cooling system. Thus, the coolant temperature rises rapidly and hot spots will exist in the cylinder head. The easiest method of checking for this condition is to carefully remove the filler cap and check for air bubbles. If air bubbles are present in the cooling system, check the precombustion chambers for tightness and, if necessary, replace the gaskets.

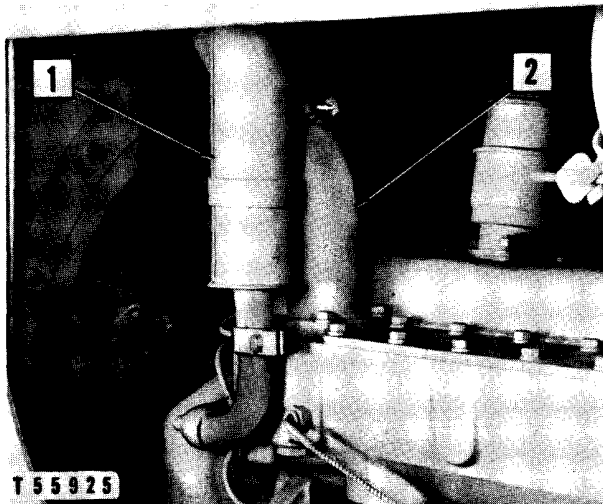
ENGINE

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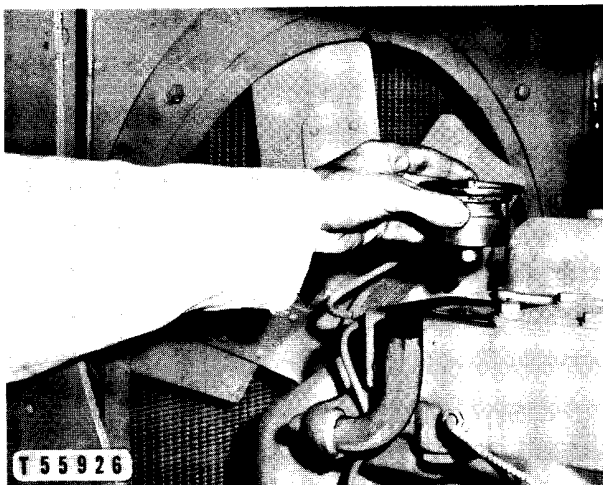
WATER TEMPERATURE REGULATOR**Water Temperature Regulator**

A water temperature regulator is used to restrict the flow of water through the radiator until the engine is at operating temperature. After this, the function of the regulator is to maintain an efficient engine operating temperature.

The temperature regulator within the diesel engine cooling system is designed to give an approximate minimum coolant temperature of 160° F. The maximum operating temperature will vary according to the air temperature and load factor, but should never exceed boiling temperature at the prevailing altitude.

REMOVAL AND INSTALLATION**REGULATOR HOUSING COVER**
☐ Remove

1-Starting engine muffler. 2-Elbow.

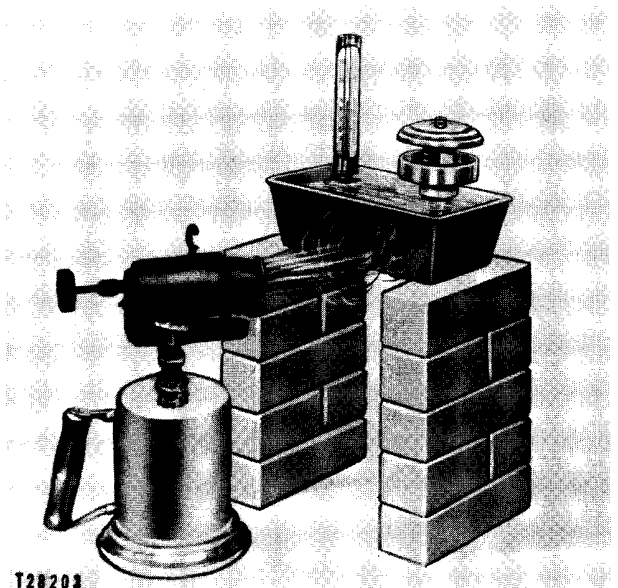
**REMOVING TEMPERATURE REGULATOR**

1. Drain engine coolant to a level below the regulator.
2. Loosen the clamp securing the hose to the elbow (2).
3. Lift out the water temperature regulator.
4. When installing the regulator, be sure it is properly seated in the bore.
5. Replace the gasket if it is damaged.

TESTING TEMPERATURE REGULATORS

The opening temperature of the regulator should be approximately 175° F., and the regulator should be fully open at 186° F. The regulator can be tested in the following manner:

1. Remove the regulator from the regulator housing.
2. Suspend the regulator in a pan of water as shown.
3. Apply heat to the pan and stir the water to maintain uniformity.
4. Observe the opening temperature of the regulator.

**TYPICAL SET-UP FOR TESTING WATER TEMPERATURE REGULATORS**

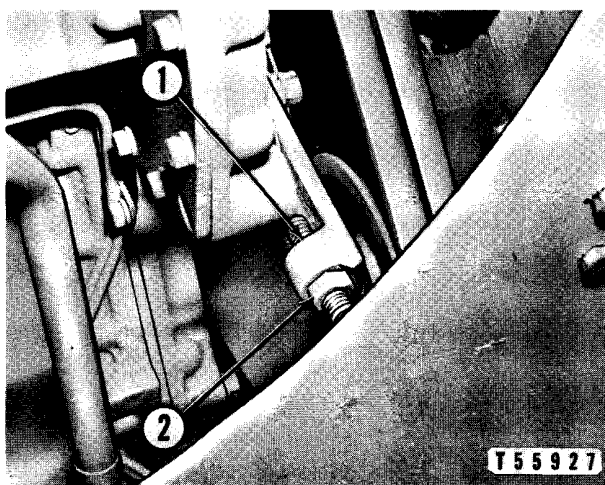
Fan Group

The fan is driven by two V-belts from a pulley on the crankshaft. The belts and pulleys should be kept clean. Accumulations of dirt and grease will cause overheating and excessive wear.

FAN BELT REPLACEMENT (Engine Not Equipped With Generator)

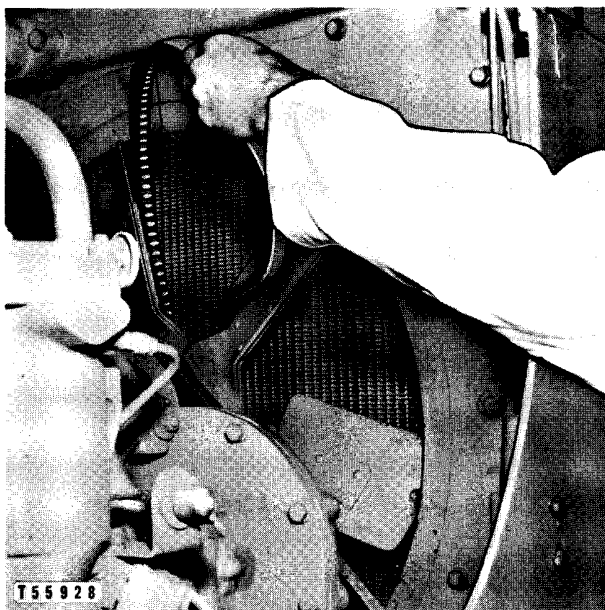
Loosen the bolts on the idler pulley adjusting bracket (1).

Loosen the locknut (2), unscrew the adjusting screw and slide the idler pulley toward the engine.



PREPARING TO REMOVE THE FAN BELT
1-Adjusting bracket. 2-Locknut.

Remove the old belts and install the new ones as shown.

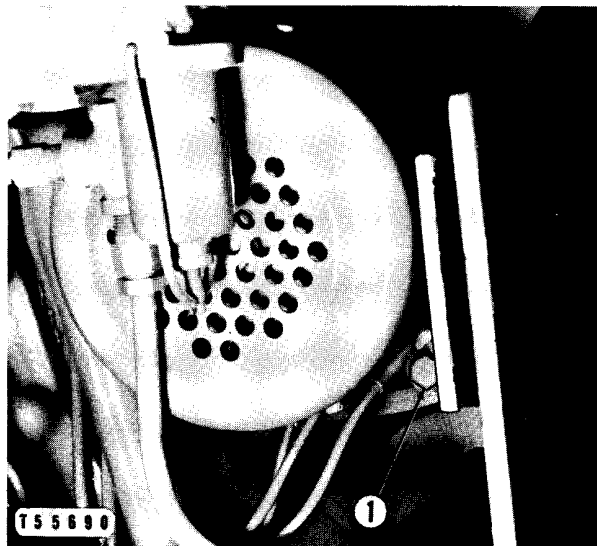


REPLACING THE FAN BELT

Adjust the fan belt tension as described in the Operation and Maintenance Instructions. See the topic, SPECIFICATIONS, for the correct fan belt tension.

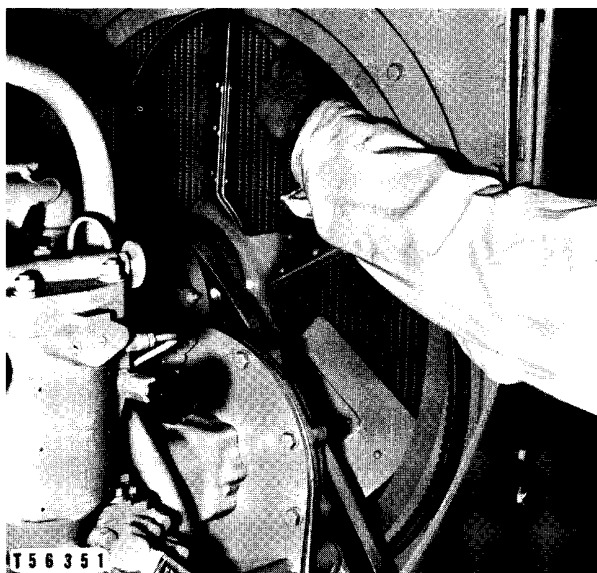
FAN BELT REPLACEMENT (Engine Equipped With Generator)

Loosen the bolt (1) and slide the generator toward the engine.



PREPARING TO REMOVE FAN BELT
1-Bolt.

Remove the old belts and install the new ones as shown.



REPLACING THE FAN BELT

Adjust the fan belt tension as described in the Operation and Maintenance Instructions. See the

**ENGINE
FAN GROUP**

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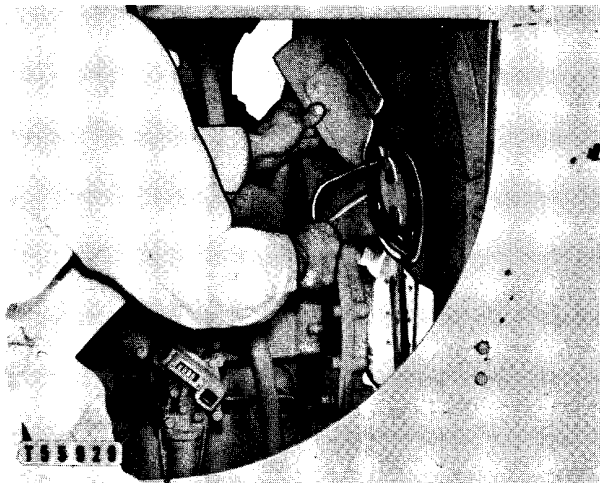
topic, SPECIFICATIONS, for the correct fan belt tension.

**FAN ASSEMBLY REMOVAL AND
INSTALLATION**

The fan is bolted to the water pump pulley.

Remove the fan belts. See the covering topic.

Remove the bolts securing the fan to the water pump pulley and remove the two halves of the fan as illustrated.

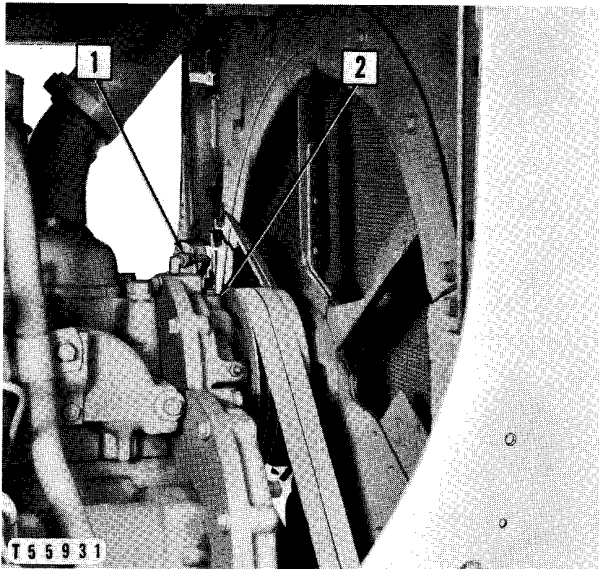
**REMOVING FAN**

Water Pump

The centrifugal type water pump is mounted on the front face of the cylinder block. The water pump is driven by two V-belts which also drive the fan.

REMOVAL AND INSTALLATION

1. Drain the coolant from the cooling system.
2. Remove the fan. See the topic, FAN REMOVAL AND INSTALLATION.

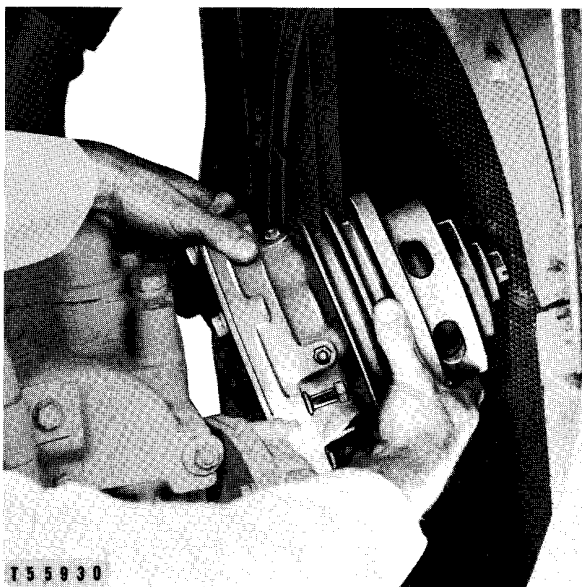


PREPARING TO REMOVE WATER PUMP

☐ Remove

1-Tube. 2-Water pump.

3. Disconnect the radiator lower hose from the water pump (2).

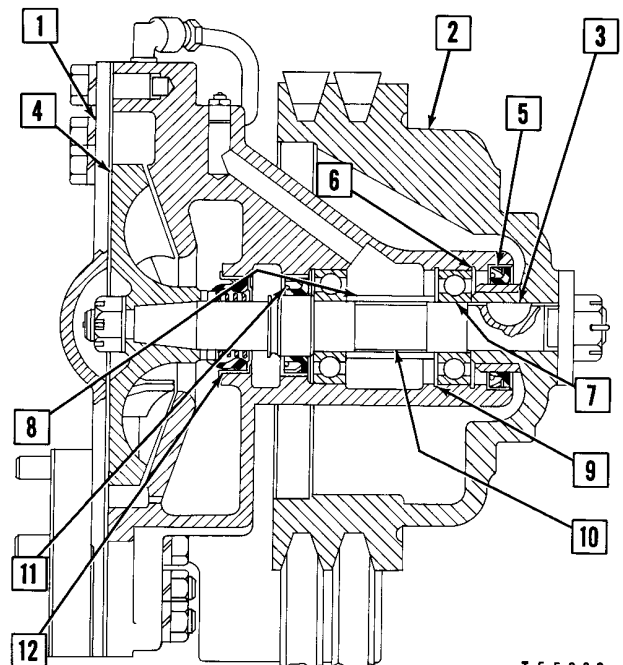


REMOVING WATER PUMP

4. Remove the bolts which secure the water pump to the cylinder block and remove the water pump as illustrated.
5. Prior to installation, inspect and replace all damaged gaskets.
6. See the topics, FAN BELT REMOVAL AND INSTALLATION, and SPECIFICATIONS, for the correct fan belt installation and adjustment.

DISASSEMBLY AND ASSEMBLY

1. Use an 8H695 Puller and an 8B7560 Step Plate to pull the pulley (2) from the impeller shaft as shown.
2. Use an 8H695 Puller and an 8B7560 Step Plate to pull impeller assembly (4) from the impeller shaft as shown.
3. Using a soft hammer, tap on the impeller end of the impeller shaft until the front bearing (7) is out of the housing.
4. Use an 8B7548 Push Puller with an 8H663 Bearing Puller Attachment and 8B7560 Step Plate to pull the front bearing from the shaft as shown.
5. After the bearing and shaft assembly has been removed, the bearing can be pressed off the shaft.



PREPARING TO DISASSEMBLE THE WATER PUMP

☐ Remove

1-Plate. 2-Pulley. 3-Key. 4-Impeller assembly. 5-Seal. 6-Lock ring. 7-Bearing. 8-Spacer. 9-Lock ring. 10-Shaft and bearing assembly. 11-Seal. 12-Carbon faced seal.

ENGINE WATER PUMP

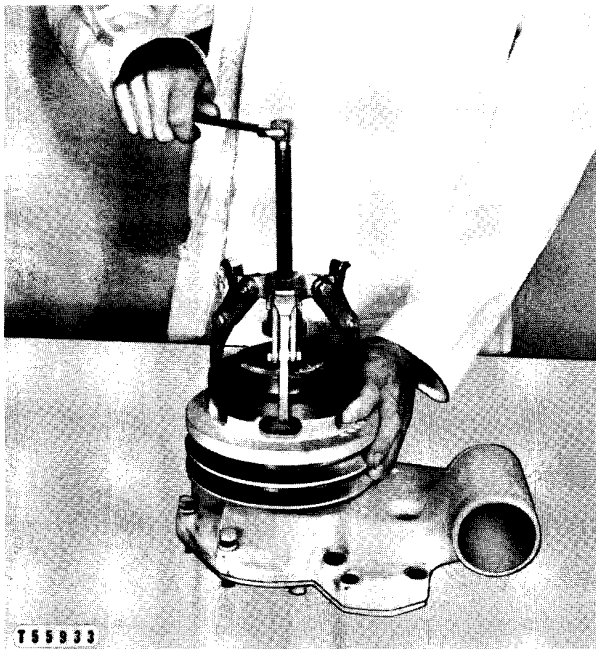
ISSUED 9-62

- When pressing the bearing onto the shaft, place the shielded side of the bearing so it will be toward the outside of the water pump housing.

NOTE

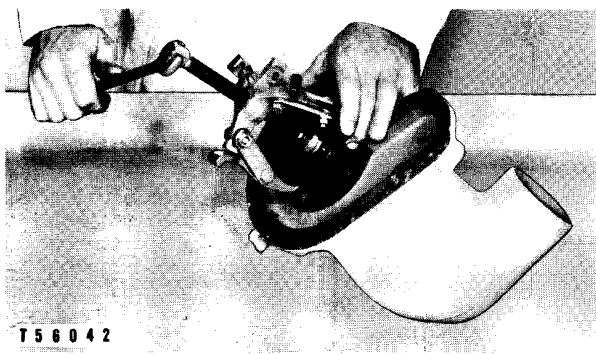
To avoid damaging the carbon-faced, self-adjusting seal, it should be installed after the shaft and bearings are assembled in the housing.

- Install the seal (11) in the housing. Install the seal so the lip will be toward the impeller.



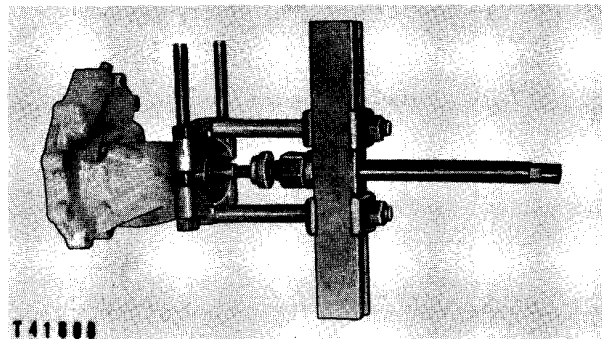
PULLEY REMOVAL

- Pack the rear bearing with ball and roller bearing lubricant and install the shaft and bearing assembly into the water pump housing.



IMPELLER REMOVAL

- Pack the bearing compartment with ball and roller bearing lubricant.



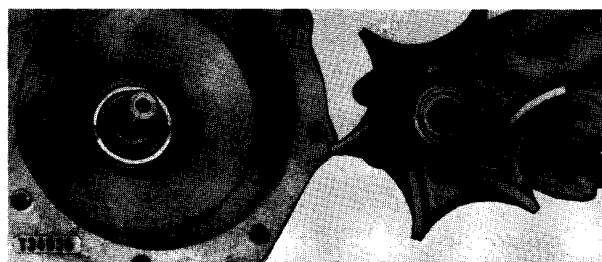
WATER PUMP BEARING REMOVAL

- Install the seal (5) with the wiping edge toward the inside.
- Prior to installing the pulley make certain the key is in the shaft. Also make certain the mounting bolts are in place in the water pump housing.
- See the topic, SPECIFICATIONS, for the proper pulley retaining nut torque.
- To install the impeller (4), place it on the shaft and torque the retaining nut. See the topic, SPECIFICATIONS, for the correct torque.

WATER PUMP SEAL REPLACEMENT

Water leaking from the drain opening on the bottom of the pump housing indicates that the water seal assembly should be replaced.

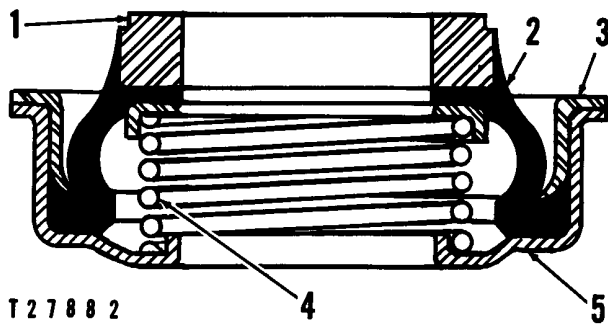
Inspect the ceramic bearing surface on the impeller which contacts the carbon thrust washer. The contact surface must be smooth and free of roughness or nicks.



INSPECTING BEARING SURFACE ON IMPELLER

The seal assembly consists of a carbon thrust washer (1), a flexible seal (2), and a seal spring (4) all retained in a seal cup (5) by the seal retainer (3). The carbon thrust washer (1) is cemented to the flexible seal (2). The entire assembly must be replaced as a unit and is pressed in the water pump housing.

It is not necessary to completely disassemble the pump to replace the water seal. The seal can be re-



T 2 7 8 8 2

SEAL ASSEMBLY

1-Carbon thrust washer. 2-Flexible seal. 3-Seal retainer.
4-Seal spring. 5-Seal cup.

moved by bending the flange on the seal and prying out, being careful not to damage the shaft or the bore in the housing.

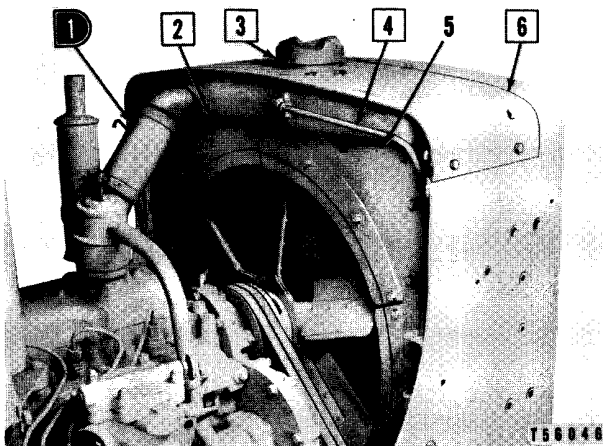
When installing a new seal assembly, coat the seal cup and housing bore with a suitable compound or lubricant to avoid shearing the brass cup and also to provide a positive seal against water leakage. Make sure the brass cup bottoms squarely in the bore so the carbon washer will bear evenly against the ceramic seal on the impeller.

After the seal is installed, see that the carbon thrust washer moves freely under finger pressure.

Radiator

REMOVAL AND INSTALLATION

1. Drain the coolant from the radiator.
2. Remove the hood. See the topic, HOOD REMOVAL.
3. Remove the fan. See the topic, FAN REMOVAL.



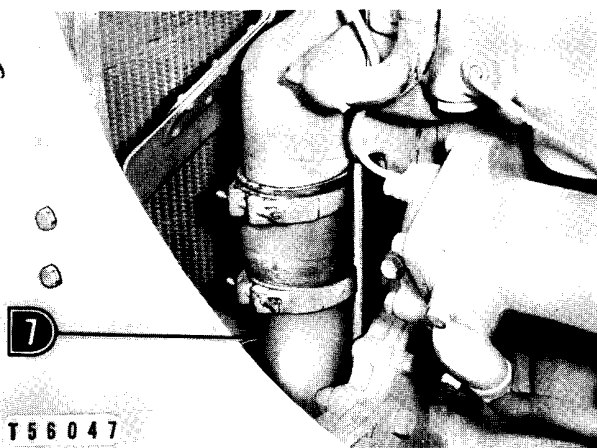
PREPARING TO REMOVE RADIATOR

☐ Remove ☒ Disconnect

1-Hose. 2-Left shield assembly. 3-Cap. 4-Tube. 5-Right shield assembly. 6-Radiator top guard.

NOTE

Remove the right shield assembly (5) before the left shield assembly (2).

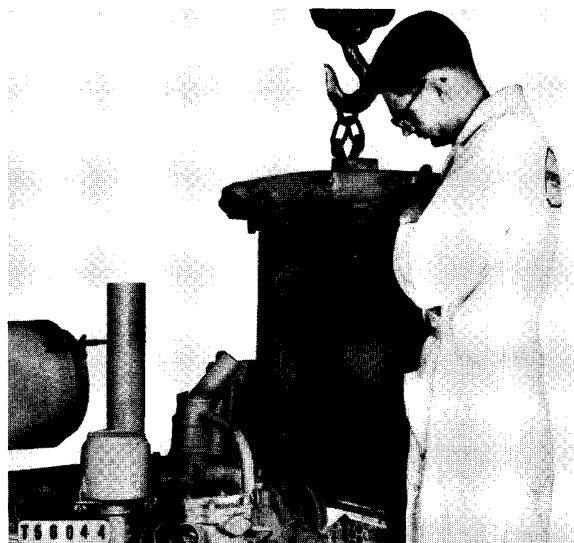


DISCONNECTING LOWER PORTION OF RADIATOR

☒ Disconnect

7-Elbow.

4. Remove the bolts that secure the upper tank to the radiator guard.
5. Attach a suitable lifting hoist and lift out the radiator as illustrated.



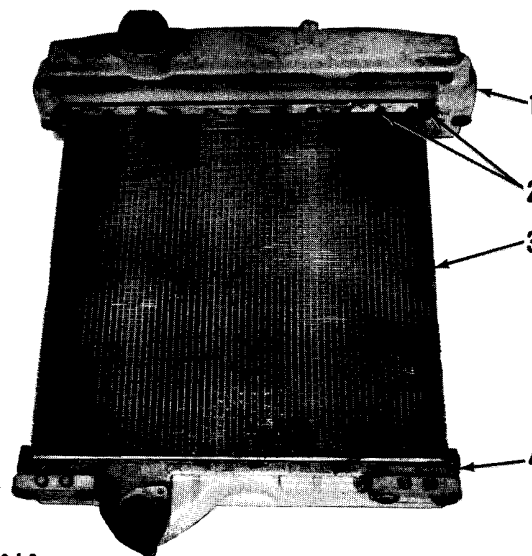
REMOVING THE RADIATOR

DISASSEMBLY AND ASSEMBLY

With the radiator removed from the engine, the radiator can be disassembled in the following manner:

1. Remove the top tank (1) by removing the bolts holding the reinforcing strip (2) and the radiator core (3) to the top tank.
2. Remove the bottom tank (4) in the same manner as the top tank.
3. Hole spacing in the header plates of the radiator core is such that the core can be reversed to extend the service life.

Clean the radiator core of all accumulations of debris between the fins and tubes with water or compressed air. Such accumulations decrease the efficiency of the cooling system and may cause the engine to overheat.



RADIATOR ASSEMBLY

1-Tank. 2-Strips. 3-Core. 4-Tank.

ENGINE RADIATOR

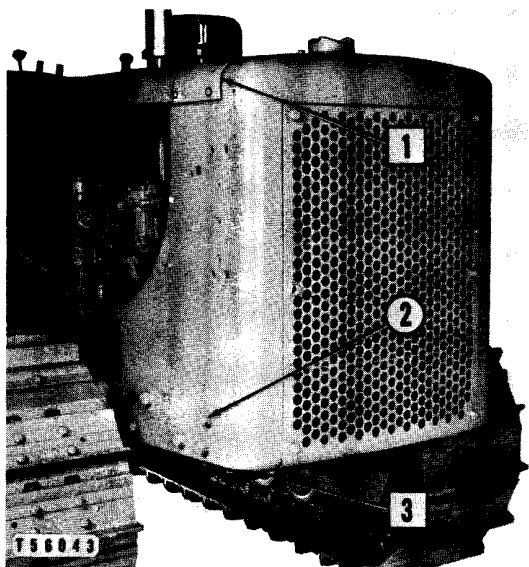
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The inside of the radiator cores, bottom tank, and top tank should be cleaned with commercial radiator cleaner according to the instructions furnished by the manufacturer.

Care should be taken, when assembling the radiator, to see that a perfect seal exists between the radiator core and the top and bottom tanks.

RADIATOR AND GUARD REMOVAL

1. Drain the coolant from the radiator.
2. Remove the hood. See the topic, HOOD REMOVAL.

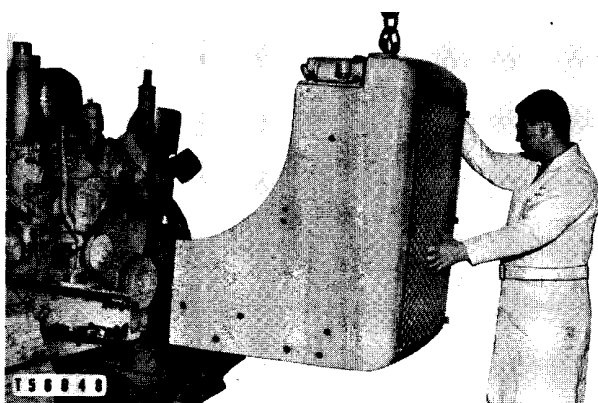


PREPARING TO REMOVE RADIATOR AND GUARD

☐ Remove

- 1-Radiator top guard. 2-Radiator guard bolts (twelve).
3-Radiator bottom guard.

3. Disconnect the radiator inlet and outlet elbows from the radiator.
4. Attach a suitable lifting device and hoist and support the weight of the guard and radiator. The guard and radiator weigh approximately 600 lbs.



REMOVING THE RADIATOR AND GUARD

5. Remove the bolts (2) that secure the radiator guard assembly to the frames.
6. Move the radiator guard assembly forward to remove it.

SEALED PRESSURE OVERFLOW

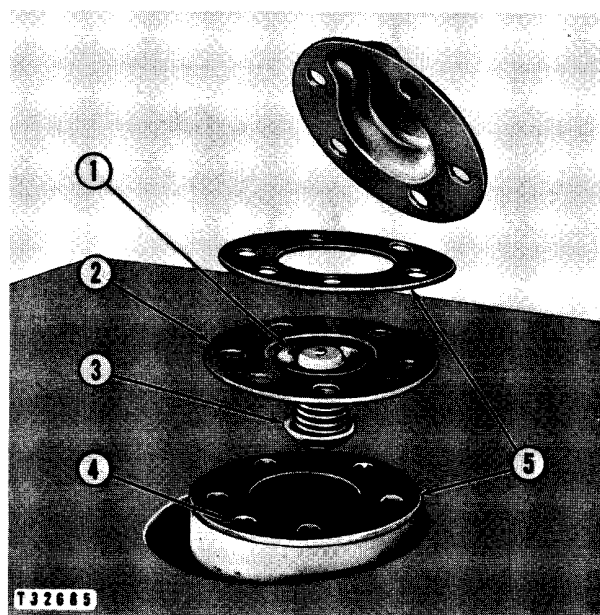
A sealed pressure overflow assembly is used to prevent loss of coolant through the radiator overflow tube when operating at an angle.

Due to expansion of the coolant by rising temperature, a pressure will be built up in the cooling system each time the engine is started. When the pressure rises above 6 PSI, the pressure relief valve (1) opens to relieve the pressure, or, if the cooling system has been overfilled, allow some coolant to escape through the radiator overflow tube. The valve closes when the temperature of the coolant levels off and remains closed unless there is a further increase in pressure.

When the temperature of the coolant falls, a vacuum will result in the cooling system. The vacuum release valve (3) opens when the vacuum reaches 1 PSI and lets in air through the overflow tube. This vacuum release valve also functions when draining the cooling system.

The relief assembly can be removed by taking out the machine screws that hold the seal (2) in place and removing the housing and seal plate. The gaskets (5) should be replaced if leaking.

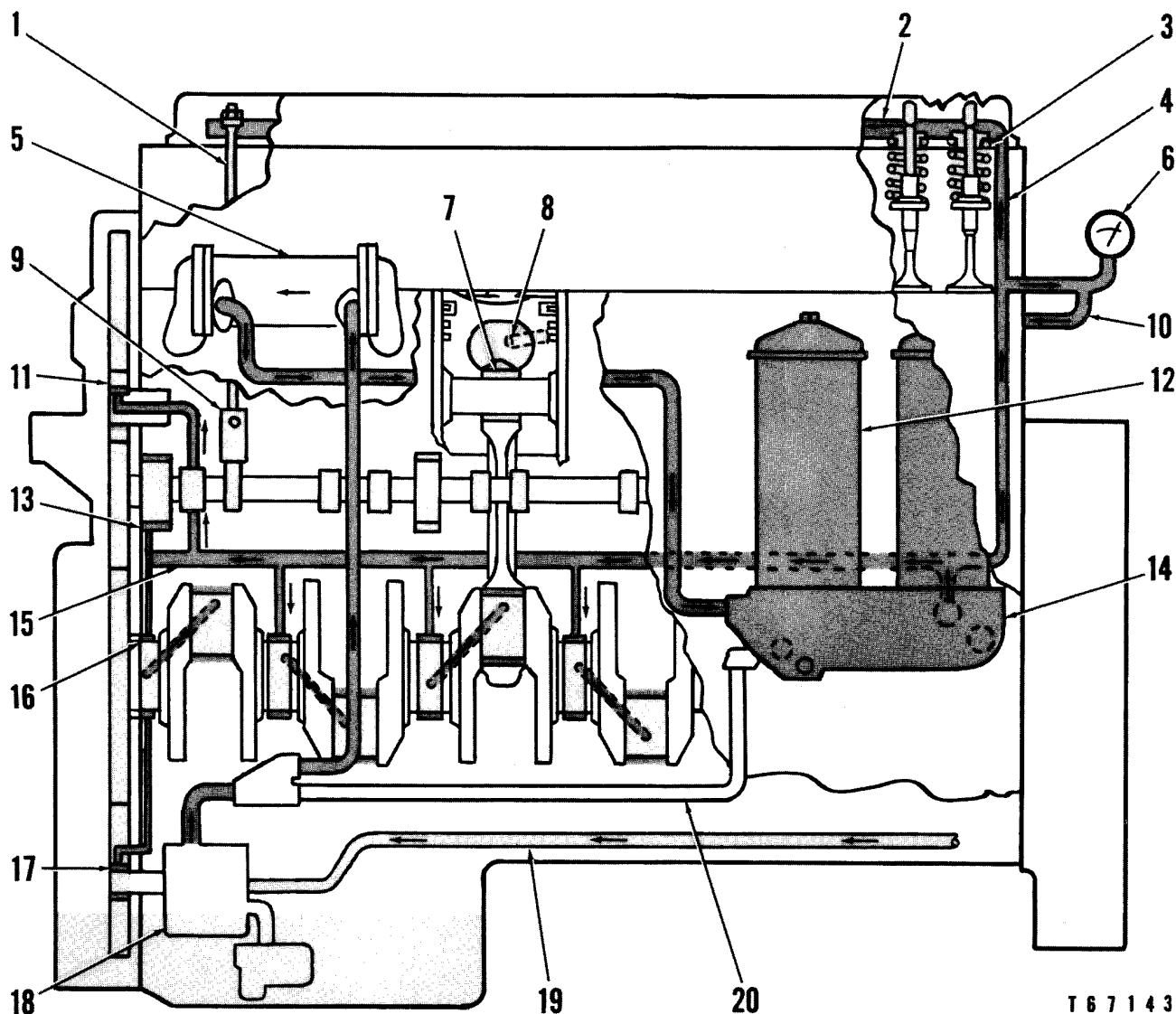
When reinstalling the gaskets (5), one on each side of the seal, make sure the correct hole in the gaskets aligns with the opening (4) to the overflow tube.



SEALED PRESSURE OVERFLOW

- 1-Pressure relief valve. 2-Seal. 3-Vacuum release valve.
4-Opening to overflow tube. 5-Gaskets.

Lubricating System



T 6 7 1 4 3

FLOW OF LUBRICATING OIL (SCHEMATIC)
(Left Side View of Engine)

- | | | | |
|--------------------------------|--|---|--------------------------------------|
| 1-Push rod. | 7-Piston pin bearing. | 11-Accessory drive idler gear stub shaft bearing. | 16-Crankshaft main bearings (seven). |
| 2-Passage in rocker arm shaft. | 8-Oil control ring passage. | 12-Oil filter (two). | 17-Oil pump idler gear bearing. |
| 3-Valve spring. | 9-Valve lifter. | 13-Camshaft front bearing. | 18-Oil pump. |
| 4-Passage to rocker arm shaft. | 10-External supply line to governor and fuel injection pump. | 14-Oil filter base. | 19-Oil scavenge tube. |
| 5-Oil cooler. | | 15-Oil manifold. | 20-Oil cooler by-pass line. |
| 6-Oil pressure gauge. | | | |

The oil pump (18) contains a scavenge pump and a pressure pump. Oil that does not return to the oil pan is collected through the scavenge tube (19) and dumped into the oil pan.

Oil is pumped from the oil pan to the oil cooler (5) and to the oil filters (12), except on initial starting where the oil is by-passed through the external line (20). See the topic, OIL FILTER. After the oil passes through the filters (12), it enters the oil distribution manifold (15). The oil manifold supplies lubricating oil to the main bearings (16), camshaft front bearing (13), and to the passage (4).

The intermediate and rear camshaft bearings are splash lubricated. The connecting rod bearings receive lubrication from the same source as the main bearings, through drilled passages in the crankshaft.

The passage (4), which supplies oil to the rocker arm shaft from the oil distribution manifold (15), is located in the rear of the cylinder block. Pressure oil to the oil pressure gauge (6) is supplied by this passage. The fuel injection pump and governor, which are mounted on the right side of the engine, receive lubrication through the external line (10).

ENGINE LUBRICATING SYSTEM

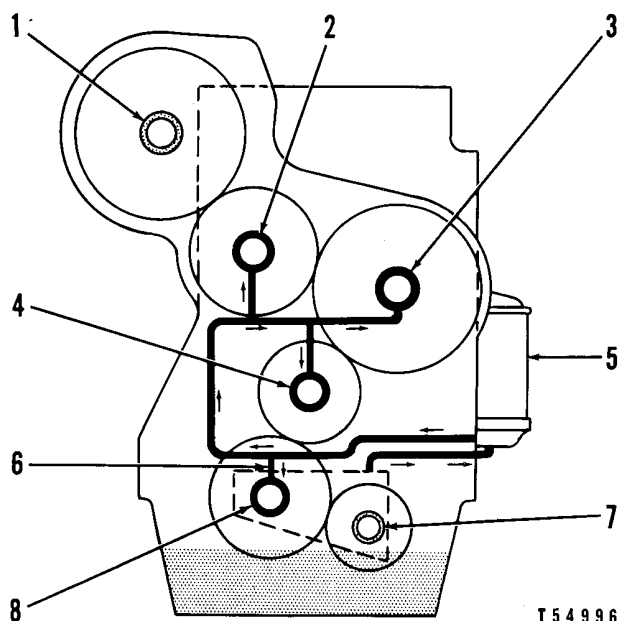
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TIMING GEAR LUBRICATION

All of the timing gear bearings are lubricated by passages in the block with the exception of the oil pump idler gear bearing (8) and the accessory drive gear bearing (1).

The oil pump idler gear bearing (8) is lubricated by pressure filtered oil through the supply line (6).

The accessory drive gear bearing (1) is lubricated by drainage from the governor housing, fuel pump housing and the accessory drive housing.



TIMING GEAR LUBRICATION

1-Accessory drive gear bearing. 2-Accessory drive idler gear bearing. 3-Camshaft gear bearing. 4-Crankshaft gear bearing. 5-Oil filter. 6-Oil pump idler gear bearing supply line. 7-Oil pump drive gear bearing. 8-Oil pump idler gear bearing.

OIL PRESSURE

When the engine is warm and running at rated speed, the gauge should register in the "operating range." A lower pressure reading is normal at low idling speeds.

If for any reason the oil gauge registers no pressure, the engine should be stopped immediately until the difficulty can be determined and corrected.

A small orifice in the gauge connection prevents rapid gauge fluctuation. This orifice should be checked for dirt, in the event the gauge becomes inoperative.

If the oil pressure gauge indicates low or no oil pressure, check for the following:

1. Low oil level.
2. Clogged oil filter.
3. Defective oil gauge, or a clogged or broken tube to the gauge.
4. Clogged suction bell screen.
5. Leaking connections.
6. Loose bearings.
7. Pressure regulating valve worn or stuck open, thus allowing oil to be by-passed into the oil pan.
8. Worn oil pump gears.
9. Clogged oil cooler, possibly causing excessively high oil temperatures, resulting in low viscosity and consequently low oil pressure.

Oil Pan and Oil Pan Plate

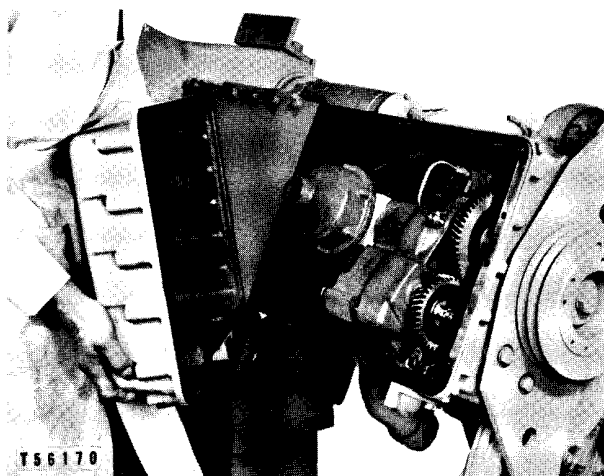
OIL PAN REMOVAL AND INSTALLATION

The oil pan can be removed without removing the engine from the tractor. The engine is shown mounted on the 9F3829 engine positioning stand.

NOTE

In order to remove the oil pan plate, the engine must be removed from the tractor.

1. Drain the lubricant from the oil pan.
2. Loosen the bolts which secure the pan to the cylinder block, plate and timing gear cover.
3. Separate the oil pan gasket from the pan so that the gasket will remain with the oil pan plate.

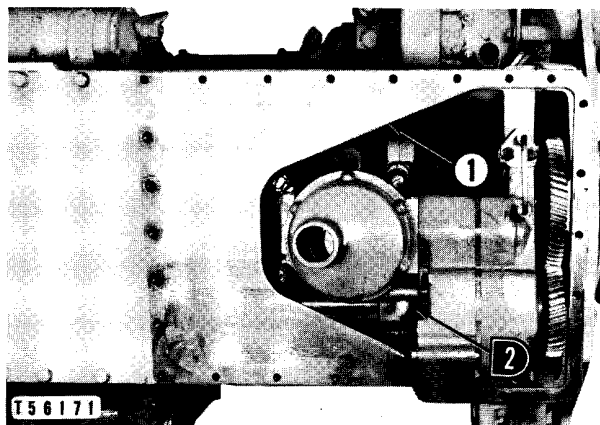


REMOVING THE OIL PAN

4. Inspect the gasket for damage.
5. Install the pan and complete the installation.
6. Refill the oil pan with lubricant in accordance with the Operation and Maintenance Instructions.

OIL PAN PLATE REMOVAL AND INSTALLATION

1. Remove the engine from the tractor. See the covering topic. The engine is shown mounted on the 9F3829 engine positioning stand.
2. Remove the oil pan. See the covering topic.

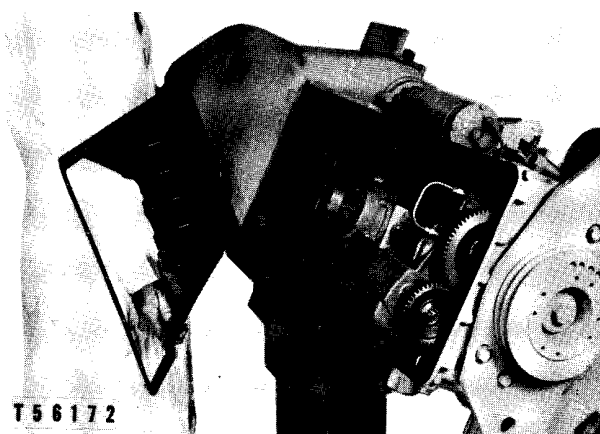


PREPARING TO REMOVE OIL PAN PLATE

D Disconnect

1-Oil pan plate. 2-Scavenge elbow.

3. Remove the bolts which secure the oil pan plate (1) to the cylinder block.
4. Remove the oil pan plate as shown.
5. Prior to installation, inspect and replace all damaged gaskets.



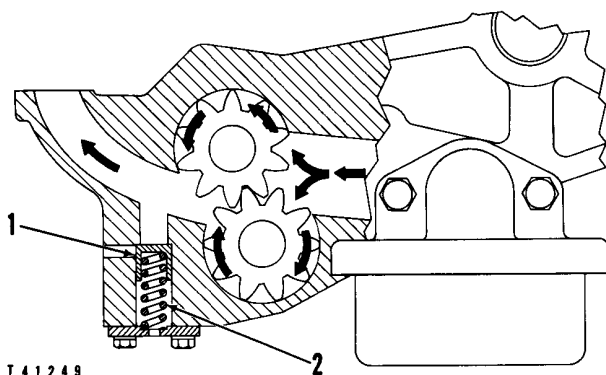
REMOVING THE OIL PAN PLATE

Oil Pump

INTRODUCTION

The oil pump is mounted on the bottom face of the cylinder block at the front of the engine. The pump is a two-section, positive displacement, gear-type pump. The rear set of gears scavenges oil from the rear of the engine and dumps it into the oil pan sump. The front set of gears supplies pressure lubrication to the engine through passages in the cylinder block.

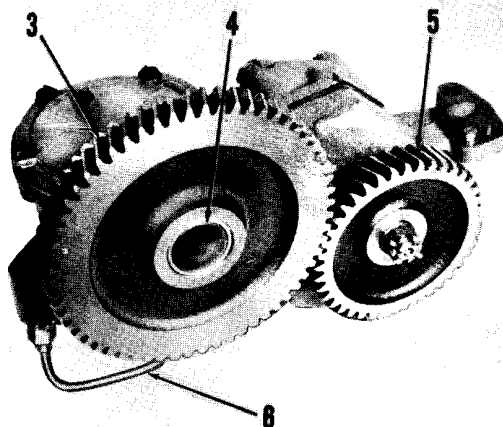
The pump is the outlet pressure control by-pass valve type. A by-pass valve is contained in the pump support body. This valve consists of a spring (2) and plunger (1) which limit the maximum oil pressure to 35-45 PSI. When the pressure of the oil overcomes the force of the spring, the plunger is unseated, dumping oil, thus maintaining a uniform maximum oil pressure.



BY-PASS VALVE (SCHEMATIC)
1-Plunger. 2-Spring.

NOTE

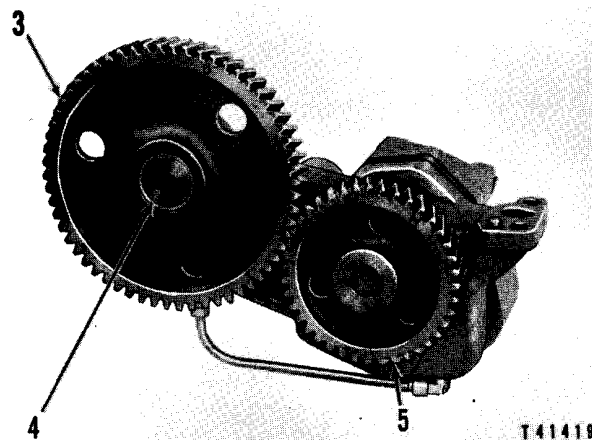
On later engines, a tube (6), connected to an oil



OIL PUMP DRIVE
(Later engines)
3-Idler gear. 4-Stub shaft. 5-Oil pump drive gear. 6-Tube.

passage in the cylinder block and to the oil pump idler gear stub shaft, furnishes lubrication for the idler gear bearings.

The oil pump drive gear (5) is driven by the idler gear (3) which is a slip fit on the stub shaft (4). The stub shaft is pressed into the oil pump body assembly. The idler gear (3) is held in place on the stub shaft by a boss on the inside of the timing gear housing. The idler gear is driven by the crankshaft gear.

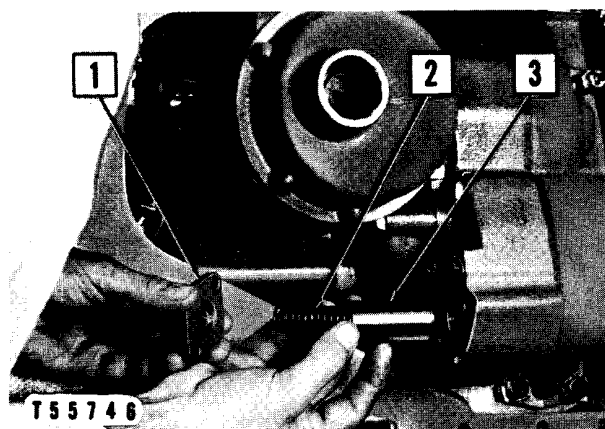


OIL PUMP DRIVE
(Earlier engines)
3-Idler gear. 4-Stub shaft. 5-Oil pump drive gear.

NOTE

On earlier engines, a tube from the pressure side of the pump to the oil pump idler gear stub shaft furnishes lubrication for the idler gear bearings.

BY-PASS VALVE



REMOVING BY-PASS VALVE
☐ Remove
1-Flange. 2-Spring. 3-Plunger.

ENGINE OIL PUMP

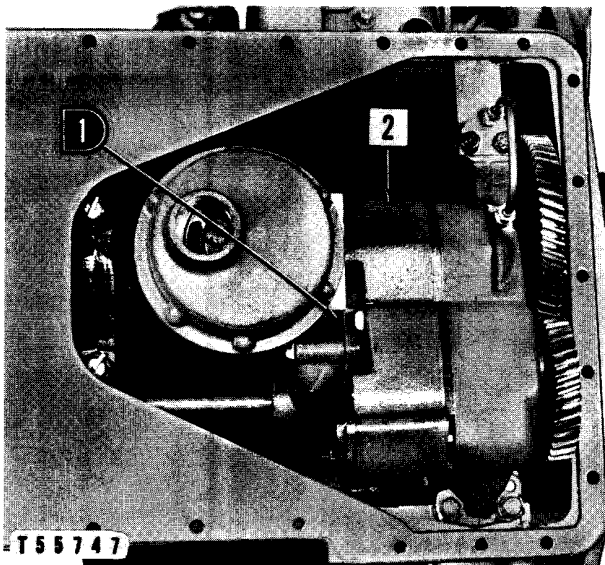
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The by-pass valve can be removed without removing the oil pump. Drain the oil and remove the oil pan.

The valve may fail to function properly if foreign material gets between the plunger and the seat. After cleaning the contact surfaces, check that the seat is smooth and flat, and that the sealing surfaces of the plunger are in good condition. If the valve is functioning properly, the oil pressure should vary only slightly as the engine speed changes.

REMOVAL AND INSTALLATION

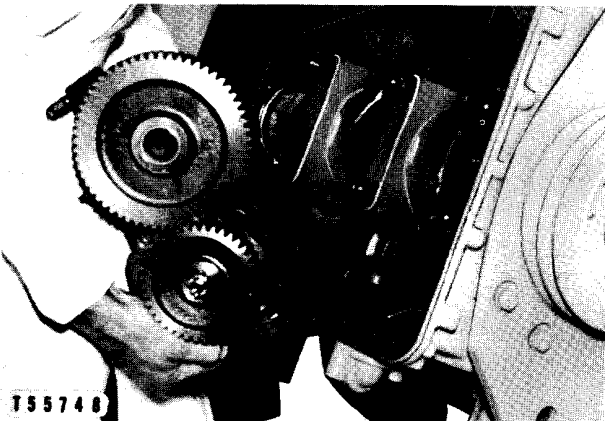
1. Drain the oil and remove the oil pan.
2. Install a new scavenge inlet elbow O-ring seal upon assembly.



PREPARING TO REMOVE THE OIL PUMP

☒ Disconnect ☐ Remove

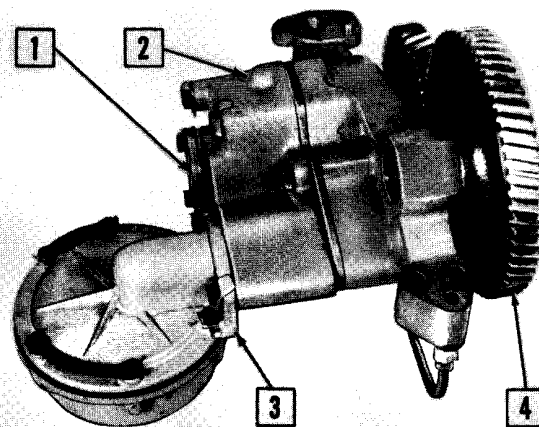
1-Scavenge inlet elbow. 2-Oil pump.



REMOVING OIL PUMP

DISASSEMBLY

See the topic, BY-PASS VALVE, for an alternate removal of the flange (1), spring and plunger (8).

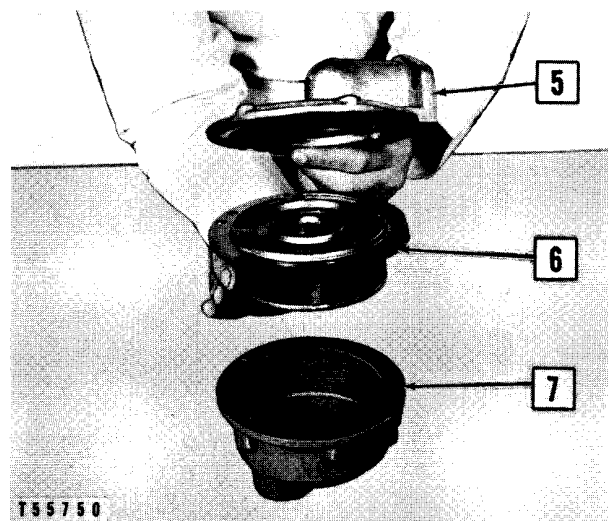


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PREPARING TO DISASSEMBLE OIL PUMP

☐ Remove

1-Flange over by-pass plunger and spring. 2-Scavenger pump body. 3-Suction bell and adapter. 4-Idler gear.



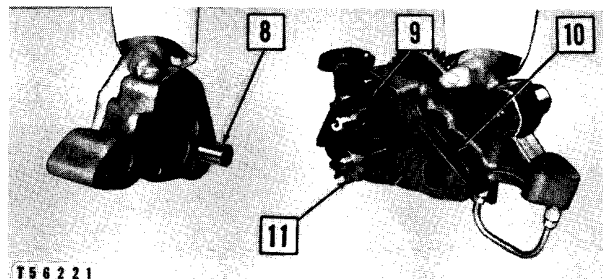
T55750

DISASSEMBLY OF SUCTION BELL

☐ Remove

5-Bell. 6-Screen assembly. 7-Cover.

1. Clean the screen (6) and inspect for damage.



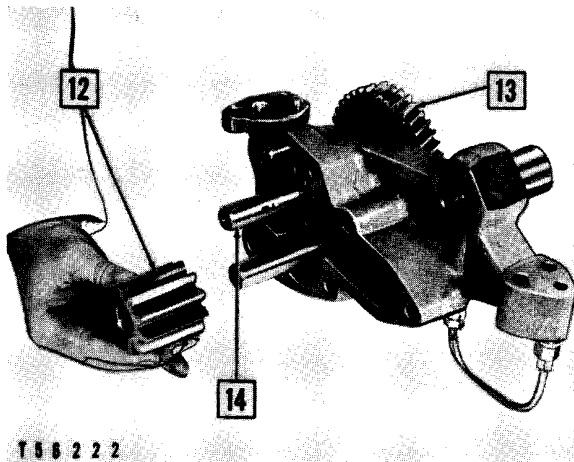
T56221

DISASSEMBLY OF SCAVENGE OIL PUMP

☐ Remove

8-Plunger. 9-Drive gear. 10-Spacer. 11-Driven gear.

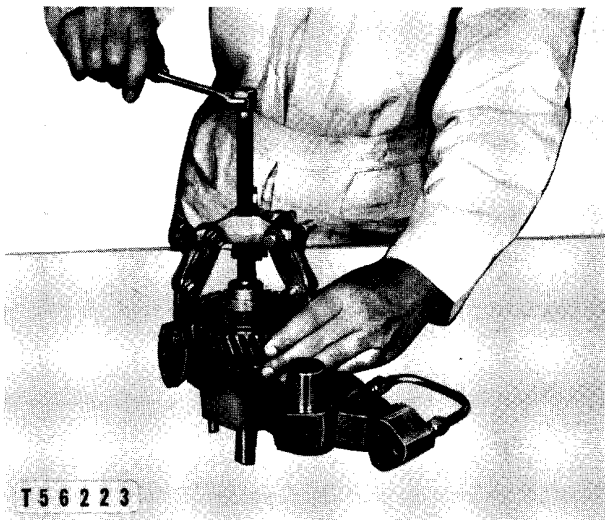
2. Prior to installation, place new gaskets on either side of the screen.



DISASSEMBLY OF OIL PUMP

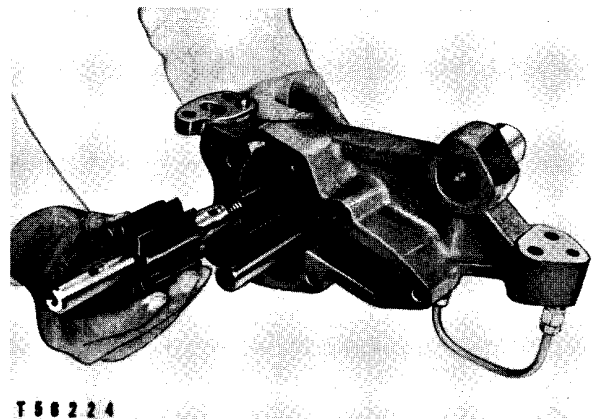
☐ Remove

12-Driven gear. 13-Drive gear. 14-Shaft assembly.



PULLING THE OIL PUMP DRIVE GEAR

3. Use an 8H695 Puller and 8B7560 Step Plate to remove the drive gear. When replacing the gear, torque the retaining nut to the value given in the topic, SPECIFICATIONS.
4. Before the shaft assembly can be removed from the pump body, it is necessary to remove the key.
5. Inspect the bearings in the pump body assembly for excessive wear and replace if necessary. See the topic, SPECIFICATIONS.
6. Ordinarily, oil pump gears should not have to be replaced unless they have worn sufficiently to cause a considerable drop in oil pressure or unless they have been damaged.



REMOVING THE SHAFT ASSEMBLY

7. Replace all worn or damaged parts, and assemble the pump in reverse order of disassembly. Check that the shafts turn freely with no binding or drag on the gears.

NOTE

If the gears bind, loosen the nuts and bolts slightly and relocate the pump bodies by tapping them lightly until the shafts turn freely.

Oil Coolers

ENGINE OIL COOLER

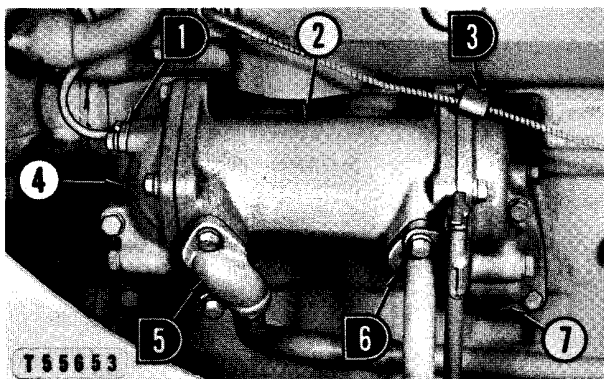
The engine oil cooler is mounted horizontally on the left front of the diesel engine block.

A portion of the coolant from the water pump circulates through the core of the cooler before entering the engine block.

Oil enters the cooler at the bottom, circulates around the tubes in the core and returns to the engine.

Removal and Installation

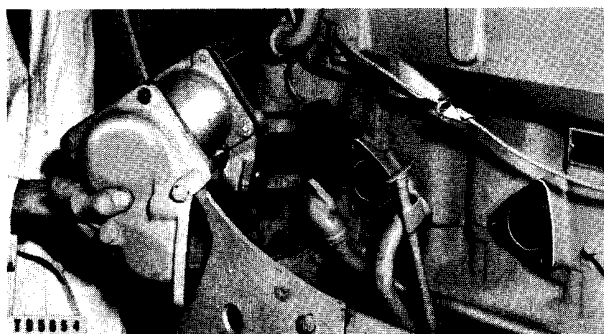
Drain the cooling system.



PREPARING TO REMOVE OIL COOLER

□ Remove ● Disconnect

1-Air vent line. 2-Engine oil cooler. 3-Clip. 4-Bonnet.
5-Oil inlet tube. 6-Oil outlet tube. 7-Bonnet.



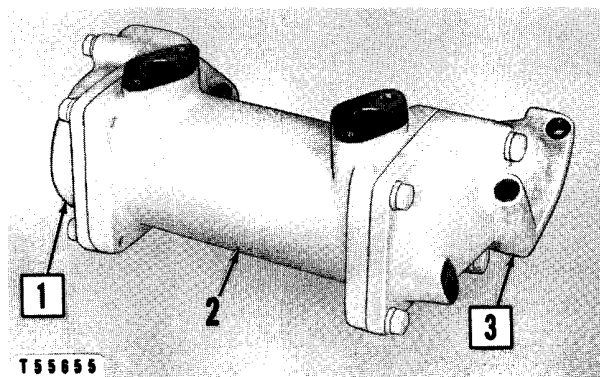
REMOVING THE OIL COOLER

Remove the bolts securing the bonnets (4) and (7) to the diesel engine block and remove the oil cooler as shown.

Check and replace, if necessary, all gaskets before installing the oil cooler on the engine.

Disassembly and Assembly

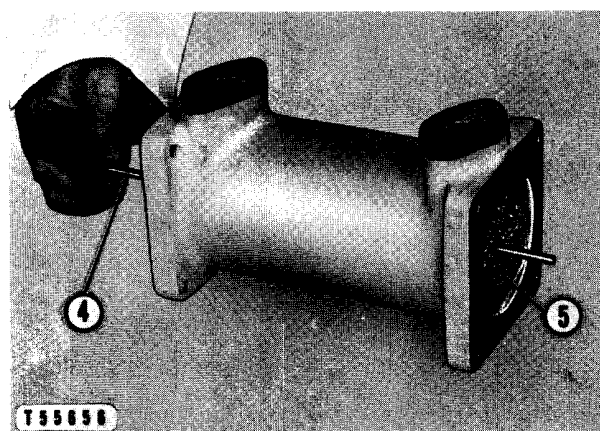
Remove the bonnets (1) and (3) from the engine oil cooler (2).



OIL COOLER DISASSEMBLY

□ Remove

1-Bonnet. 2-Cooler assembly. 3-Bonnet.

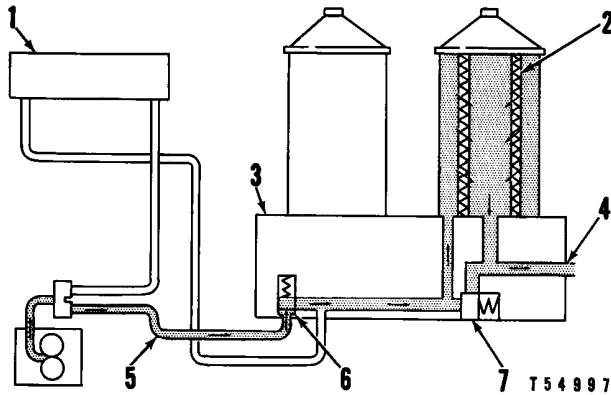


CLEANING OIL COOLER TUBES

4-Rod. 5-Tubes.

Accumulations of sediment can be removed from the tubes in the core (5) with a rod (4) 13/64" in diameter as shown.

Oil Filter

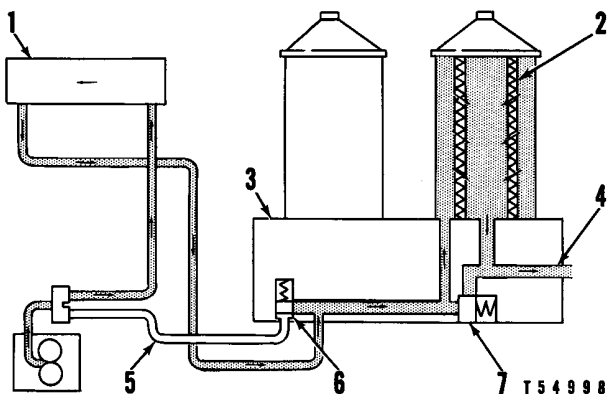
FLOW OF OIL THROUGH THE OIL COOLER,
BASE AND OIL FILTER

FLOW OF OIL (COLD START)

1-Oil cooler. 2-Oil filter. 3-Oil filter base. 4-Outlet passage to engine manifold. 5-External oil cooler by-pass line. 6-Oil cooler by-pass valve. 7-Oil filter by-pass valve.

An oil filter by-pass valve (7) and oil cooler by-pass valve (6) are incorporated into the oil filter base. An external oil cooler by-pass line (5) is provided.

On cold starts, the oil cooler by-pass valve (6) opens, eliminating the oil cooler (1) restriction. The oil from the pump is directed through the external line (5), past the oil cooler by-pass valve (6) and into the oil filter base (3). The oil then passes through the oil filters (2) and on through the outlet passage (4) to the engine oil manifold, providing lubrication to the engine components.

FLOW OF OIL (NORMAL OPERATION)
(Nomenclature same as previous illustration)

As the oil temperature increases, the oil cooler by-pass valve (6) closes, and the oil is then forced to flow through the oil cooler (1) before entering the filter base (3).

The oil filter by-pass valve (7) opens when the filters (2) become restricted and allows oil to by-pass

the filters (2) and flow directly to the outlet passage (4). Unless the filters are restricted or oil viscosity is extremely high, only filtered oil is furnished to the engine bearings.

The by-pass valves assure a supply of lubricating oil to the engine anytime the filter, cooler, or both become clogged.

NOTE

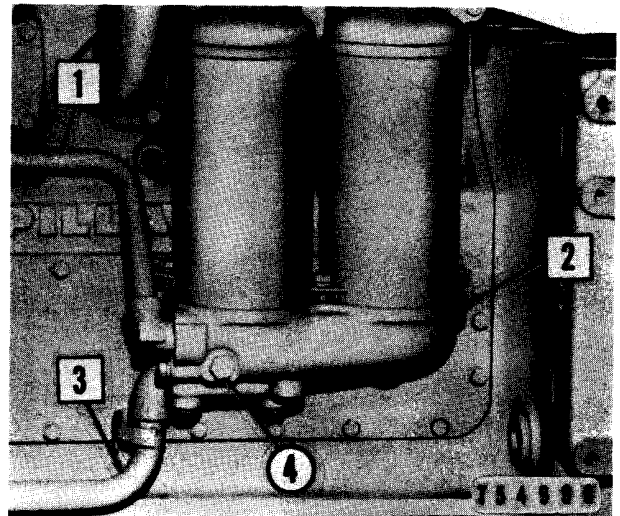
Earlier machines did not have the oil cooler by-pass valve (6) or the external oil cooler by-pass line (5).

REMOVAL AND INSTALLATION

Drain the filter housing by removing the drain plug (4).

Inspect the O-ring seal at the oil distribution manifold and replace if damaged.

Replace any damaged gaskets or seals. Coat the O-ring seals on the oil distribution manifold, oil cooler outlet line, and oil cooler by-pass line with liquid soap to ease installation.



OIL FILTER REMOVAL

☐ Remove

1-Oil cooler outlet line. 2-Oil filter base assembly.
3-Oil cooler by-pass line. 4-Drain plug.

DISASSEMBLY AND ASSEMBLY

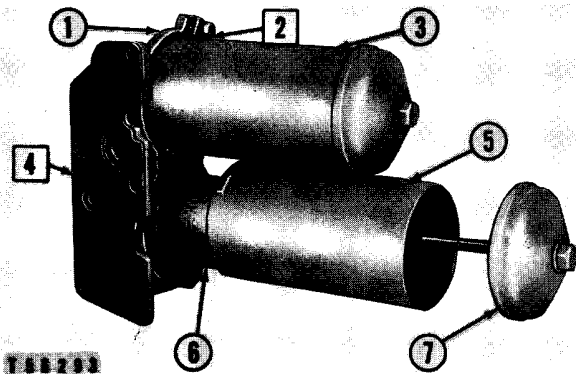
NOTE

Earlier machines used an O-ring seal in place of the gasket (4) to provide a seal between the filter base and the side of the engine.

1. Unscrew the cover assembly (7) and remove the filter element (6) and case (5) from the filter base (1).

ENGINE OIL FILTER

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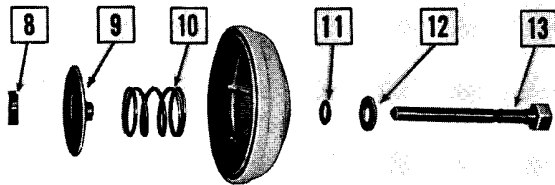
OIL FILTER DISASSEMBLY

☐ Remove

1-Filter base. 2-Bolt and clamp. 3-Case assembly.
4-Gasket. 5-Case. 6-Element. 7-Cover assembly.

NOTE

The case assembly (3) is disassembled in the same manner.

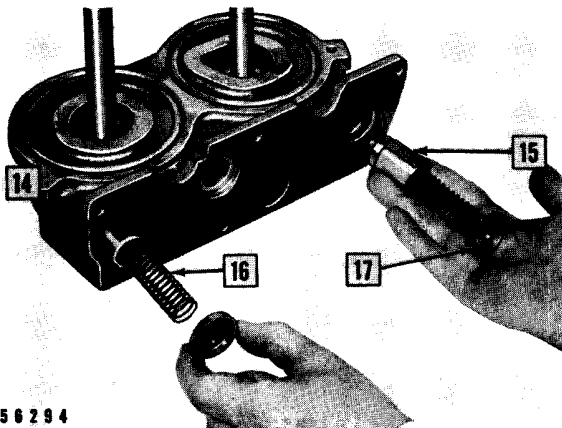


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COVER DISASSEMBLY

☐ Remove

8-Clip. 9-Retaining plate. 10-Spring. 11-O-ring seal.
12-Gasket. 13-Screw.



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FILTER BASE DISASSEMBLY

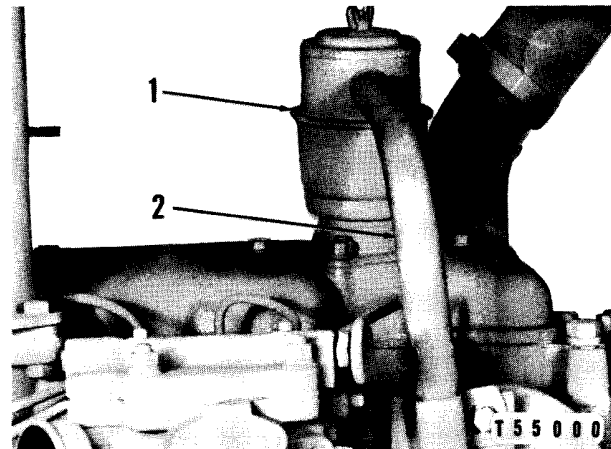
☐ Remove

14-Oil cooler by-pass valve plunger. 15-Oil filter by-pass
valve plunger. 16-Springs (two). 17-Plugs (two).

3. Inspect the oil cooler and oil filter by-pass valve plungers (14) and (15) and the surfaces against which they seat in the filter base. Replace the plungers if they worn or damaged.
4. Replace any damaged seals or gaskets.
5. Assemble in the reverse procedure of disassembly.

CRANKCASE BREATHER

The crankcase breather (1) is located on the valve cover of the diesel engine. A screen under the crankcase breather cap can be removed for cleaning. The breather assembly can be removed by removing the nuts which secure the base to the valve cover and by removing the bolts which hold the crankcase fumes disposal tube (2) to the cylinder head. Replace the gasket if it is damaged.



CRANKCASE BREATHER

1-Crankcase breather. 2-Crankcase fumes
disposal tube.

2. Clean the filter parts thoroughly.

Fuel System

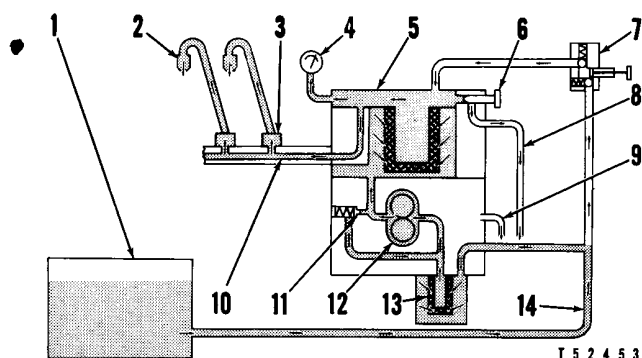
The fuel transfer pump (12) delivers fuel from the supply tank (1) to the fuel filter housing (5). The fuel is filtered prior to reaching the fuel filter housing by the primary fuel filter (13).

A spring-loaded by-pass valve (11) in the fuel transfer pump limits the maximum pressure of the fuel within the fuel filter housing (5). When the pressure of the fuel within the housing exceeds the valve setting, the by-pass valve is forced open thus allowing the fuel to be by-passed to the inlet side of the fuel transfer pump (12).

A drain line (9) is provided to drain, from the fuel transfer pump (12), any leakage along the shaft in the pump.

After passing through the fuel filter, the fuel flows into the fuel manifold (10) which distributes fuel to the fuel injection pumps (3).

A cam-actuated injection pump for each cylinder forces fuel through an injection line to the proper injection nozzle (2). A definite quantity of fuel is forced into the precombustion chamber by the injection pump at the proper time.

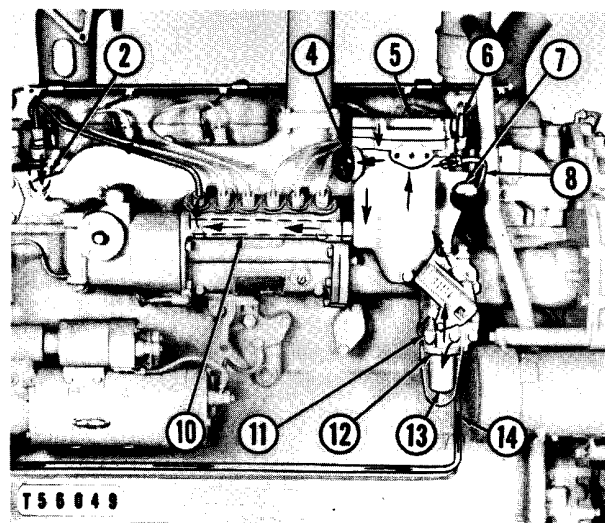


FLOW OF FUEL (SCHEMATIC)

- | | |
|--------------------------|-------------------------|
| 1-Fuel supply tank. | 8-Vent line. |
| 2-Fuel injection nozzle. | 9-Drain line. |
| 3-Fuel injection pump. | 10-Fuel manifold. |
| 4-Fuel pressure gauge. | 11-By-pass valve. |
| 5-Fuel filter housing. | 12-Fuel transfer pump. |
| 6-Vent valve. | 13-Primary fuel filter. |
| 7-Fuel priming pump. | 14-Fuel supply line. |

One air vent valve (6) is provided to vent the air from the fuel system. It is located on the filter housing cover between the filter and the injection pumps. When the vent valve is opened and fuel is caused to flow through the system by the action of either the transfer pump (12) or a priming pump (7), air escapes through the vent line (8).

When the fuel system is vented, the vent valve (6) should be left open until a solid stream of fuel flows out the vent line (8), indicating that all air has been bled from the system.



FUEL SYSTEM

(Nomenclature Same as Previous Illustration)

The fuel pressure gauge (4) is connected to the fuel filter housing. Lack of pressure on the gauge indicates difficulty in the supply side of the system. This could be caused by an empty diesel fuel tank, a closed shut-off valve at the tank, clogged filter, dirt under the by-pass valve, open vent valve, a faulty transfer pump or gauge.

To eliminate the possibility of a clogged filter causing a low fuel pressure gauge reading, install a new filter element.

Air in the system or water in the fuel can also be responsible for the engine firing irregularly or failing to start.

If, after eliminating the above possibilities, there still seems to be something wrong with the fuel system, check the injection side of the system. Fuel injection pumps and fuel injection valves can be checked as described later.

Many times the fuel system is blamed when the fault lies elsewhere. A smoky exhaust may be caused by faulty fuel injection valves. It can also be caused by a dirty air cleaner, overloading at high altitude, or badly worn piston rings.

Irregular firing can be caused by faulty fuel injection valves. It can also be caused by the burning of excessive lubricating oil escaping past worn piston rings, or new rings which have not become fully seated.

Fuel Supply Equipment

FUEL TRANSFER PUMP

The gear-type fuel transfer pump is driven by a gear machined on the accessory drive shaft.

Removal and Installation

1. Shut off the fuel at the fuel supply tank.
2. Remove the Service Meter. See the topic, SERVICE METER REMOVAL AND INSTALLATION.

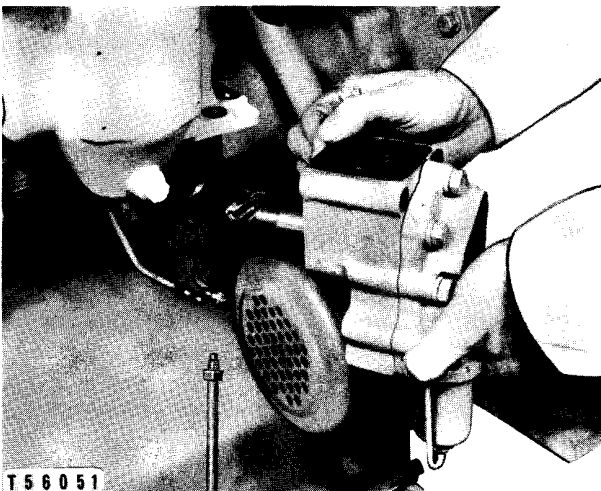


PREPARING TO REMOVE THE FUEL TRANSFER PUMP

 Disconnect

1-Drain line. 2-Priming pump supply line.
3-Fuel supply line.

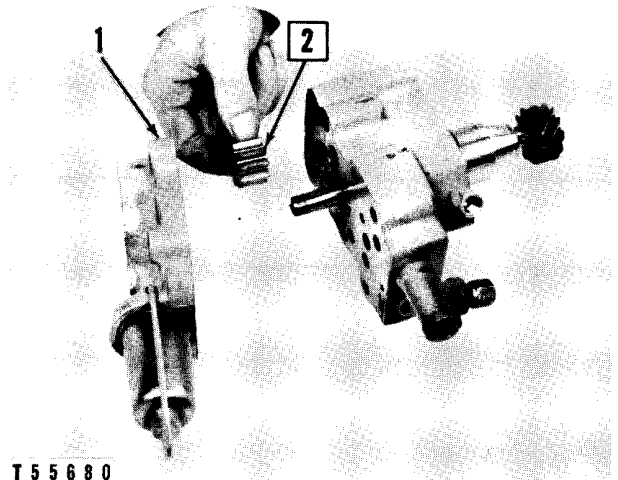
3. Remove the bolts which secure the fuel transfer pump to the adapter housing and the bolts which secure the fuel filter housing to the pump.



REMOVING THE FUEL TRANSFER PUMP

4. Pull the fuel transfer pump outward from the accessory drive housing as illustrated.
5. Prior to installation, inspect the gaskets and replace if damaged.

Disassembly and Assembly



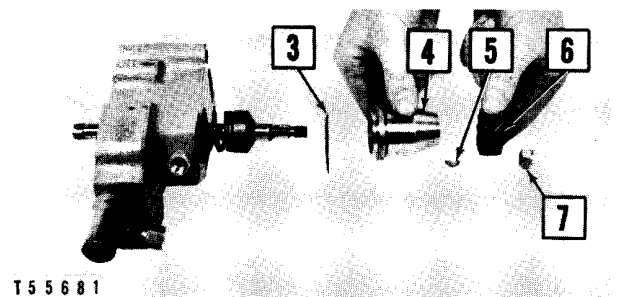
T 5 5 6 8 0

DISASSEMBLY OF FUEL TRANSFER PUMP

☐ Remove

1-Cover. 2-Idler gear.

1. Use an 8H695 Puller and a suitable step plate to remove the gear (6) from the shaft.

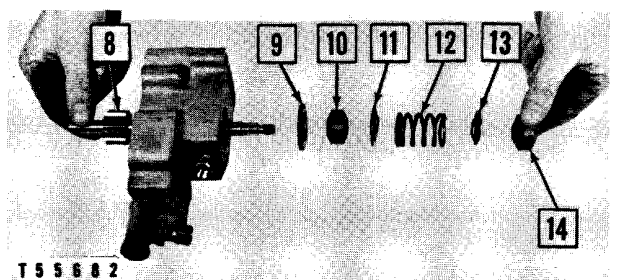


T 5 5 6 8 1

FUEL TRANSFER PUMP DISASSEMBLY

☐ Remove

3-Gasket. 4-Seat assembly. 5-Key. 6-Gear. 7-Nut.



T 5 5 6 8 2

FUEL TRANSFER PUMP DISASSEMBLY

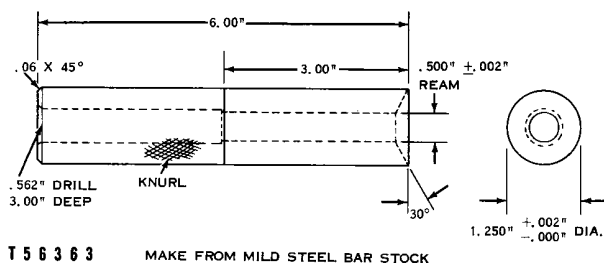
☐ Remove

8-Drive shaft assembly. 9-Gasket. 10-Seal. 11-Retainer.
12-Spring. 13-Retainer. 14-Seal.

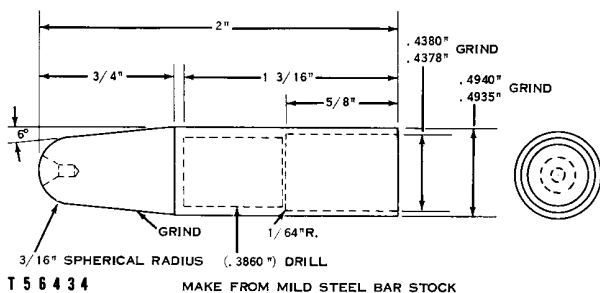
ENGINE FUEL SUPPLY EQUIPMENT

ISSUED 9-62

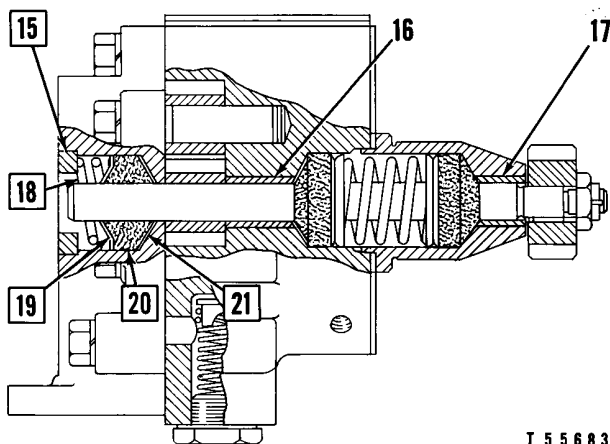
2. The bearing (16) in the pump body and the bearing (17) in the seat assembly can be pressed out and replaced if worn or damaged. Install the bearings with tapered edges toward the seal.
3. The seals (10), (14) and (20) should be soaked prior to installation in a solution of 1/2 SAE 30 oil and 1/2 diesel fuel long enough to soften them. The seals should be tamped firmly into the pump body bore and around the shaft with a tamping tool machined as illustrated. A guide should be used over the end of the transfer pump shaft to guide the seals over the threads and sharp



SEAL TAMPING TOOL



SEAL AND TAMPING TOOL GUIDE



FUEL TRANSFER PUMP BEARINGS

☐ Remove

15-Retainer. 16-Bearing. 17-Bearing. 18-Spring.
19-Retainer. 20-Seal. 21-Gasket.

corners of the shaft. The guide should be machined as illustrated. The guide also should remain in place when using the tamping tool.

4. When assembling the pump, install the gear with the grooved side facing the nut (7) and tighten the nut. See the topic, SPECIFICATIONS, for proper nut torque.
5. When assembling the pump, apply a thin film of Permatex No. 3 or equivalent to the surface between the pump body and pump cover.

CAUTION

Do not allow Permatex to enter pump.

6. Apply a thin film of Permatex No. 3 or equivalent to the outside diameter of the spring retainer (15). The retainer must be flush with the cover to the tolerance given in the topic, SPECIFICATIONS.

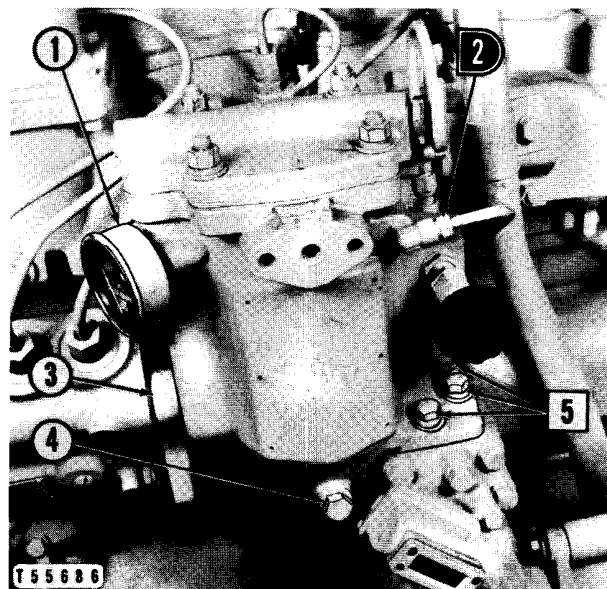
FUEL FILTER HOUSING

Removal and Installation

Drain the fuel filter housing by removing the drain plug (4).

NOTE

If the engine is equipped with a fuel priming pump, the pump and filter housing are removed as a unit.



PREPARING TO REMOVE FUEL FILTER HOUSING

☐ Remove ☒ Disconnect

1-Fuel pressure gauge. 2-Drain line. 3-Adapter.
4-Drain plug. 5-Fuel filter housing retaining bolts.

ENGINE FUEL SUPPLY EQUIPMENT

ISSUED 9-62

Remove the fuel filter housing by sliding it toward the front of the engine to clear the adapter (3).

Inspect and replace all damaged seals and gaskets.

FUEL BY-PASS

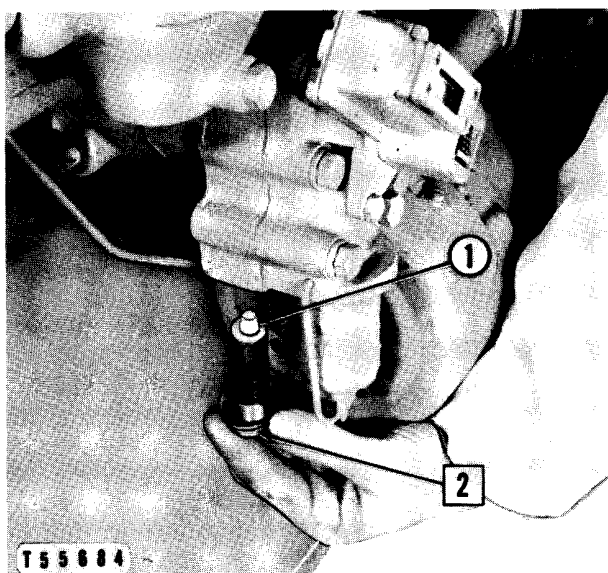
The fuel transfer pump supplies more fuel than is required for engine operation. The fuel by-pass valve prevents excessive pressure within the fuel system. The by-pass valve may fail to function properly if dirt gets between the plunger and the seat. If the plunger is not seating properly, there will be little, if any, fuel pressure. If the plunger sticks closed, the pressure will be excessive but will not show on the gauge.

The by-pass valve is installed in the fuel transfer pump body.

Removal and Installation

Clean the contact surfaces. See that the seat is smooth and flat, and that the sealing surface of the plunger (1) is in good condition.

See the topic, SPECIFICATIONS for correct spring length and pressure.



REMOVING THE BY-PASS VALVE

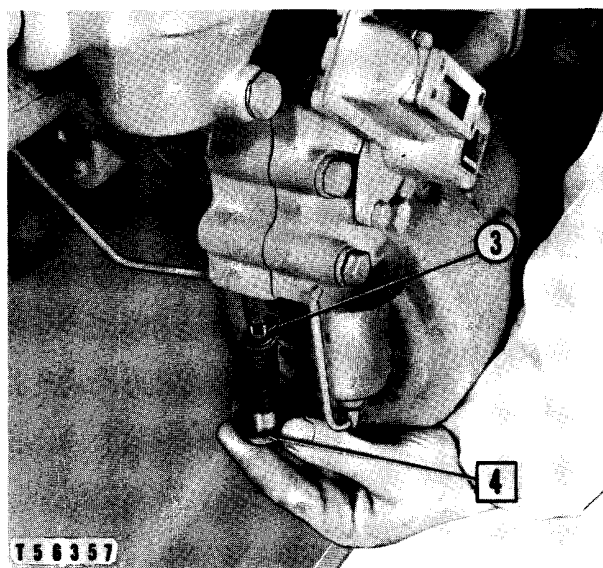
☐ Remove

1-Plunger. 2-Bolt.

Carefully install the plunger and spring into the fuel transfer pump body.

A resilient sealing ring is molded to the shoulder of the plunger to form the sealing surface of the new style plunger. The new plunger is adaptable to all earlier models. When converting to the new style

plunger, add a slight chamfer to the edge of the housing bore that accepts the nose of the by-pass valve plunger.



REMOVING NEW STYLE BY-PASS VALVE

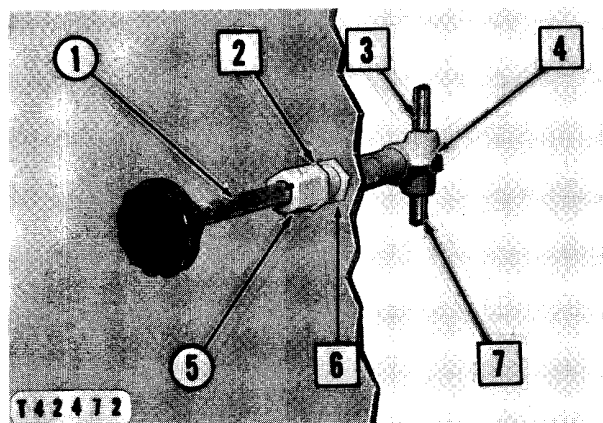
☐ Remove

3-Plunger assembly. 4-Bolt.

FUEL PRIMING PUMP

Removal and Installation

Loosen the nut (5) and remove the nut and plunger (1) as an assembly.



FUEL PRIMING PUMP REMOVAL

☐ Remove

1-Plunger. 2-Nut. 3-Outlet line. 4-Pump housing.
5-Nut. 6-Nut. 7-Inlet line.

Disassembly and Assembly

The outlet check valve may be removed by removing screw (3).

ENGINE

FUEL SUPPLY EQUIPMENT

ISSUED 9-62

The inlet check valve may be removed by removing screw (2).

When assembling the plunger to the pump housing, inspect the O-ring seals (1) and check valves and replace if necessary.

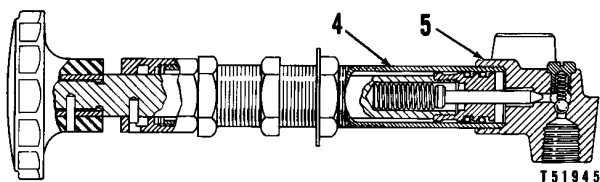


T42473

FUEL PRIMING PUMP DISASSEMBLY

1-O-ring seals. 2-Screw. 3-Screw.

Other parts can be replaced if necessary. The barrel (4) is threaded into the body (5).



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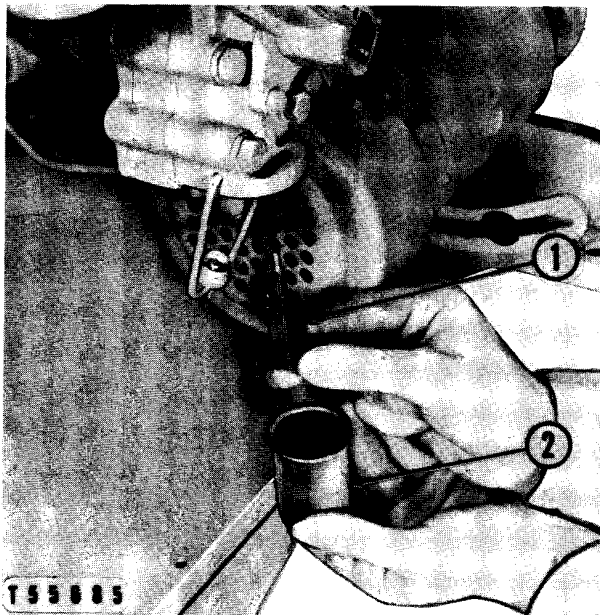
FUEL PRIMING PUMP DISASSEMBLY

4-Barrel. 5-Body.

PRIMARY FUEL FILTER

Removal and Installation

1. Shut off the fuel valve at the fuel supply tank.



T55885

PREPARING TO REMOVE THE PRIMARY FUEL FILTER

1-Element assembly. 2-Bowl.

ENGINE

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FUEL INJECTION EQUIPMENT**Fuel Injection Equipment**

The most likely causes for faulty fuel injection performance are:

1. Air in the fuel system.
2. Low fuel supply.
3. Water in the fuel.
4. Clogged fuel filter.
5. Insufficient fuel transfer pump pressure.

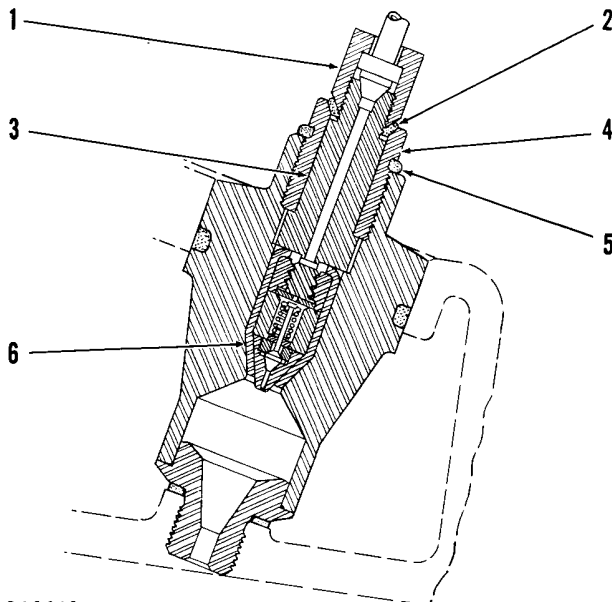
If these conditions are checked and corrected and the engine still does not perform properly, the fuel injection equipment should be checked.

CHECKING FUEL INJECTION EQUIPMENT

Before removing a fuel injection pump or valve for testing on an engine that is missing or puffing black smoke at the exhaust, a simple check can be made to determine which cylinder is causing the difficulty. With the engine operating at a speed which makes the defect most pronounced, momentarily loosen the fuel line nut on the injection pump sufficiently to "cut out" the cylinder. Check each cylinder in the same manner. If one is found where loosening makes no difference in the irregular operation of the engine, or causes puffing of black smoke at exhaust to cease, probably the pump and valve for only that cylinder need be tested.

FUEL INJECTION VALVE

The function of the fuel injection valves is to inject and atomize the fuel from the fuel injection pumps



T 35615

FUEL INJECTION VALVE CROSS SECTION
1-Fuel line assembly. 2-Seal. 3-Body. 4-Nut.
5-Seal. 6-Nozzle assembly.

into the precombustion chambers. Irregular engine operation and smoking may be caused by an improperly operating injection valve.

The fuel injection valve assembly is installed in the precombustion chamber. The fuel injection valve assembly consists of the body (3), nut (4), and the capsule type nozzle assembly (6). Only the nozzle assembly need be replaced.

The nozzle assembly (6) and body (3) are held in place by the nut (4), and they can be removed by disconnecting the fuel line assembly (1) and removing the nut and seals (2) and (5).

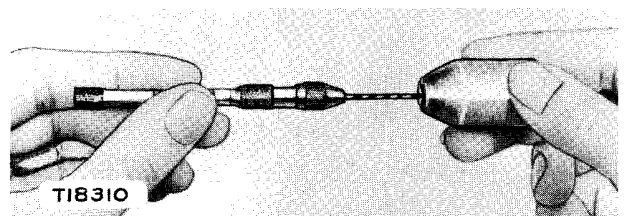
The body and the nozzle assembly can then be lifted out. The nozzle assembly is only finger-tight on the body.

When installing a fuel injection valve, always check the seats of both the nozzle and the precombustion chamber.

Tighten the nut (4) to the torque value listed in the topic, SPECIFICATIONS, to assure proper seating of the nozzle assembly. Excessive tightness will damage the nozzle. Excessive looseness will allow the nozzle to leak and in some instances cause the nozzle case to bulge or split.

Cleaning Fuel Injection Valve

The operation of the engine will be affected if the orifice in the nozzle becomes partially filled with carbon. To correct this, remove the nozzle assembly from the valve body and clean the fuel discharge hole using the drill given in the topic, SPECIFICATIONS, and a 5B1254 Chuck, which are part of the 5B1401 Cleaning Tool Group. If the valve does not operate properly after cleaning, the nozzle should be checked for leakage.



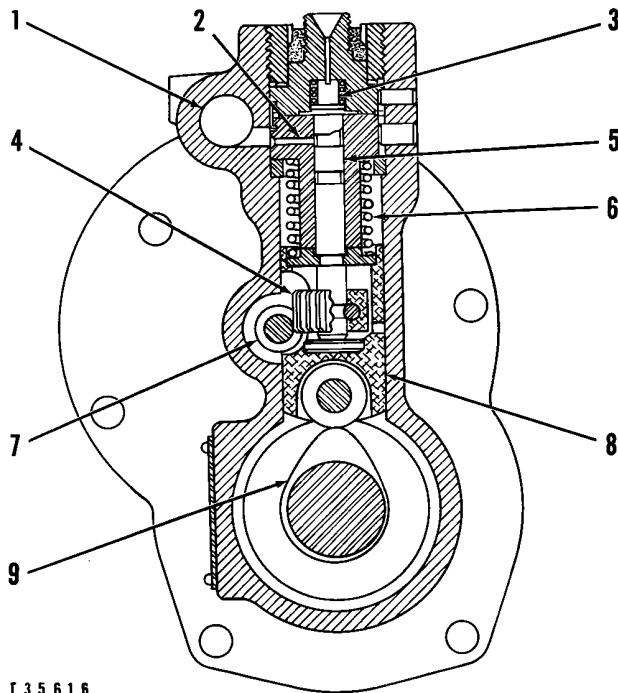
CLEANING FUEL INJECTION NOZZLE ASSEMBLY

Checking Fuel Injection Valve

The condition of a capsule-type nozzle assembly can be tested out of the engine on the Caterpillar Diesel Fuel Injection Test Apparatus. The spray characteristic, valve unseating pressure, and the rate of leakage of the nozzle assembly can be determined.

FUEL INJECTION PUMPS

Fuel enters the fuel injection pump housing through passage (1) and enters the fuel injection pump body through the inlet port (2). The injection pump plungers (5) and the lifters (8) are lifted by the cam lobes on the camshaft (9) and always make a full stroke. The lifters are held against the cam lobes by the springs (6). Each pump measures the amount of fuel to be injected into its respective cylinder and delivers it to the fuel injection nozzle.



FUEL INJECTION PUMP

1-Fuel passage. 2-Inlet port. 3-Check valve. 4-Gear segment. 5-Pump plunger. 6-Spring. 7-Fuel rack. 8-Lifter. 9-Camshaft.

The amount of fuel pumped per stroke is varied by turning the plunger in the barrel. The plunger is turned by the governor action through the circular rack (7) which turns the gear segment (4) on the bottom of the pump plunger.

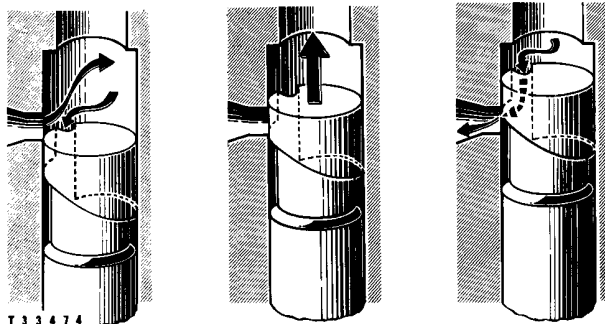


FIG. A
Port Uncovered
Fuel Enters Barrel

FIG. B
Port Covered
Injection Begins

FIG. C
Port Uncovered
Injection Ends

Figures A, B and C illustrate the functioning of an injection pump as the plunger makes a stroke.

In Fig. A the plunger is down and the inlet port (2) is uncovered. Fuel flows into the space above the plunger through the slot and into the recess around the plunger.

In Fig. B the plunger has started up and the port is covered. The fuel is trapped and will be forced through the check valve (3), fuel line, and injection valve as the plunger moves upward.

In Fig. C the plunger has risen until the port is uncovered by the recess in the plunger. The fuel can now escape back through the port into the fuel manifold and injection will cease.

It will be noted that the recess in the pump plunger forms a helix around the upper end of the plunger. Figures D, E and F illustrate how rotating the pump plunger affects the quantity of fuel injected.

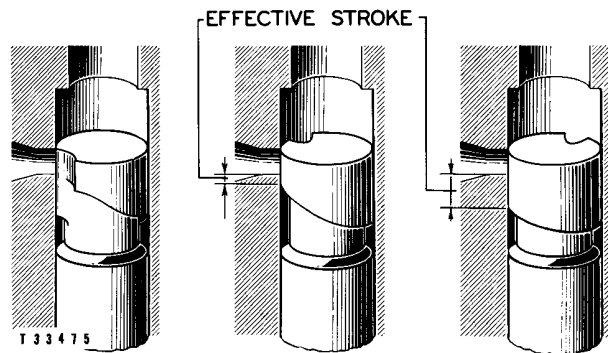


FIG. D
SHUT OFF

FIG. E
IDLING
POSITION

FIG. F
FULL LOAD
POSITION

In Fig. D the plunger has been rotated into the shut-off position. The slot connecting the top of the plunger with the recess is in line with the port; therefore, no fuel can be trapped and injected.

In Fig. E the plunger has been rotated into the idling position. The narrow part of the plunger formed by the helix will cover the port for only a short part of the stroke. This permits only a small amount of fuel to be injected per stroke.

In Fig. F the plunger has been rotated into the full load position. The wide part of the plunger formed by the helix covers the port for a longer part of the stroke. This permits a larger amount of fuel to be injected per stroke.

Worn fuel injection pumps will result in loss of power and hard starting. These same conditions may be present if the piston rings and cylinder liners are badly worn. However, in the case of worn piston rings and liners, the hard starting and loss

ENGINE

FUEL INJECTION EQUIPMENT

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of power will be accompanied by poor compression, a smoky exhaust and excessive blow-by gases from the crankcase breather.

Ordinarily, if one fuel injection pump on an engine is worn, it will be found that all of the injection pumps are worn and need replacing. Failure to replace all of the worn injection pumps may result in erratic and irregular engine operation.

FUEL INJECTION PUMP HOUSING

The fuel injection pump housing is bolted to the accessory drive housing. The fuel injection camshaft is mounted in the lower section of the housing and is driven by an off-center tang on the end of the accessory drive shaft.

The fuel injection pump camshaft turns in three bearings, one at each end of the camshaft, and one at the center. The camshaft determines the vertical movement of the fuel pump plungers and also drives the governor weights.

The fuel injection pump camshaft is splash lubricated by the oil from the governor housing. The cam lobes dip into the oil and in turn lubricate the lifters. Lubricating oil drains forward through the accessory drive housing and into the diesel engine crankcase.

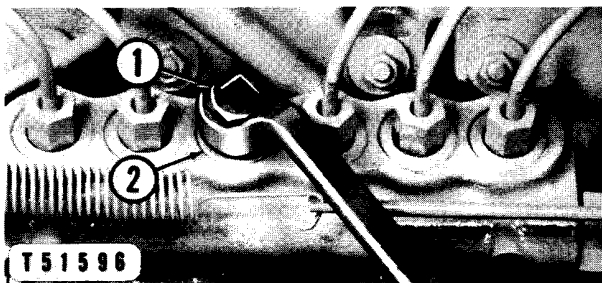
Removal and Installation

The fuel injection pump housing and the governor can be removed as a unit, or the governor can be removed separately. See the topic, ACCESSORY DRIVE REMOVAL.

When installing the pump housing, align the off center tang on the accessory drive shaft with the slot in the end of the camshaft.

Fuel Injection Pump Removal

Disconnect or remove the fuel injection line.

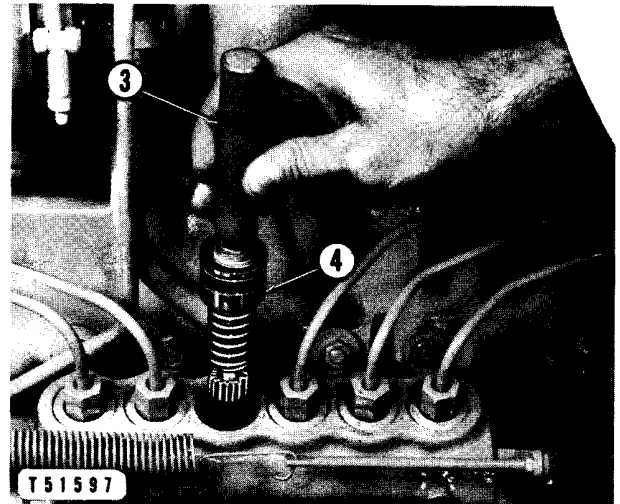


PREPARING TO REMOVE INJECTION PUMP
1-1M6952 Wrench. 2-Bushing.

Remove the cap and the felt washer under the cap.

Use a 1M6952 Wrench (1) to remove the fuel injection pump retaining bushing (2) from the fuel injection pump housing.

Use a 1M6954 Extractor (3) to remove the fuel injection pump (4) from the housing.



REMOVING INJECTION PUMP
3-1M6954 Extractor. 4-Fuel injection pump.

Fuel Injection Pump Disassembly and Assembly

1. Remove the seal (1).

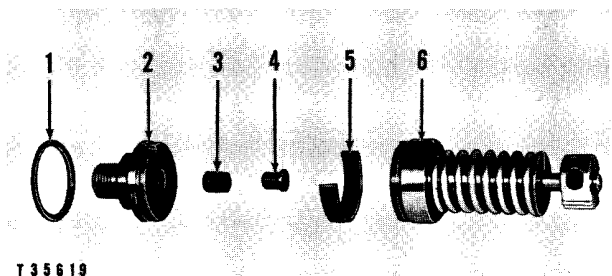
CAUTION

All disassembly and assembly should be done with clean hands on a clean, lint free cloth.

2. Remove the retaining ring (5) and separate the bonnet (2) and barrel assembly (6).

NOTE

Considerable care should be taken not to drop and lose the check valve (4) and spring (3) retained in the bonnet.



FUEL INJECTION PUMP DISASSEMBLY
1-Seal. 2-Bonnet. 3-Spring. 4-Check valve.
5-Ring. 6-Barrel assembly.

3. Separate the spring (8), washer (9) and plunger (10) from the barrel (7).

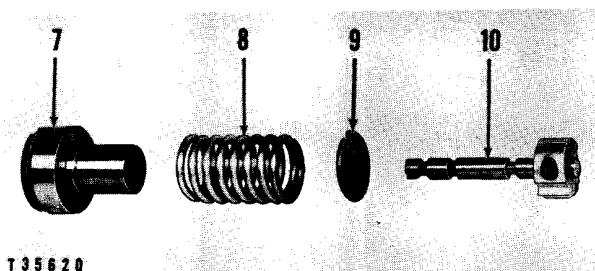
ENGINE FUEL INJECTION EQUIPMENT

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4. Remove washer and spring from the plunger.

CAUTION

Considerable care should be exercised when removing the spring and washer to prevent damage to the plunger surfaces.



FUEL INJECTION PUMP DISASSEMBLY

7-Barrel. 8-Spring. 9-Washer. 10-Plunger assembly.

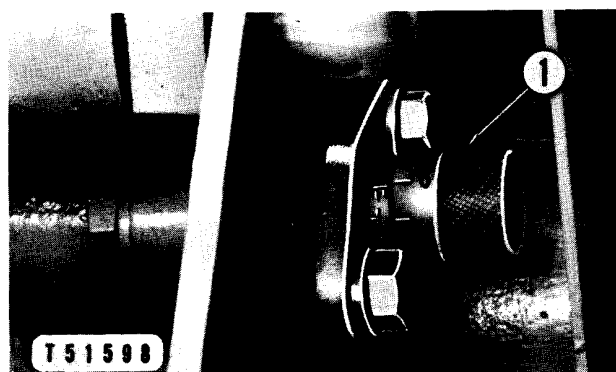
5. Clean all parts with clean diesel fuel and assemble, replacing any worn or damaged parts.

CAUTION

The barrel and the plunger assembly are matched, and the individual parts are not interchangeable with other barrels or plunger assemblies. Use extreme care in inserting the plunger into the bore of the barrel.

Fuel Injection Pump Installation

1. Remove the cover on the end of the accessory drive housing and position the fuel rack by using the 8M530 Rack Setting Gauge (1). Bolt the gauge to the fuel injection pump housing as shown. When the rack is properly positioned, the reading on the gauge should be .000".

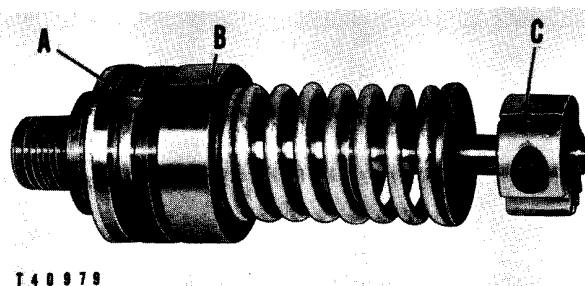


POSITIONING RACK
1-8M530 Rack Setting Gauge.

2. Align the notches (A) and (B) on the bonnet and the barrel with slot (C) on the gear segment.

NOTE

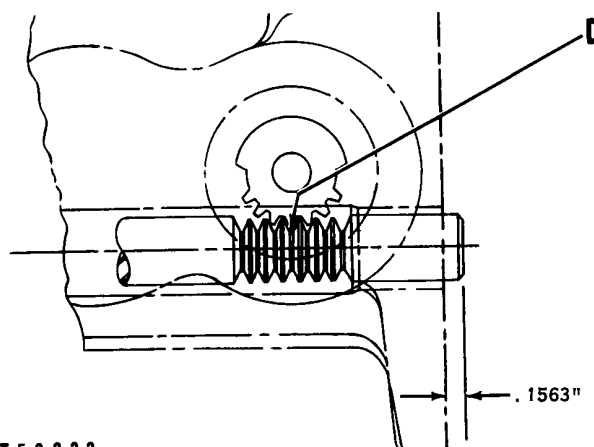
The procedure thus far has aligned the center notch on the rack for each pump with the center tooth of the gear segment on the injection pump plunger.



ALIGNING THE INJECTION PUMP

A-Notch in bonnet. B-Notch in barrel. C-Slot in gear segment.

3. Insert the assembly into the bore of the housing with the extractor. The notches in the bonnet and the barrel must engage with the two locating dowels in the bore of the pump housing. These dowels align the pump barrel fuel inlet port with the fuel manifold outlet port.



CORRECT POSITION OF GEAR SEGMENT AND RACK
D-Fourth notch on fuel rack.

4. After each plunger and barrel assembly is installed, place a new rubber seal over the bonnet and start the retainer bushing into the housing. Push the plunger and barrel assembly into the housing until the dowel is engaged with the bonnet and screw the retainer bushing finger tight.

CAUTION

After the plunger and barrel assembly is installed, leave the 1M6954 Extractor in position to hold the barrel from turning until the retainer bushing has been screwed in finger tight. If the retainer bushing will not screw down and seat properly in the housing, do not force it by using a wrench. The bushing will not seat because the notches in the barrel are not properly aligned with the dowels in the pump housing. Damage

ENGINE**FUEL INJECTION EQUIPMENT**

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to the dowel pins can result from forcing the bushing.

5. Tighten the bushing to the torque value listed in the topic, SPECIFICATIONS.

NOTE

Prior to installation of the plunger and barrel assembly, make certain the lifter for the assembly to be installed is at the bottom of its stroke.

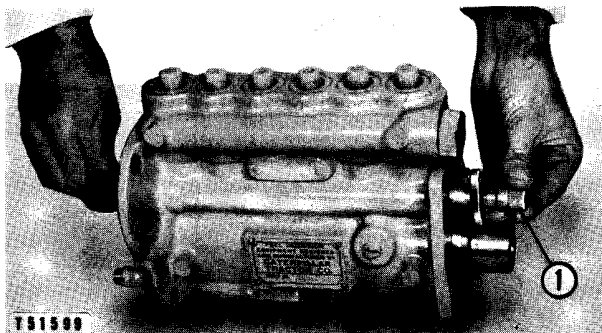
CAUTION

With a torque value of less than that listed in the specifications, the pump will leak. If the torque value is greater, the housing can be damaged.

6. The rack travel must be measured to make certain that the plunger gear segment is properly meshed with the fuel rack. The gear segment and rack are so constructed that the maximum travel of the rack in either direction can be obtained only when the 4th tooth of the gear segment is meshed with the 4th notch on the rack. If the gear segment is misplaced in either direction, the rack travel will be reduced in both directions. It is, therefore, necessary to measure the rack travel in only one direction to make certain that the gear segment is properly meshed with the fuel rack. This is measured by using the 8M530 Rack Setting Gauge (1). The rack is moved to its extreme fuel on position. A minimum reading of + .312" on the gauge indicates that the gear segment is properly positioned with the rack. The rack collar must be removed in order to obtain this much rack travel.

CAUTION

An injection pump improperly installed in the "fuel ON" side position of its gear segment can cause an engine to overspeed with resultant serious damage to the engine and driven equipment.



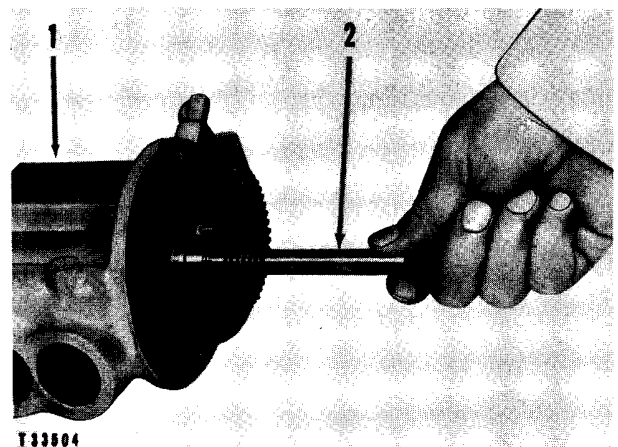
CHECKING RACK TRAVEL
1-8M530 Rack Setting Gauge.

7. If this measurement is not obtained, the gears are improperly positioned and the procedure must be repeated.
8. Remove the Rack Setting Gauge, install the felt washers and connect the fuel injection lines.

Fuel Rack Removal and Installation

The fuel rack can be removed from the fuel injection pump housing with the pump housing on the engine if desired.

Remove the fuel injection pumps and the governor weights. See the covering topics.



FUEL RACK REMOVAL
1-Fuel injection pump housing. 2-Fuel rack.

Remove the fuel rack (2) from the injection pump housing (1) as shown.

Install in the reverse order of removal.

Fuel Rack Bearing Removal and Installation

The fuel rack moves in two bearings, one pressed into each end of the fuel injection pump housing.

See the topic, SPECIFICATIONS, for bearing clearances.

The bearing on the accessory drive end of the fuel injection pump housing is a straight-type bearing and can be replaced if worn or damaged without following any marking procedure. The bearing on the governor end of the pump housing must be installed as illustrated.

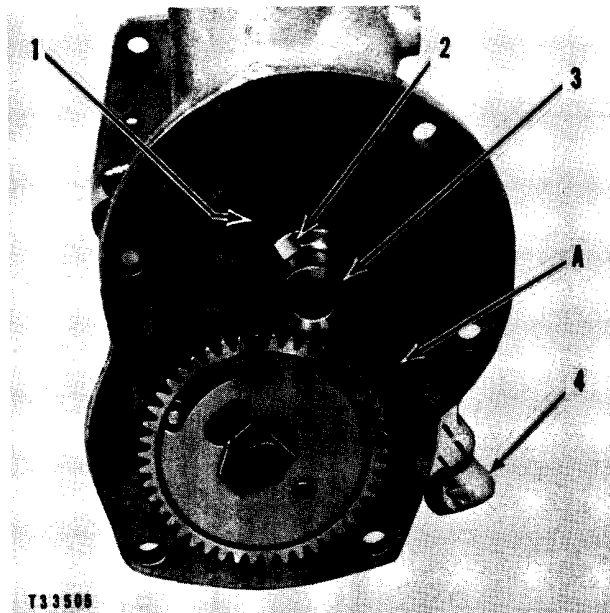
1. Remove the fuel injection pump housing, governor, fuel injection pumps and fuel rack. See the covering topics.
2. Place two scribed marks (1) as shown above the groove (2) on the bearing (3) to align the groove on the bearing with the drilled lubrication passage (A) from the fitting (4).

NOTE

Wire the spacer and the lifter together and tag them to identify with the pump bore from which they were removed.

3. Install the lifter assembly and the spacer, tilting the spacer to one side of the pump bore to clear the pump locating dowels in the fuel injection pump housing.
4. For timing dimension setting, see the topic, FUEL PUMP TIMING DIMENSION SETTING.

Fuel Injection Pump Camshaft



FUEL RACK BEARING REMOVAL

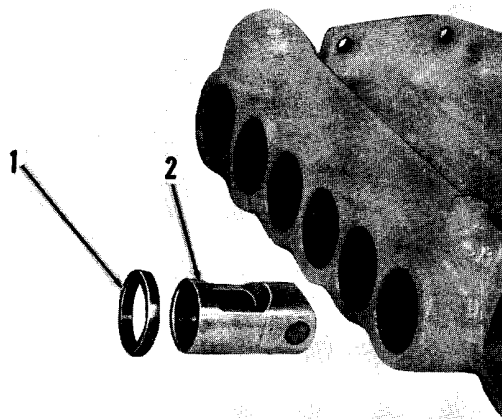
1-Scribe marks. 2-Lubrication groove. 3-Bearing.
4-Fitting. A-Lubrication passage.

3. Remove the bearing (3).
4. Remove the bearing on the accessory drive end of the fuel injection pump housing by driving it out with a punch inserted through the housing bore for the bearing (3).

Fuel Pump Lifter Removal and Installation

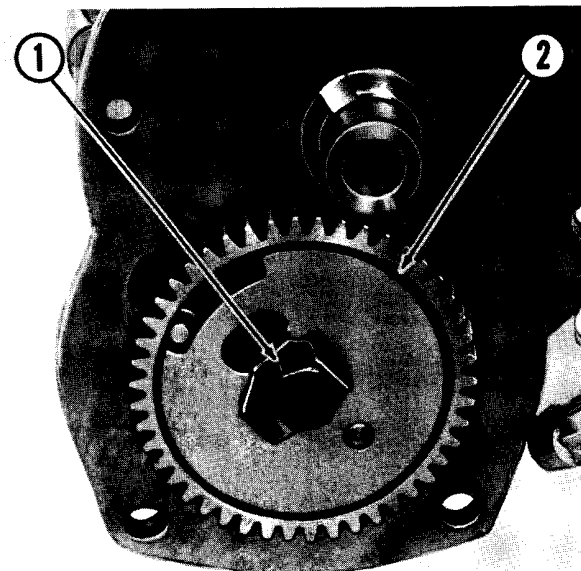
The fuel pump lifters can be removed with the fuel injection pump housing installed on the engine if desired.

1. Remove the fuel injection pumps and the fuel rack. See the covering topics.
2. Remove the spacer (1) and lift out the lifter assembly (2).



LIFTER REMOVAL

1-Spacer. 2-Lifter assembly.

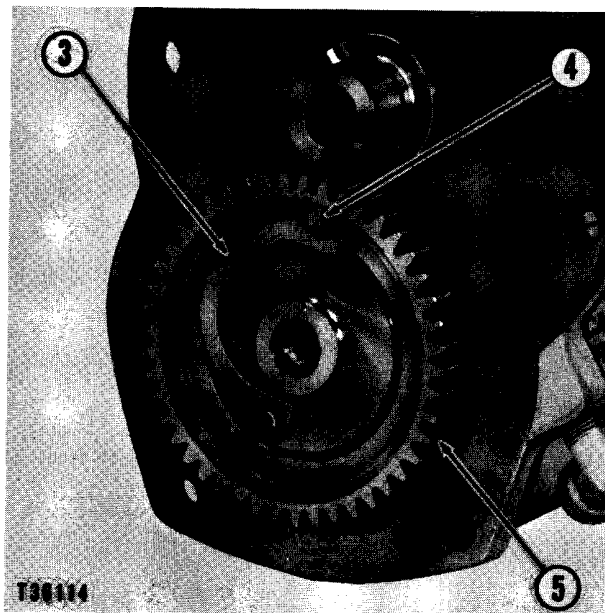


T33506

PREPARING TO REMOVE CAMSHAFT

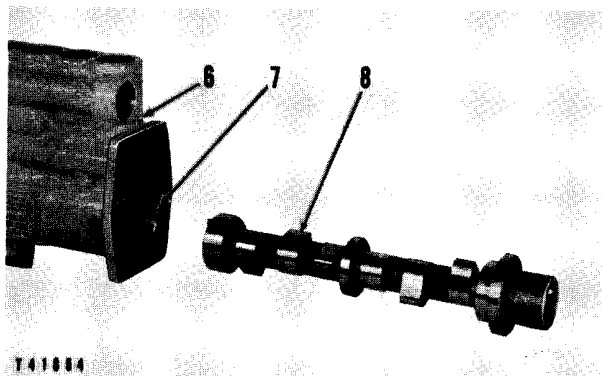
1-Bolt. 2-Plate.

1. Remove the fuel injection pump housing, governor weights, fuel injection pumps, fuel rack, and lifters. See the covering topics.
2. Bend the lock and remove the bolt (1) and the plate (2).
3. Remove the spring (3), floating pin (4) and the ring gear (5).
4. Carefully remove the camshaft (8) from the fuel injection pump housing (6) at the accessory drive end of the housing.
5. Inspect the camshaft bearing (7) and replace if worn or damaged. See the topic, SPECIFICATIONS, for bearing clearance. Install the bearings flush with the housing face.



GEAR REMOVAL
3-Spring. 4-Pin. 5-Gear.

6. Install the camshaft in the reverse order of removal.



CAMSHAFT REMOVAL
6-Pump housing. 7-Bearing. 8-Camshaft.

FUEL PUMP TIMING DIMENSION SETTING (ON ENGINE)

The timing dimensions should be checked and re-set, if necessary, to account for worn timing gears or worn pump lifters. See the topic, FUEL INJECTION PUMP LIFTER WASHER AND PUMP PLUNGER INSPECTION.

The timing dimensions can be set in the following manner:

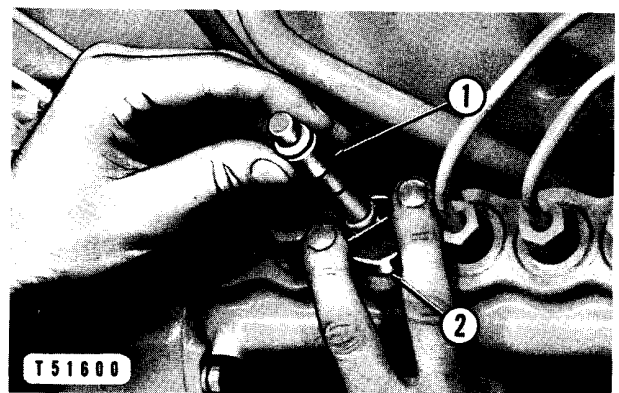
1. Remove the fuel injection line.
2. Remove the fuel injection pump for the respective timing dimension to be checked. See the topic, FUEL INJECTION PUMP REMOVAL.

3. Turn the crankshaft in the direction of engine rotation to top center (TC) on the compression stroke of the cylinder for which the timing dimension is to be checked. For the location of the flywheel pointer, see the topic, FLYWHEEL AND FLYWHEEL HOUSING.

NOTE

If the top center mark on the flywheel is turned past the pointer, turn the flywheel backward approximately 60°. Then turn the crankshaft again in the direction of crankshaft rotation until the top center mark aligns with the pointer.

4. Install the 2M5219 Gauge (2) in the pump bore. Check the timing dimension using a depth micrometer (1) with a 4" - 5" rod. See the topic, SPECIFICATIONS, for correct timing dimension.



CHECKING TIMING DIMENSION SETTING
1-Depth micrometer. 2-2M5219 Gauge.

5. See the topic, SPECIFICATIONS.

NOTE

It is important when checking and setting the timing dimension that the crankshaft be turned in the direction of engine rotation. After the timing dimension has been checked or set according to specifications, turn the crankshaft a few degrees in the direction of engine rotation. Again check the reading on the micrometer. This reading should be less than the measurement when checked with the crankshaft at top center thus indicating the lifter is rising and was checked at the correct position.

6. If all the timing dimensions are to be checked or reset, continue the procedure in the normal firing order of the engine.
7. After checking the timing dimensions, see the topic, FUEL INJECTION PUMP INSTALLATION.

FUEL PUMP TIMING DIMENSION SETTING (OFF ENGINE)

The following procedure for checking the timing dimensions with the pump housing off the engine will result in correct fuel injection timing if the pump housing is installed on a new engine, or, an engine with new timing gears and coupling. This method does not compensate for wear that may be present on timing gears and coupling in an engine that has seen considerable service. Therefore, to insure correct fuel injection timing, it is very important to recheck one of the timing dimensions with the pump housing installed on the engine on which it is intended to remain. Follow the procedure described under the topic, FUEL PUMP TIMING DIMENSION SETTING (ON ENGINE).

1. Install the 2M5220 Pointer Assembly by placing the two dowels on the pointer assembly in the dowel holes on the fuel injection pump housing.
2. Place the timing plate on the drive end of the camshaft and align the coupling with the camshaft. Use a bolt and washer to secure the plate and coupling to the camshaft. This allows the camshaft to be rotated by rotating the timing plate.
3. Refer to the table in the topic, SPECIFICATIONS, and select the timing plate setting for the timing dimension being checked or set. Set the timing plate to this setting by rotating it counterclockwise until the proper degree setting lines up with the edge of the boss on the pointer assembly. Lock in position with the lockscrew.
4. If all the timing dimensions are to be checked or reset, continue the same procedure in the firing order of the engine. Recheck each timing dimension after the adjustment has been made to make sure the dimension is correct.

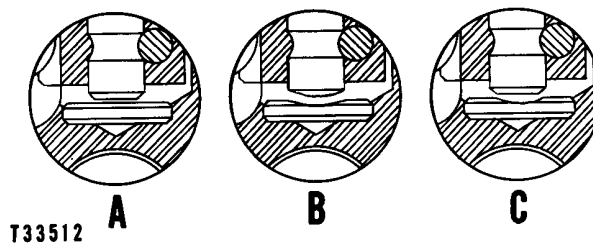
FUEL INJECTION PUMP LIFTER WASHER AND PUMP PLUNGER INSPECTION

The timing dimension should be checked to compensate for wear in the timing gears or lifters to assure that the point of the fuel injection is correct. If the timing dimension is too small, injection will begin early, and, if too great, injection will be late.

When pump plunger wear becomes excessive, the lifter washer may also be worn in such a manner that it will not make full contact with the end of a new plunger. To avoid rapid wear on the end of the new plunger, lifter washers showing visible wear should always be replaced.

Fig. A illustrates the contact surfaces of a new pump plunger and a new lifter washer. In Fig. B the

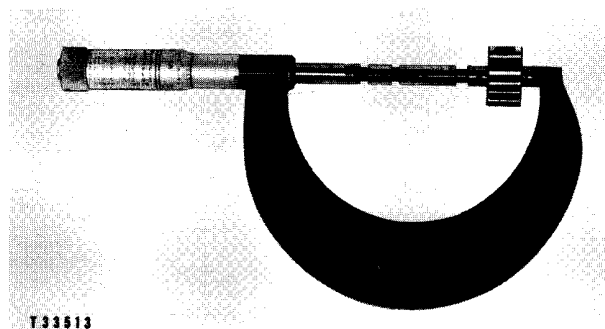
pump plunger and lifter washer have worn considerably. Fig. C shows how the flat end of a new plunger makes poor contact with a worn lifter washer, resulting in rapid wear to both parts.



WEAR BETWEEN WASHER AND PLUNGER

A pump can maintain a satisfactory discharge rate and yet be unserviceable because of delayed timing resulting from wear on the lower end of the plunger. When testing a pump which has been in use for a long time, the length of the plunger should be checked and the pump discarded if the plunger wear exceeds the value listed in the topic, SPECIFICATIONS. The length of new plungers is also given in the specifications. The length should be checked with a micrometer as shown.

Also, inspect the plunger for wear on its upper diameter. The performance of pumps worn in this manner can be checked in the Instructions for Fuel Injection Test Apparatus.



CHECKING LENGTH OF PLUNGER

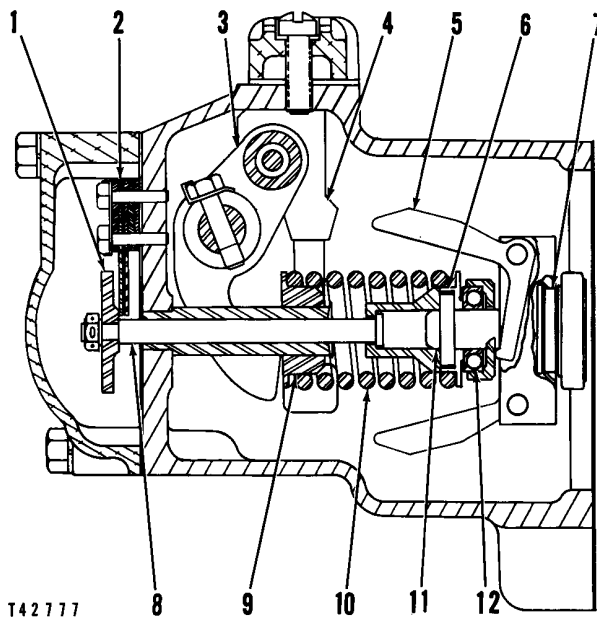
Governor

The governor is located on the right side of the engine and is mounted on the rear face of the fuel injection pump housing. It is driven by a gear on the fuel injection pump camshaft.

OPERATION AND LUBRICATION

The operator selects a desired engine speed by moving the governor control lever. The governor maintains this speed nearly constant even though the load varies.

The governor regulates the amount of fuel injected into the cylinders. The force of the governor spring (10) tends to move the fuel rack (11) forward in the direction of more fuel. The centrifugal force of the rotating weights (5) tends to move the rack to the rear in the direction of less fuel.



T42777

GOVERNOR CROSS SECTION - RIGHT SIDE VIEW

1-Collar. 2-Torque spring. 3-Lever assembly. 4-Shoulder.
5-Weight. 6-Spring seat. 7-Bearing. 8-Bolt. 9-Spring seat.
10-Governor spring. 11-Fuel rack. 12-Thrust bearing.

When the operator moves the governor control lever to increase the engine speed setting, the lever assembly (3) rotates and pushes against the spring seat (9), compressing the governor spring. The spring force moves the fuel rack to increase the engine speed.

As the engine speed increases, the centrifugal force of the rotating weights (5) causes their levers to act against the thrust bearing (12) and governor spring seat (6), thus relieving the force of the gover-

nor spring and moving the rack in the direction of less fuel until a state of equilibrium is reached.

When the engine encounters a load, the speed decreases. The centrifugal force of the governor weights decreases. The spring, opposed by a lesser force, moves the rack in the direction to give the engine more fuel which increases the power. The engine speed will then increase until the force of the governor weights again balances the force of the compressed governor spring.

This sequence is reversed when the load on the engine is decreased.

A torque spring (2) and collar (1) limit the distance the rack assembly can travel. The collar is secured to the bolt (8) which acts against the spring seat (6) which is fastened to the fuel rack (11). As the rack moves forward to increase the fuel being delivered to the cylinders, the collar makes contact with the torque spring and the full load position is reached. Under conditions of lugging, the rack continues to move forward a slight amount and compresses the torque spring, thus delivering an additional quantity of fuel to the cylinders in an attempt to bring the engine out of the lug condition.

See the topic, ADJUSTMENTS, for the high and low idle speed adjustments.

A spring-loaded plunger within the lever assembly bears against the shoulder (4) of the low idle adjusting screw when the governor control lever is in the lower idle position. In order to shut the engine down, it is necessary to force the plunger to ride over the shoulder on the screw.

The governor is lubricated by the engine lubricating system. Oil from the diesel engine oil manifold is directed to a passage in the cylinder block. An external line directs the oil from the passage in the rear of the block to a passage in the fuel injection pump housing which mates with a drilled passage in the governor bearing (7). The various other parts of the governor are lubricated by splash.

The oil drains through a passage in the fuel injection pump housing mounting bracket into the diesel engine crankcase.

REMOVAL AND INSTALLATION

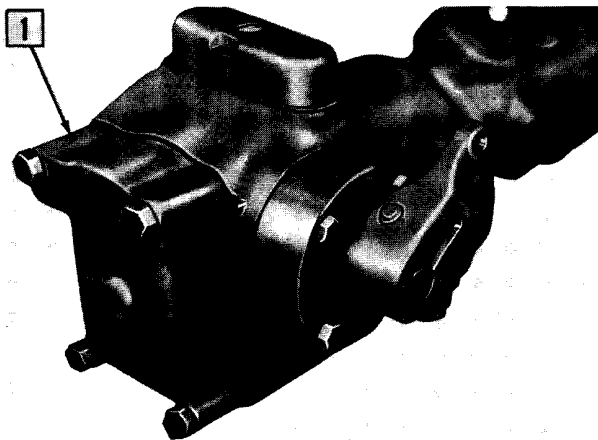
Governor Housing and Control Mechanism

The governor can be disassembled mounted on the engine or removed from the engine.

Remove the governor and fuel injection pump housing as a unit as outlined in the topic, ACCESSORY DRIVE REMOVAL AND INSTALLATION.

ENGINE GOVERNOR

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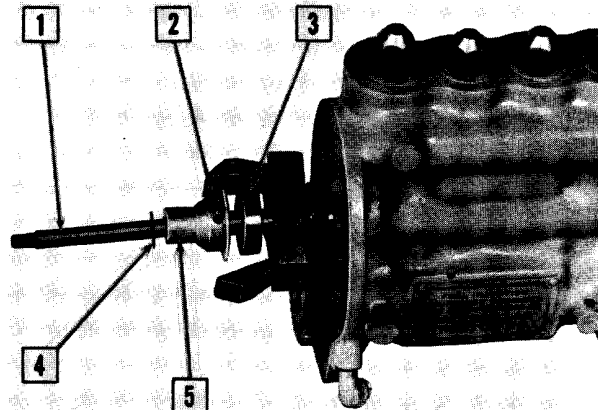


T 56559

PREPARING TO REMOVE GOVERNOR HOUSING

☐ Remove

1-Cover.



T 41921

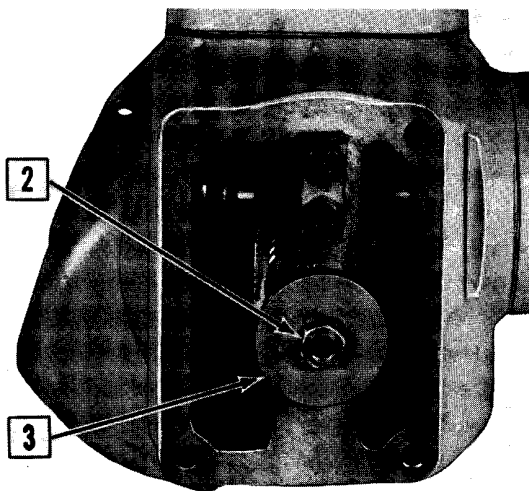
PREPARING TO REMOVE GOVERNOR WEIGHT

☐ Remove

1-Bolt. 2-Pin. 3-Thrust bearing. 4-Bumper spring. 5-Seat.

NOTE

The bumper spring (4) is installed with the concave face toward the seat (5). This arrangement permits the outside edge of the spring to contact the seat (5).



T 43715

PREPARING TO REMOVE GOVERNOR HOUSING

☐ Remove

2-Nut. 3-Collar.

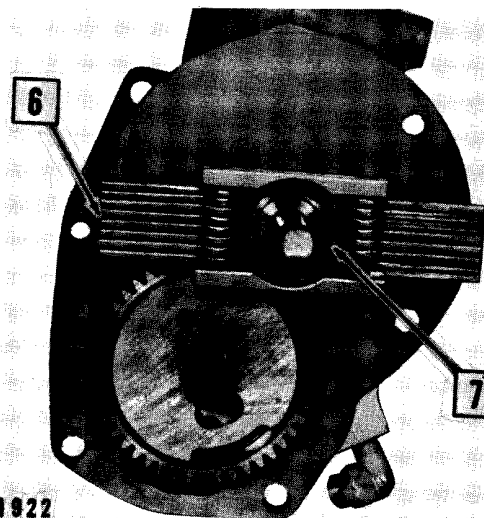
NOTE

Insert a pin through the hole in the full load stop bolt to prevent the bolt from turning while removing and installing the nut (2) and collar (3).

Remove the bolts securing the governor to the fuel injection pump housing and slide the governor housing away from the fuel injection pump housing.

Governor Weight Removal

Separate the governor housing from the fuel injection pump housing as outlined in the covering topic.



T 41922

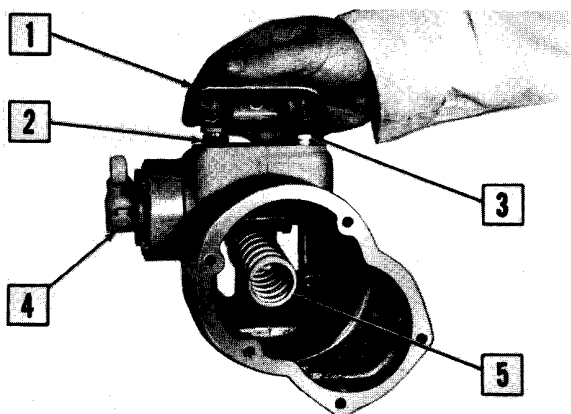
WEIGHT REMOVAL

☐ Remove

6-Weight assembly. 7-Snap ring.

GOVERNOR DISASSEMBLY

1. Remove the seat behind the spring (5).
2. Remove the bolts securing the cover assembly (6) to the governor housing and remove the cover assembly (6), control shaft assembly (9) and the governor shaft assembly (10).
3. The gasket between the cover assembly (6) and the governor housing should be inspected, and replaced if damaged.

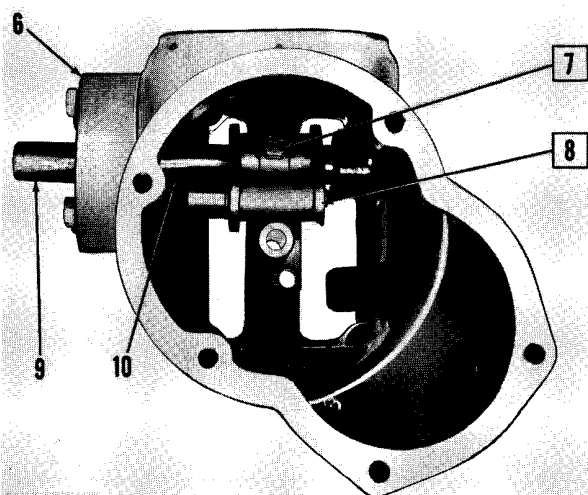


T 56560

DISASSEMBLING THE GOVERNOR

☐ Remove

1-Cover. 2-High idle screw. 3-Low idle screw. 4-Lever.
5-Governor spring.

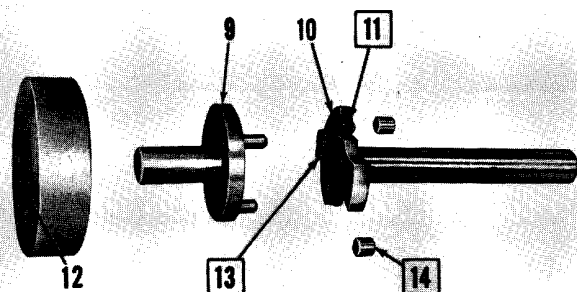


T 52465

COVER ASSEMBLY AND LEVER ASSEMBLY REMOVAL

☐ Remove

6-Cover assembly. 7-Bolt. 8-Lever assembly. 9-Control
shaft assembly. 10-Governor shaft assembly.



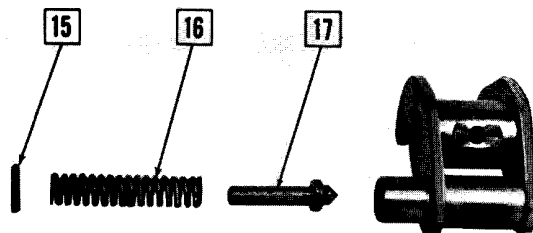
T 52466

DISASSEMBLING COVER AND SHAFT ASSEMBLIES

☐ Remove

9-Control shaft assembly. 10-Governor shaft assembly.
11-Roller springs (two). 12-Seal. 13-Locating spring.
14-Rollers (two).

4. Separate the shaft assemblies (9) and (10) and remove the seal (12) from the cover assembly.
5. Replace the seal (12), with the lip facing in, if it is damaged.



T 52764

DISASSEMBLING LEVER ASSEMBLY

☐ Remove

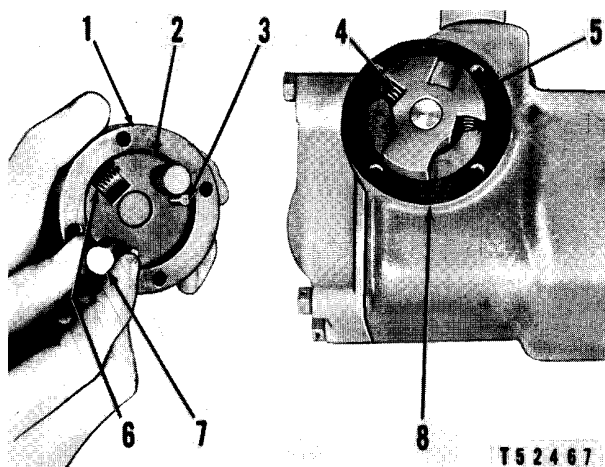
15-Pin. 16-Spring. 17-Plunger.

Control Mechanism Assembly and Installation

NOTE

Apply a light oil to the parts of the control mechanism during assembly.

1. Inspect the inside surface of the cover assembly (1). If the ramp is excessively worn, the assembly should be replaced.
2. Install the control shaft assembly (2) into the cover assembly (1).
3. Insert the roller springs (4) into the governor shaft assembly (5) as shown, using a small amount of grease to hold them in place.
4. Position the rollers (7) on the dowels (3) as shown.



T 52467

PREPARING TO INSTALL COVER ASSEMBLY

1-Cover assembly. 2-Control shaft assembly. 3-Dowel (two). 4-Roller springs (two). 5-Governor shaft assembly. 6-Locating spring. 7-Rollers (two). 8-Gasket.

ENGINE GOVERNOR

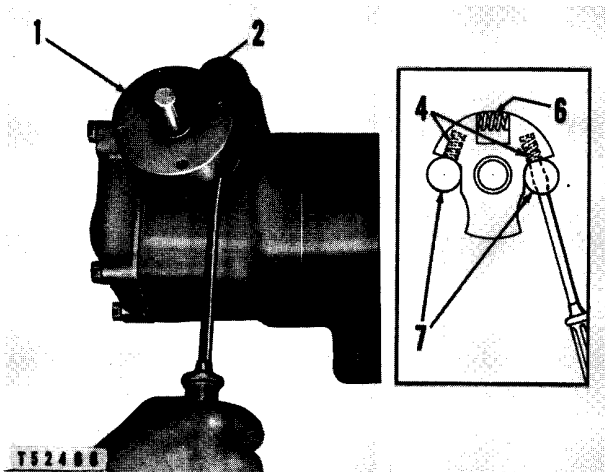
ISSUED 9-62

5. Insert the locating spring (6) into the groove in the control shaft assembly.
6. Install the gasket (8) on the governor housing as shown.
7. Install the cover assembly (1) over the governor shaft assembly (5) until the rollers (7) contact the roller springs (4).
8. Using a screwdriver inserted between the cover assembly (1) and the governor housing as shown, compress the roller springs (4) to allow the rollers (7) to position properly in front of the roller springs (4) and to permit the cover assembly to fit in place.
10. Insert and tighten the bolts securing the cover assembly to the governor housing.
11. When assembling the lever assembly to the shaft assembly, rotate the shaft so the lever (9) will be in the position shown, before installing the bolt into the lever assembly.

ADJUSTMENTS

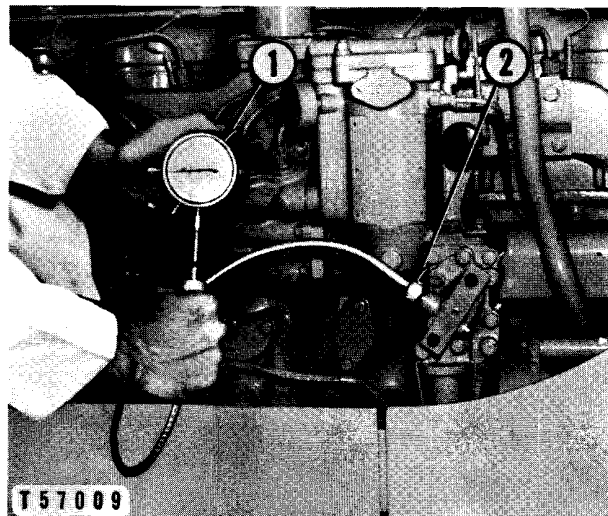
High and Low Idle Speed

To check the speed of the engine, remove the service meter and install a 7M6001 Tachometer Drive (2) as shown.



INSTALLING COVER ASSEMBLY

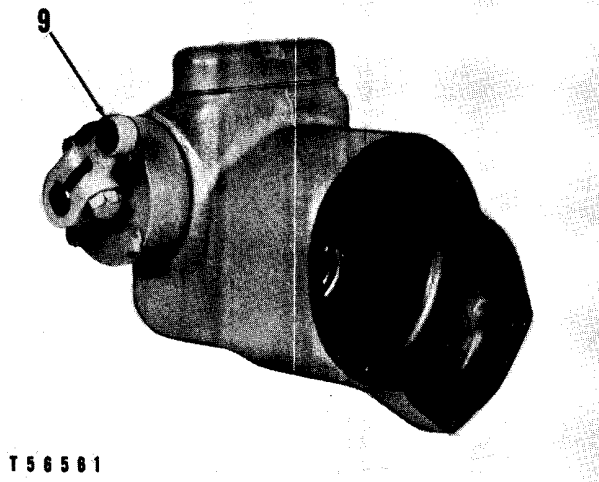
1-Cover assembly. 2-Control shaft assembly. 4-Roller springs. 6-Locating spring. 7-Rollers.



CHECKING ENGINE IDLE SPEED

1-Tachometer. 2-7M6001 Tachometer Drive.

9. Turn the control shaft assembly (2) back and forth slightly to seat the locating spring (6) in its groove properly.



LEVER ASSEMBLY INSTALLATION

9-Lever.

Install a 1:1 ratio tachometer on the flexible coupling of the tachometer drive and start the engine. The reading observed will be actual engine speed.

CAUTION

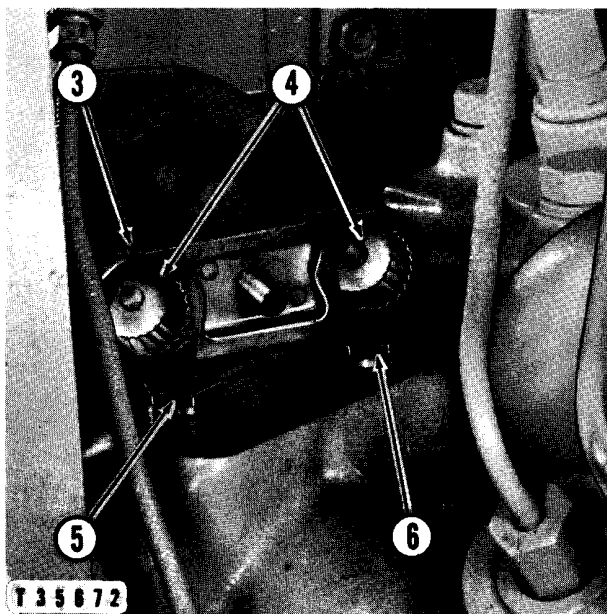
Do not attempt to install or remove the flexible coupling with the engine running.

The high and low idle speeds can be adjusted by removing the cover (3) on the top of the governor housing and turning the adjusting screws (5) and (6). The screw (6) nearest the diesel engine adjusts the low idle speed. The holes (4) in the cover are shaped to act as retainers to prevent the screws from turning after the adjustment is made.

To adjust the idle speed settings, remove the cover and turn the desired adjusting screw. Turning either adjusting screw in a clockwise direction will decrease the idling speeds.

After setting the idle speed, move the governor control lever to change the engine speed. Return it

to the desired idle position and recheck the idle speed. Repeat this procedure until the desired idle speed is obtained.



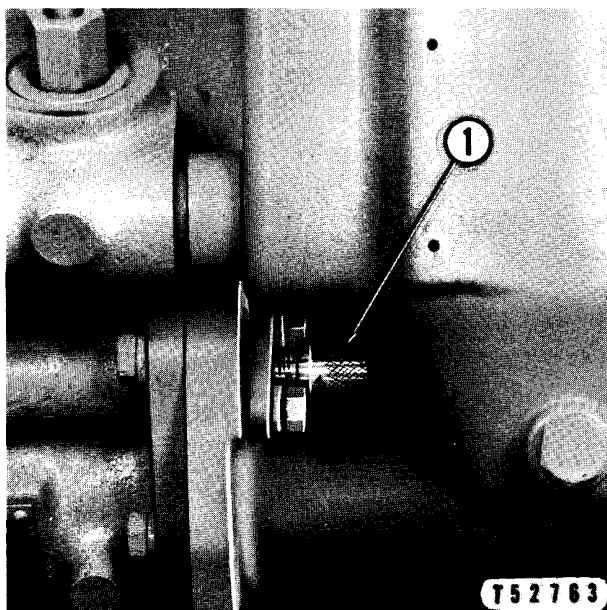
GOVERNOR ADJUSTMENTS

3-Cover. 4-Retainer holes. 5-High idle adjusting screw.
6-Low idle adjusting screw.

Fuel Rack Setting

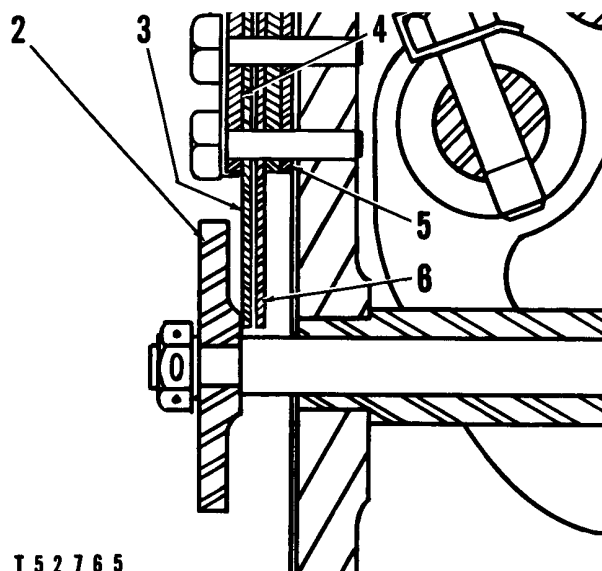
The fuel rack setting can be checked and adjusted with the fuel injection pump housing either removed or installed on the engine.

1. Remove the rack cover from the front of the fuel injection pump housing and the cover from the rear of the governor housing.



CHECKING FUEL RACK SETTING
1-8M530 Rack Setting Gauge.

2. Install an 8M530 Rack Setting Gauge (1) over the front end of the fuel rack.
3. Adjust the position of the fuel rack to obtain the desired full load rack setting. With the fuel rack in the full load position, the collar (2) should just contact the torque spring (3).
4. If the collar (2) is compressing the torque spring (3), remove shims (5), until the collar just contacts the torque spring when the fuel rack is at the full load setting for the engine. Place all shims removed from behind the stop bar (6) between the spacer (4) and spring (3) where they will be available for future use.



ADJUSTING FUEL RACK SETTING

2-Collar. 3-Torque spring. 4-Spacer. 5-Shims.
6-Stop bar.

NOTE

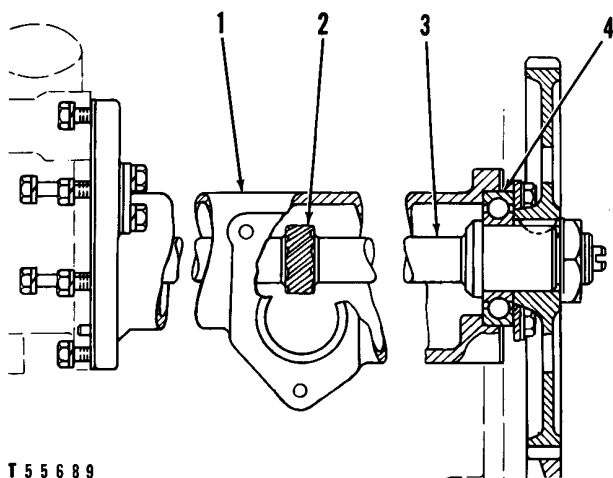
Do not place any of the removed shims between the torque spring (3) and the stop bar (6) as this will affect the torque characteristics of the engine during lug conditions. The spacer between the torque spring and the stop bar must remain in place.

5. If the collar does not contact the torque spring when the fuel rack is in the full load position, add shims (5) to obtain the proper setting.

Accessory Drive

The accessory drive is the assembly used to drive the governor, fuel injection pump camshaft, service meter and fuel transfer pump.

The accessory drive shaft (3) is supported by a ball bearing (4) at the front and enclosed by an adapter housing (1) on the right side of the engine. The adapter bolts to the timing gear housing plate at the front. The fuel injection pump housing is bolted to the rear of the adapter housing.



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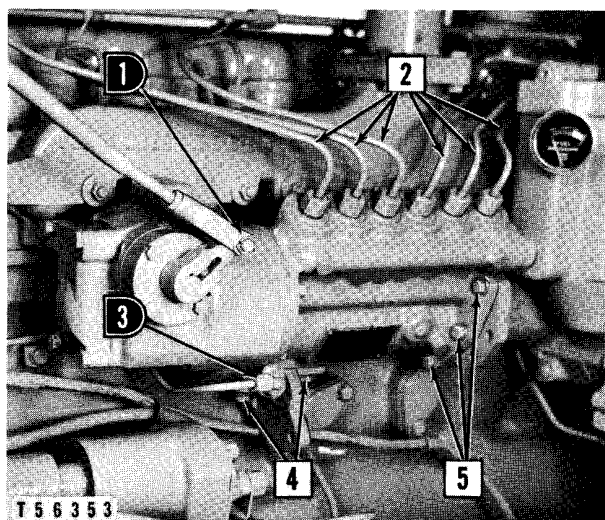
ACCESSORY DRIVE (PARTIAL CROSS-SECTION)
1-Adapter housing. 2-Gear. 3-Accessory drive shaft.
4-Bearing.

The fuel transfer pump is driven by a helical gear (2) machined on the accessory drive shaft. The service meter is mounted on the fuel transfer pump cover and is driven by the transfer pump drive shaft. The fuel injection pump camshaft is coupled to the rear of the accessory drive shaft and the governor is driven from the rear of fuel pump camshaft.

REMOVAL AND INSTALLATION

The accessory drive shaft can be removed without removing the timing gear cover. The accessory drive gear is accessible after removing the cover on the timing gear cover assembly.

1. Install caps over the fuel injection pumps, and plugs in the fuel injection lines to prevent the entry of dirt into the fuel injection system.
2. Remove the fuel injection pump housing and the governor as an assembly by sliding the assembly toward the rear of the engine.
3. Remove the fuel transfer pump and fuel filter. See the covering topics.

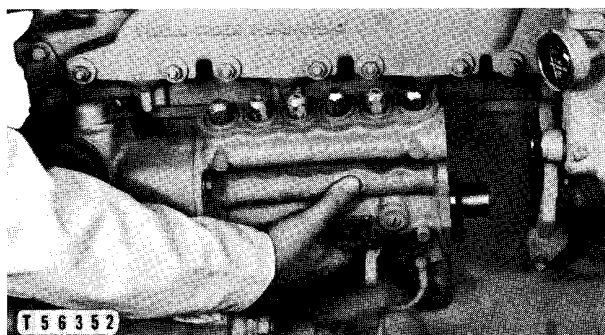


PREPARING TO REMOVE THE FUEL INJECTION PUMP HOUSING AND GOVERNOR

☐ Remove ☒ Disconnect

1-Governor control linkage. 2-Fuel injection lines. 3-Lubricant supply line. 4-Retaining bolts. 5-Fuel injection pump housing retaining bolts.

4. Remove the fan belts. See the topic, FAN BELT REPLACEMENT.
5. Remove the accessory drive gear cover which is located on the timing gear cover.
6. Position the accessory drive gear to allow access to the bolts which secure the accessory drive housing to the timing gear plate and remove the bolts.



GOVERNOR AND FUEL INJECTION PUMP HOUSING REMOVAL

7. Remove the accessory drive gear retaining nut.
8. Use a 5F7465 Puller and 4M3251 Adapter to push the accessory drive shaft out of the accessory drive gear.

NOTE

The pin (6) in the timing gear cover prevents the accessory drive gear from moving out of

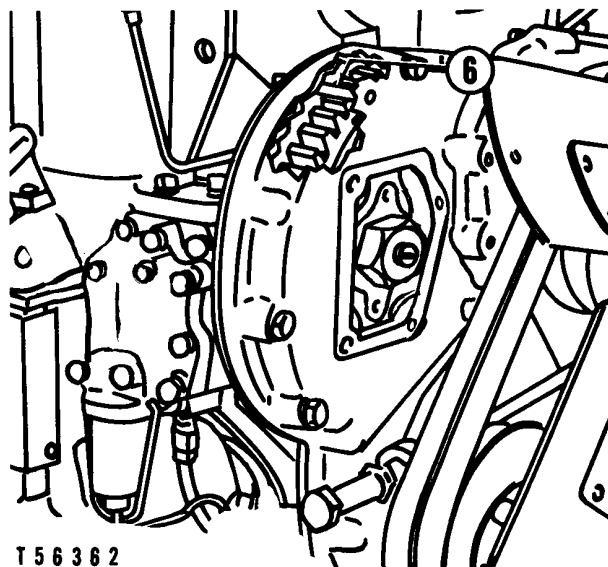
ENGINE ACCESSORY DRIVE

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REMOVING THE ACCESSORY DRIVE SHAFT
6-Pin.

engagement with the accessory drive idler gear by limiting the distance between the centers of the two gears after the accessory drive shaft has been removed.



GEAR LOCATING PIN
6-Pin.

9. Remove the accessory drive housing and shaft.
10. Prior to installation of the accessory drive housing, inspect and replace the gasket between the timing gear plate and the accessory drive housing if it is damaged.
11. Use the 4M3250 Camshaft Gear Installation

Group to pull the accessory drive shaft through the accessory drive gear.

CAUTION

Use care to prevent the bearing retainer from falling down inside the timing gear cover.

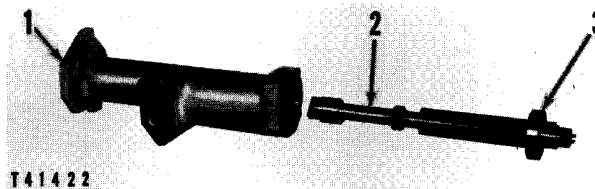


INSTALLING THE ACCESSORY DRIVE SHAFT

DISASSEMBLY AND ASSEMBLY

Remove the accessory drive shaft (2) from the adapter housing (1).

Press the bearing (3) off the shaft.



ACCESSORY DRIVE DISASSEMBLY

1-Adapter housing. 2-Accessory drive shaft. 3-Bearing.

Assemble the accessory drive.

SERVICE METER

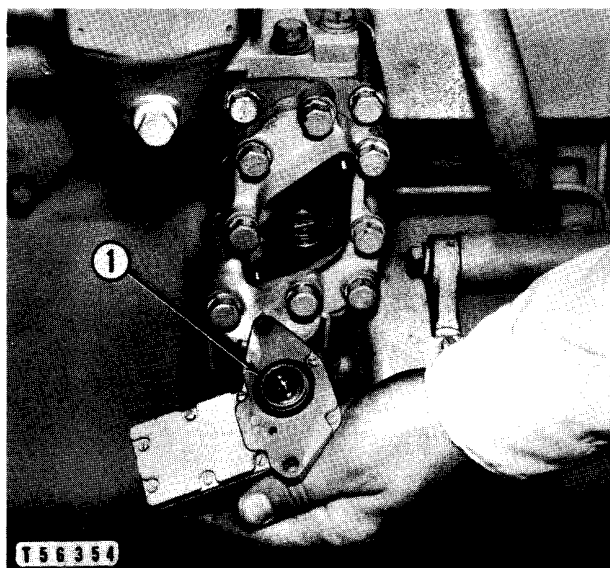
The service meter is mounted on the fuel transfer pump cover and is driven by the transfer pump drive shaft. It provides a means of determining time periods for lubrication and maintenance.

The end of the transfer pump drive shaft is grooved to drive the service meter. A tachometer drive group can be installed between the service meter and the transfer pump cover. Engine speed can be checked at this point by using a speed indicator. The speed obtained in this manner will be actual engine speed.

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**ENGINE
ACCESSORY DRIVE**

To remove the service meter, remove the bolts which secure the service meter to the transfer pump cover. Replace the seal (1) if damaged.



REMOVING THE SERVICE METER
1-Seal.

Air Induction and Exhaust System

INLET MANIFOLD

The inlet manifold is cast as an integral part of the cylinder head. It is located within the left side of the cylinder head. The inlet manifold has an external opening at the upper front of the cylinder head for mounting the air inlet elbow.

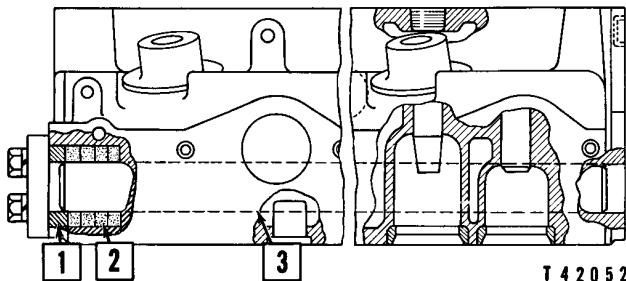
A passage (pipe) which carries starting engine exhaust gases through the inlet manifold has openings at either end of the cylinder head. The starting engine exhaust gases enter through the rear opening and are expelled from the front opening into the starting engine exhaust pipe.

The heat from these exhaust gases heats the incoming air to the diesel engine. This aids in starting the diesel engine.

INLET AIR HEATER PIPE REMOVAL AND INSTALLATION

If the inlet air heater pipe should require replacement, it can be removed in the following manner.

1. Disconnect the starting engine exhaust pipe at the rear of the diesel engine inlet manifold.
2. Disconnect the starting engine exhaust outlet pipe at the front of the diesel engine inlet manifold.



REPARING TO REMOVE INLET AIR HEATER PIPE

☐ Remove

1-Washer. 2-Packing. 3-Inlet air heater pipe.

3. The inlet air heater pipe can be driven from the front of the cylinder head toward the rear by using a bushing driver with a maximum shoulder diameter of 1.120 inches and a pilot diameter of 0.990 inch.
4. Install new packing at the time of installation.

EXHAUST MANIFOLD

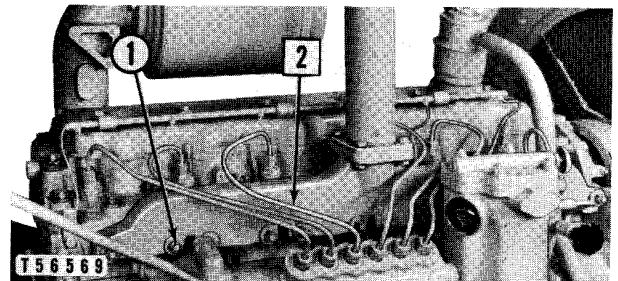
The exhaust manifold is of one-piece construction and is bolted directly to the cylinder head. Gaskets form the seal between the manifold and the cylinder head.

Removal and Installation

Place caps over the fuel injection pumps and nozzles, and plugs in each end of the fuel injection lines to prevent the entry of dirt or other foreign matter into the fuel injection system.

Remove the nuts (1) which secure the exhaust manifold to the cylinder head.

New gaskets should be used when installing the exhaust manifold.



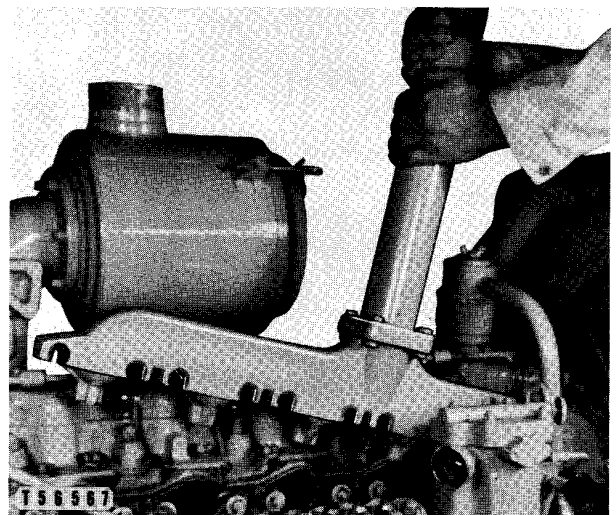
PREPARING TO REMOVE EXHAUST MANIFOLD

☐ Remove

1-Nuts (twelve). 2-Fuel injection lines (six).

NOTE

The exhaust manifold cannot be removed by lifting it straight up from the engine. It must be rotated about the stud on the right end. Then by moving the manifold towards the rear of the engine, it can be removed.



REMOVING EXHAUST MANIFOLD

AIR CLEANER

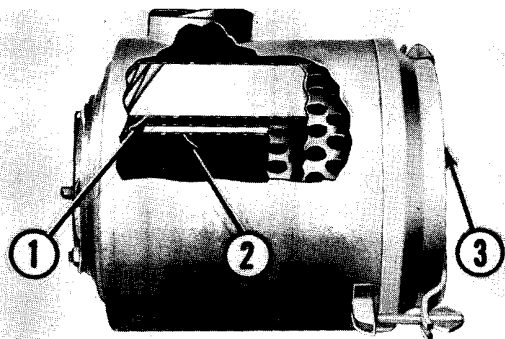
A horizontally-mounted, dry-type air cleaner containing two separate air cleaner elements, is used on this engine. The primary element (1), can be re-

ENGINE

AIR INDUCTION AND EXHAUST SYSTEM

ISSUED 9-62

moved and cleaned with air pressure, or by washing as recommended in the Operation and Maintenance Instructions. A secondary element (2) is provided in case of a rupture in the primary element.



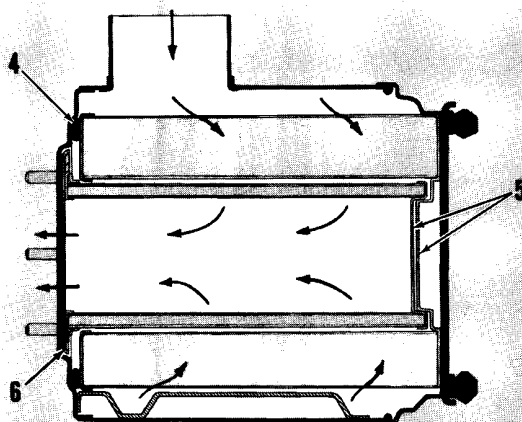
T56550

AIR CLEANER

1-Primary element. 2-Secondary element. 3-Body.

The secondary element should be removed and serviced periodically in accordance with the Operation and Maintenance Instructions. Both elements should also be replaced periodically as covered in the Operation and Maintenance Instructions.

During operation, air from the prescreener flows around the outside and through the elements to the center of the air cleaner and from there to the inlet manifold. Air is prevented from by-passing the primary element by a seal (4) which is bonded to the end of the primary element. Steel plates (5) cover and seal one end of each element. Gaskets (6) between the secondary element, body assembly and air transfer pipe prevent unfiltered air from being drawn into the engine at these points.

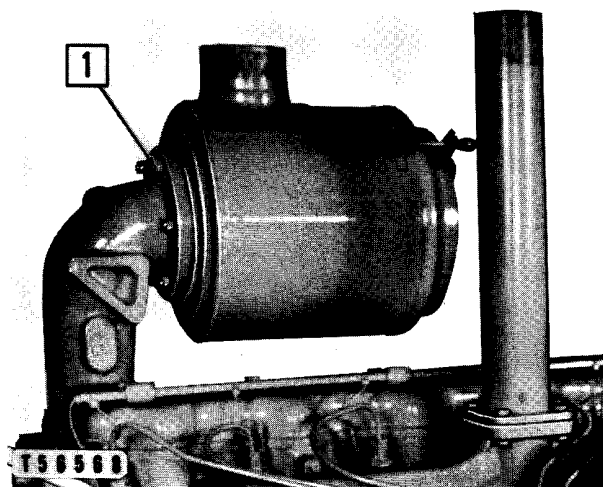


T41518

AIR CLEANER AIR FLOW
4-Seal. 5-Plates. 6-Gaskets.

Removal and Installation

Remove the prescreener from the extension on the body assembly and remove the hood.



PREPARING TO REMOVE AIR CLEANER

- ☐ Remove
1-Nut (six).

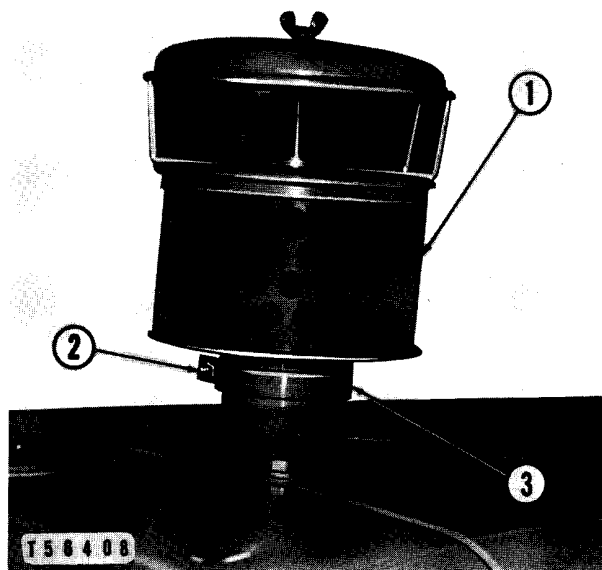
Remove the air cleaner and cover the opening in the inlet elbow.

Install in the reverse order of removal, using new gaskets between the air cleaner and the elbow.

PRESCREENER

Either a precleaner assembly or a vertically mounted dry-type air prescreener can be used on this engine. The prescreener filters atmospheric air and directs the pre-filtered air to the air cleaner. The prescreener is mounted on an extension of the air cleaner body assembly.

Removal and Disassembly



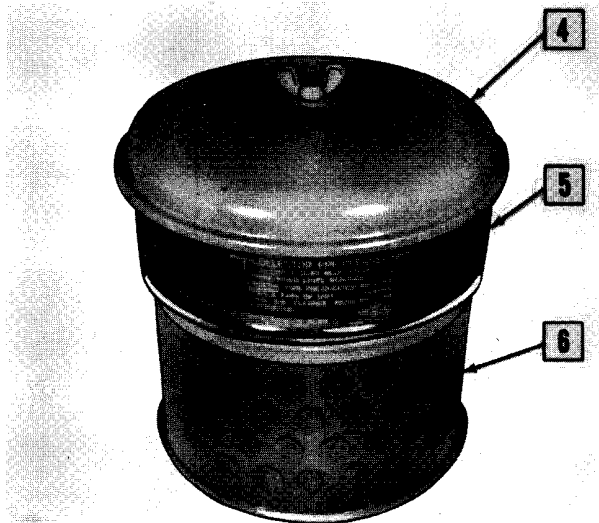
PREPARING TO REMOVE PRESCREENER
1-Prescreener. 2-Bolt. 3-Clamp.

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AIR INDUCTION AND EXHAUST SYSTEM

Loosen the bolt (2) on the clamp (3) and remove the prescreener.



T56410

PREPARING TO DISASSEMBLE PRESCREENER☐ Remove

4-Cover assembly. 5-Body. 6-Screen assembly.



T56409

PRESCREENER DISASSEMBLED

The body (5) and screen (6) can be cleaned with air pressure or by washing as recommended in the Operation and Maintenance Instructions.

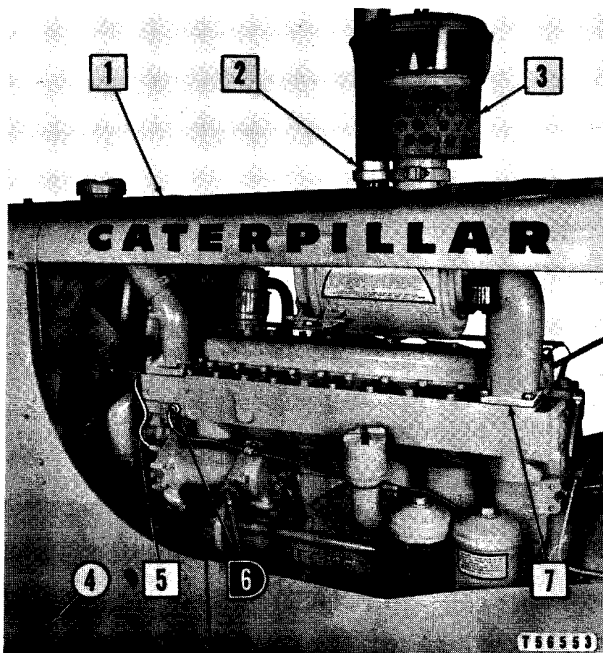
Cylinder Head

The inlet manifold is cast integral with the cylinder head on the left side. Water directors direct the flow of coolant toward the valve ports and precombustion chambers.

Rubber seals and ferrules seal the water and lubrication passages between the cylinder head and cylinder block.

CYLINDER HEAD REMOVAL (Electric Start)

1. Drain the coolant from the engine. The drain plug can be reached through the hole (4).



PREPARING TO REMOVE CYLINDER HEAD

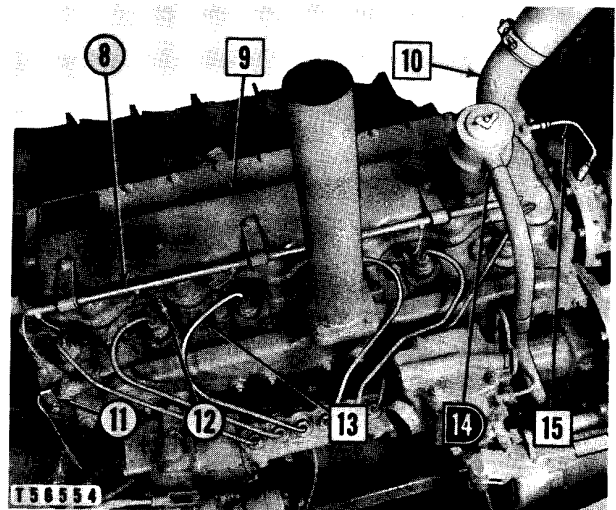
☐ Remove ☒ Disconnect

- 1-Hood. 2-Exhaust pipe. 3-Pre-screener. 4-Hole. 5-Line. 6-Heat indicator. 7-Air cleaner and air transfer pipe.

2. Disconnect wiring harness (8) from glow plugs (12) and the clip (11). After removing the lines (13), cap the injectors, pumps and lines to prevent dirt or other foreign matter from entering.
3. Remove the exhaust manifold (17). See the covering topic.

NOTE

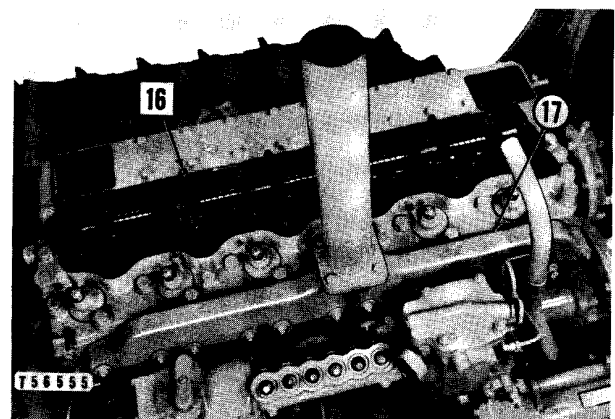
It is necessary to remove the exhaust manifold in order for one of the head bolts to be tightened to the proper torque value. Remove any glow plugs necessary to provide easier access to the bolt heads.



PREPARING TO REMOVE CYLINDER HEAD

☐ Remove ☒ Disconnect

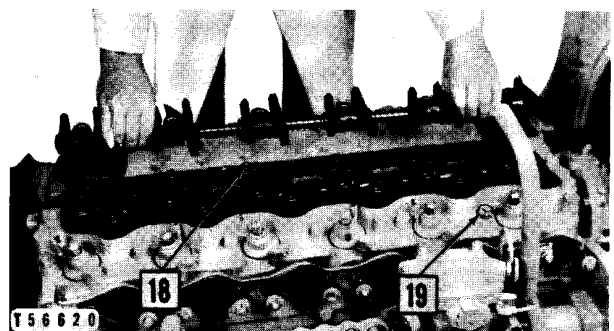
- 8-Wiring harness. 9-Valve cover. 10-Water line. 11-Clip. 12-Glow plug (six). 13-Fuel injection line (six). 14-Oil breather line. 15-Vent line.



PREPARING TO REMOVE CYLINDER HEAD

☐ Remove

- 16-Rocker arm assembly. 17-Exhaust manifold.



REMOVING ROCKER ARM ASSEMBLY

☐ Remove

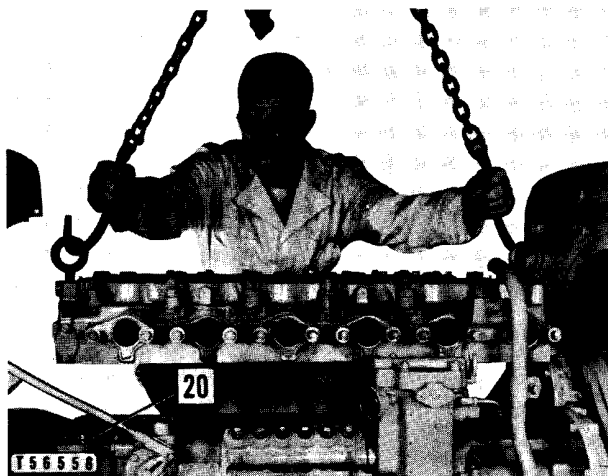
- 18-Push rods. 19-Head bolt.

4. Install two $\frac{3}{4}$ " - 10 (NC) eyebolts in the cylinder head as shown.

ENGINE CYLINDER HEAD

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5. Use a suitable hoist to remove the cylinder head from the cylinder block. The approximate weight of the cylinder head is 325 pounds.



REMOVING CYLINDER HEAD

☐ Remove

20-Seals and ferrules.

CYLINDER HEAD INSTALLATION

A new cylinder head gasket should be installed.

1. Place the gasket on the cylinder block.
2. Install the ferrules and seals.
3. Carefully lower the head on the block. Proper head alignment is assured by dowel pins located in the block.

CAUTION

When installing the head on the cylinder block, see that the ferrules and seals are properly installed and started properly into the recessed holes in the head. Rock the head gently until it seats flat on the head gasket.

4. Install the head bolts and tighten them in the numerical sequence as illustrated. See the topic, SPECIFICATIONS, for torque values.
5. Install the push rods and valve rocker arm assembly.

If, after engine run-in, the bolt torque has dropped below the value given in the topic, SPECIFICATIONS, the bolts should be retightened.

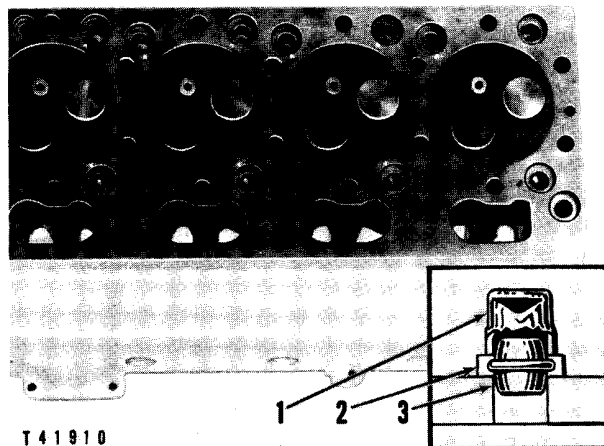
6. Set the valve clearance as given in the topic, SPECIFICATIONS.

WATER DIRECTORS

Water directors (1) are provided to direct the flow of coolant to critical areas for maximum cooling ef-

fectiveness. They are pressed into place in the head, after aligning the notch on the director with the V-mark on the head.

The seals (2) and copper ferrules (3), which seal the water passage between the head and top of the block, are replaceable. The seal is installed over the flange on the ferrule. This can be done easily if the inner surface of the seal is first coated with liquid soap.

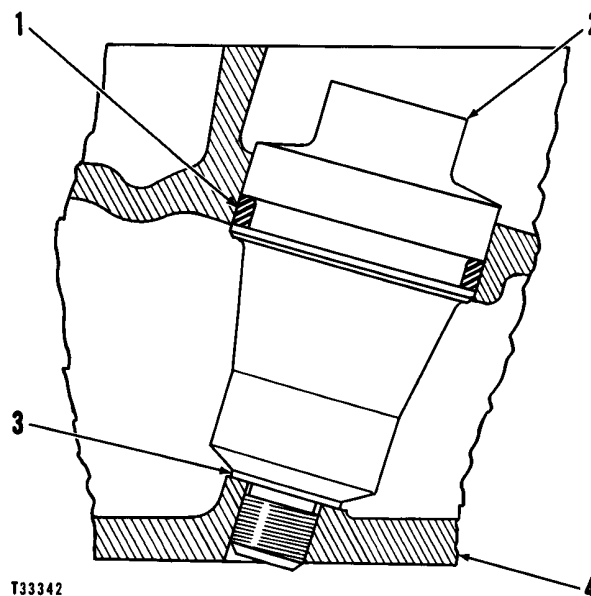


CYLINDER HEAD SHOWING WATER DIRECTORS

1-Director. 2-Seal. 3-Ferrule.

When installing the ferrules and seals, the rolled edge of the ferrule should be toward the head to facilitate installation of the head.

PRECOMBUSTION CHAMBERS



PRECOMBUSTION CHAMBER

1-O-ring seal. 2-Precombustion chamber. 3-Gasket. 4-Cylinder head.

The precombustion chamber (2) is threaded into the cylinder head (4) and sealed at the top of the head with an O-ring seal (1).

The O-ring seal prevents leakage of coolant.

The gasket (3) prevents combustion gases from entering the cooling system, as well as preventing leakage of coolant into the cylinder.

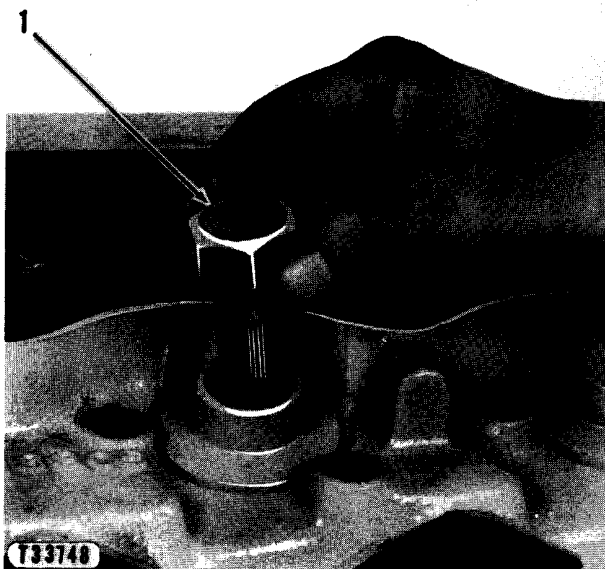
Removal and Installation

1. Remove the fuel injection valve assembly.
2. Place the 5F8353 Wrench Adapter (1) in the serrations of the precombustion chamber and remove the precombustion chamber from the head.

NOTE

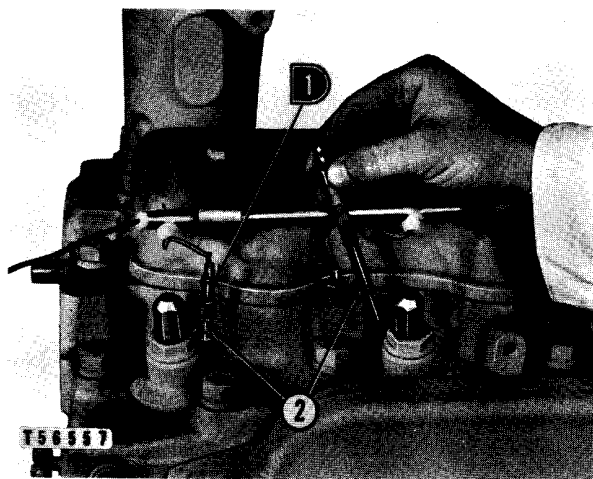
If a cylinder head has collected an excessive amount of scale or rust within the water jacket, remove the cylinder head and clean it thoroughly. This is also an indication that the entire cooling system should be cleaned.

3. Place a new gasket, coated on the bottom side with a thin layer of grease, in the head. The grease will help retain the washer while installing the precombustion chamber.
4. Coat the chamfered portion of the cylinder head and the new rubber seal with soap.
5. Insert the precombustion chamber in the head and tighten with the 5F8353 Wrench Adapter. See the topic, SPECIFICATIONS, for the correct torque value when installing the precombustion chamber.



REMOVING PRECOMBUSTION CHAMBER
1-5F8353 Wrench Adapter.

Removal and Installation of Precombustion Chambers Equipped with Glow Plugs



REMOVING GLOW PLUG

D Disconnect

1-Electrical lead. 2-Glow plugs.

1. Remove the glow plugs (2) as shown.

NOTE

The precombustion chambers (3) and the gaskets (4) must be marked so they can be installed in their original locations unless new gaskets are to be installed.

2. Remove the precombustion chambers (3). See the topic, PRECOMBUSTION CHAMBER REMOVAL AND INSTALLATION.



REMOVING PRECOMBUSTION CHAMBER
EQUIPPED WITH GLOW PLUG
3-Precombustion chamber. 4-Gasket.

**ENGINE
CYLINDER HEAD**

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NOTE

If new gaskets are to be installed, see the topic, SPECIFICATIONS, for proper positioning of the precombustion chambers.

3. Install the precombustion chamber, using the thinnest gasket, and tighten it to the torque value specified in the topic, SPECIFICATIONS.
4. Insert the glow plug. If it cannot be installed because of interference, remove the precombustion chamber and discard the gasket.
5. Reinstall the precombustion chamber using the thicker gasket and tighten it to the specified torque value.
6. Install the glow plugs and tighten to the torque value shown in the specifications.
7. Connect the electrical leads.

ENGINE VALVES AND VALVE MECHANISM

Valves and Valve Mechanism

The valves and the valve mechanism admit inlet air and release exhaust gases at precisely timed intervals during engine operation.

The engine is of the four-stroke cycle, having four separate strokes required for each cylinder to complete one cycle. The camshaft is timed to the crankshaft and turns at one-half engine speed.

The inlet valve is open during the inlet stroke of the piston to permit air to enter the cylinder. After the compression and power strokes, the exhaust valve is opened to permit the piston, in moving up on the exhaust stroke, to force the burned gases from the cylinder.

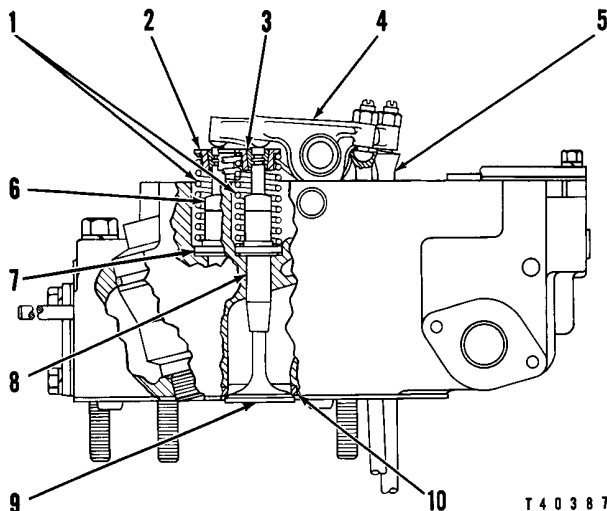
Properly adjusted valves will operate for many hours before they need to be reconditioned. Eventually, however, the valve faces and seats can become pitted which ultimately allows compression pressure losses.

Valve rotators cause the valves to rotate while the engine is operating. The rate of rotation is approximately three degrees each time the valve is opened.

This rotation decreases the possibility of excessive carbon deposit build up on the valves.

NOTE

Naturally aspirated engines have no valve stem guide seals (6) and the valve rotators (7) are replaced by spacers.



VALVES AND VALVE MECHANISM

1-Valve springs. 2-Retainer. 3-Lock. 4-Rocker arm. 5-Push rod. 6-Valve guide seal. 7-Valve rotator. 8-Valve guide. 9-Valve. 10-Valve seat insert.

To determine whether a valve rotator is operating, observe the valve spring retainer. If the retainer does not rotate during operation, replace the valve rotator as a unit.

VALVE ROCKER ARM ASSEMBLY

Removal

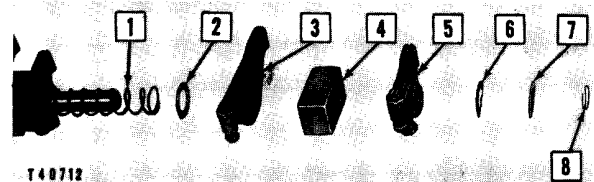
Remove the rocker arm cover.

NOTE

When replacing the rocker arm cover, torque the retaining bolts to 5 lb. ft.

Remove the bolts which secure the rocker arm assembly to the cylinder head. Remove the rocker arm assembly.

Disassembly



VALVE ROCKER ARM DISASSEMBLY

☐ Remove

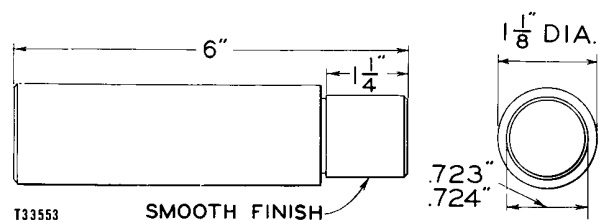
1-Spring. 2-Washer. 3-Rocker arm. 4-Bracket. 5-Rocker arm. 6-Spring. 7-Washer. 8-Ring.

Remove the remaining brackets and rocker arms from the rocker arm shaft.

The bearings in the rocker arms should be replaced if the clearance between the bearings and the shaft exceeds the value listed in the topic, SPECIFICATIONS.

Assembly

In order to properly install new rocker arm bearings, a driver similar to the one shown in the illustration should be used.



TOOL FOR ALIGNING ROCKER ARM BEARINGS

NOTE

Be sure to align the oil holes in the bearing and the rocker arm before pressing the bearing into the rocker arm.

Assemble the rocker arm shaft.

Installation

Install the rocker arm assembly. Make sure the compression release shaft is located properly to al-

ENGINE

VALVES AND VALVE MECHANISM

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low the pin in the rocker arm bracket to locate properly.

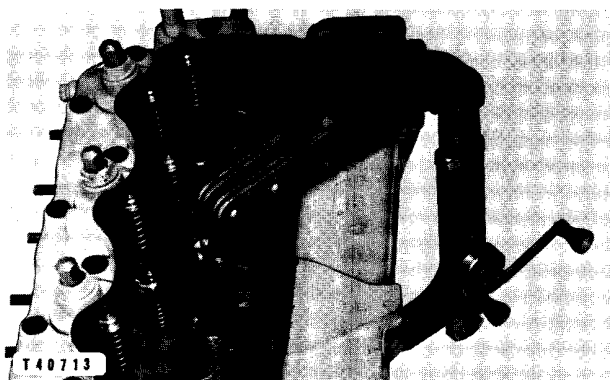
Tighten the rocker arm bolts to the torque values listed in the specifications.

VALVES

Removal

Remove the cylinder head as described in the covering topic.

Compress the valve springs and remove the locks. Release the compressor and remove the retainers, springs, and the valve rotators or spacers if so equipped. The valves can then be removed.



COMPRESSING VALVE SPRING

The illustration shows a standard valve spring compressor being used.

Cleaning

After removing a cylinder head from the diesel engine and the valve assemblies from the head, carefully scrape all carbon accumulations from the parts. Be sure to clean the valves thoroughly, as well as the valve stems, valve guides and valve ports.

Valve Inspection and Reconditioning

The valves should always be carefully inspected. If the valve faces are pitted or making poor contact with the valve seat, they should be refaced in a valve refacing machine. If the valves are deeply pitted, badly warped or worn, they should be replaced.

In refacing, be sure there is sufficient metal left on the head of the valve to prevent "dishing" of the valve in service.

The wear of the valve stems can be checked by the use of a 0" to 1" micrometer. The valve stem should be measured in three places. Use the measurement near the top of the valve stem, where the valve stem does not touch the guide, as the original valve stem diameter.

CAUTION

In handling valves, caution should be exercised to prevent nicking or scratching the radius between the valve face and stem. A very small nick can cause the valve head to break off during service.

See the topic, SPECIFICATIONS.

Checking Valve Seats

Coat the valve face with Prussian blue and rotate the valve in the valve seat. Remove the valve and examine the contact pattern on both valve and seat. A line of contact near the top and around the entire circumference of the valve seat should indicate line contact with the valve.

The valve seats can be ground with a valve seat grinding tool. Care should be exercised in its use, as too much material may be removed quickly and unknowingly.

After the valve seats have been ground until they are smooth and concentric with the valve guides, all parts should be cleaned thoroughly.

Valve Installation

Exhaust valves marked "EX" on the valve head should be installed in the exhaust ports. Inlet valves marked "IN" on the valve head should be installed in the inlet ports.

Lubricate the stems, guides and seals for initial starting.

Insert the valve through the valve guide and install the spring and the spring retainer. Compress the spring with a compressor, such as shown in the topic, VALVE REMOVAL. Insert the locks, large end down, and tap the retainer lightly as the spring compressor is removed, to make sure the locks are seated properly in the retainer.

NOTE

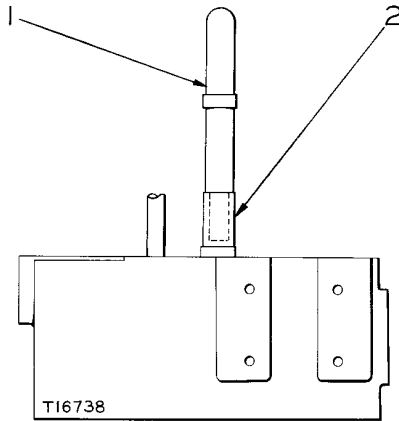
Install the spring with the painted end up.

VALVE GUIDES

The inlet and exhaust valves operate in replaceable valve guides. After the valves have been removed, clean the valve stems and the valve guides and inspect the valve guide seal, if so equipped.

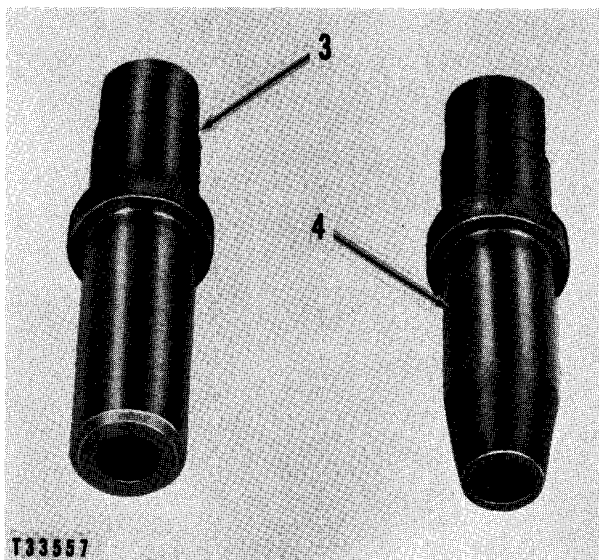
The valve guide wear should be checked with a gauge or by the pilots furnished with some makes of valve seat regrinding equipment. Generally the pilots are supplied in graduated sizes. Use a micrometer to measure the diameter of the largest pilot

that will pass through the guide. This dimension will indicate the wear in the guide, excluding possible out-of-roundness.



INSTALLING VALVE GUIDE
1-4H447 Driver. 2-Valve guide.

The valve guide (2) can be pressed or driven in or out of the head by the 4H447 Driver (1) after removing the valve guide seal. On earlier engines, the inlet valve guide (4) can be identified from the exhaust valve guide (3) by the tapered end as shown. On later engines, the same guide is used for both inlet and exhaust valves.



VALVE GUIDES
3-Exhaust valve guide. 4-Inlet valve guide.

The guides should be pressed into place carefully with the type of driver or inserting tool shown in the accompanying illustration, to prevent damage to the guides. A reamer run through the guides after they are installed will insure correct valve stem clearance. See the topic, SPECIFICATIONS, for the

correct reamer size. Install the seals after the guides are in place.

VALVE TIMING

The valve timing is controlled by the setting of the timing gears. Timing is shown in the topic, TIMING GEARS.

VALVE LIFTERS

The valve lifters are located in the left side of the cylinder block. Valve lifters for all cylinders except No. 1 can be removed from the engine in the following manner.

NOTE

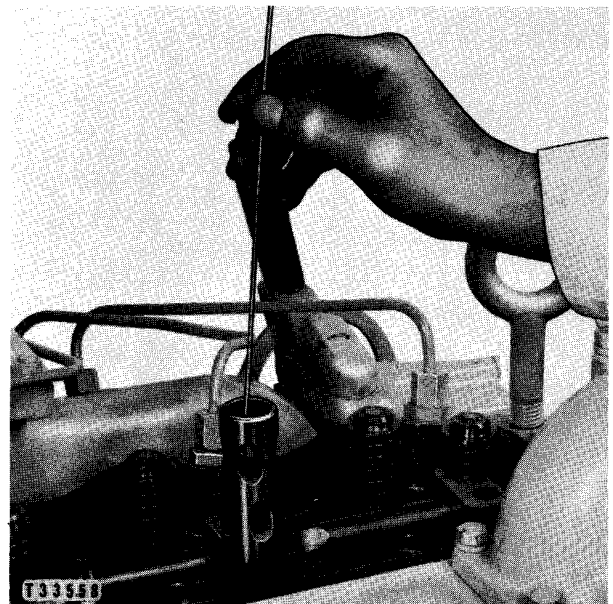
It is necessary to remove the cylinder head to remove the valve lifters for cylinder No. 1 on some engines.

Some lifters have a wider base and are not removable from the top of the engine. The camshaft must be removed prior to their removal.

Remove the valve rocker arm assembly. See the topic, VALVE ROCKER ARM REMOVAL.

Remove the push rods.

Using a suitable piece of wire, approximately 15" long with a 1/2" hook bent in one end, remove the valve lifters as shown.



REMOVING VALVE LIFTER

COMPRESSION RELEASE MECHANISM

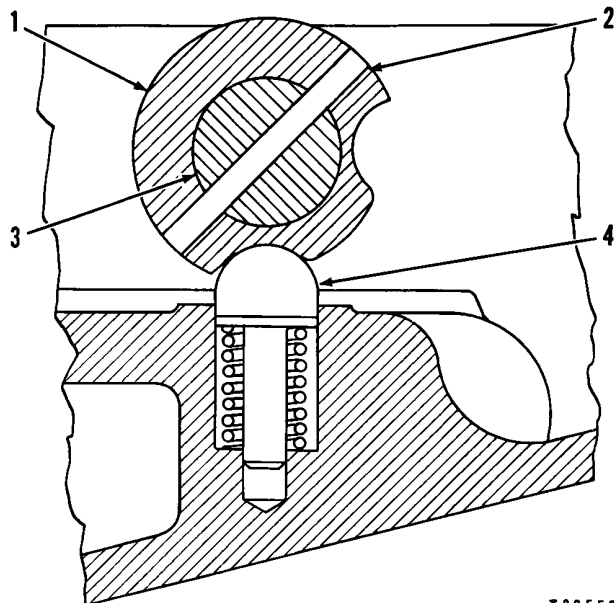
The compression release mechanism opens the inlet valve on each cylinder, allowing the diesel engine to be easily cranked.

ENGINE

VALVES AND VALVE MECHANISM

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The compression release shaft is locked in the RUN or START position by a spring loaded plunger (4) which engages in a stop (1) held in place by spring pin (2) on the compression release shaft (3).

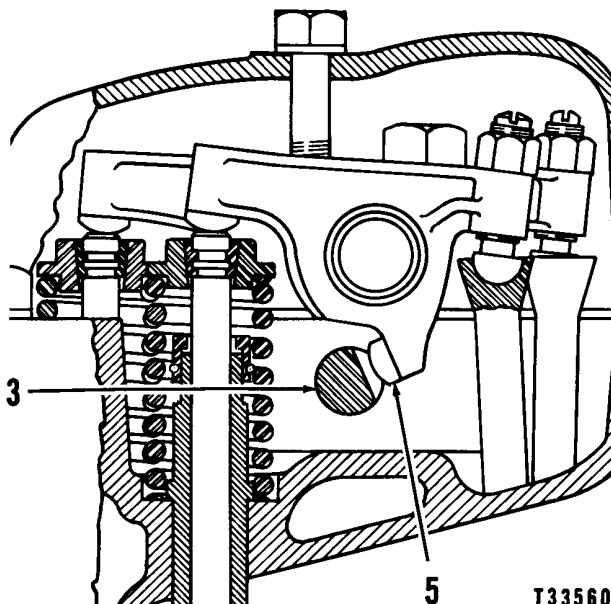


T33559

COMPRESSION RELEASE MECHANISM

1-Stop. 2-Spring pin. 3-Shaft. 4-Plunger.

Each inlet valve rocker arm has a button (5) which comes in contact with the compression release shaft (3) when the inlet valve is closed and the compression release is in the START position. When the compression release is in the RUN position, flat sections on shaft allow normal operation of the rocker arms.



T33560

COMPRESSION RELEASE MECHANISM

3-Shaft. 5-Rocker arm button.

When the shaft is rotated to the START position, the high portions of the shaft contact the inlet valve rocker arm buttons and hold the inlet valves open.

Removal and Installation

1. Remove the rocker arm assembly.
2. Disconnect the compression release lever.
3. Remove the spring pin from the stop on the compression release shaft.
4. Slide the shaft out of the head.
5. Prior to installation inspect the seal on the shaft and replace it if it is damaged.

Front Support

REMOVAL AND INSTALLATION

Remove the crankshaft pulley. See the covering topic.

Support the front of the engine by some means other than the front support itself.

Remove the bolts which secure the front support to the bushing assembly.

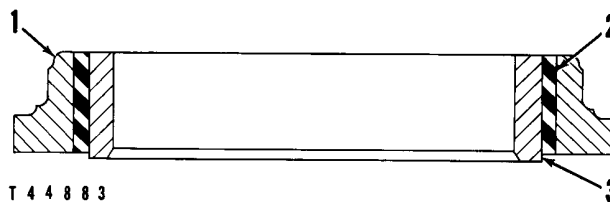
NOTE

Tag the shims from each side of the front support so they will be installed in the same position.

BUSHING ASSEMBLY REMOVAL AND INSTALLATION

The bushing assembly is installed by a special process that gives a very tight fit between the rubber ring (2), support (1) and steel inner ring (3). The bushing assembly is serviced only as a unit.

The steel inner ring (3) is a .003" to .006" press fit on the timing gear cover. Should it become necessary to replace the bushing assembly for any reason, it is necessary to ruin the three parts of the assembly in the process.



BUSHING ASSEMBLY

1-Support. 2-Rubber ring. 3-Steel inner ring.

Pull, burn off or cut off the support (1), remove the rubber ring (2), and then burn or cut the steel inner ring off the timing gear cover.

As the new bushing assembly is pressed on the timing gear cover, it should be kept in its proper position relative to the engine front support.

CAUTION

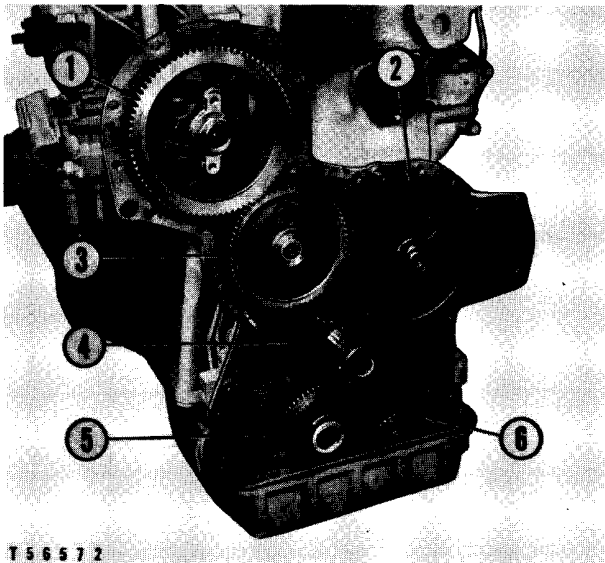
Be sure the inside of the timing gear cover is adequately supported prior to installation of the bushing assembly.

NOTE

If excessive heat is applied to the timing gear cover during the cutting operation, the oil seal in the timing gear cover should be replaced. See the topic, TIMING GEAR COVER REMOVAL AND INSTALLATION.

Timing Gears

The timing gears are located at the front of the engine and drive the accessories and keep the rotation of the crankshaft, camshaft and fuel injection pump camshaft in correct relationship to each other.



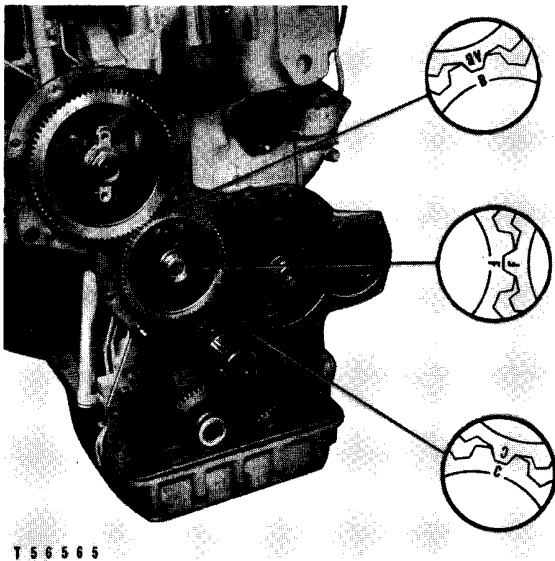
T 56572

TIMING GEARS

1-Accessory drive gear. 2-Camshaft gear. 3-Accessory drive idler gear. 4-Crankshaft gear. 5-Oil pump idler gear. 6-Oil pump drive gear.

TIMING GEARS AND TIMING MARKS

1. Remove the timing gear cover. See the topic, TIMING GEAR COVER REMOVAL.



T 56566

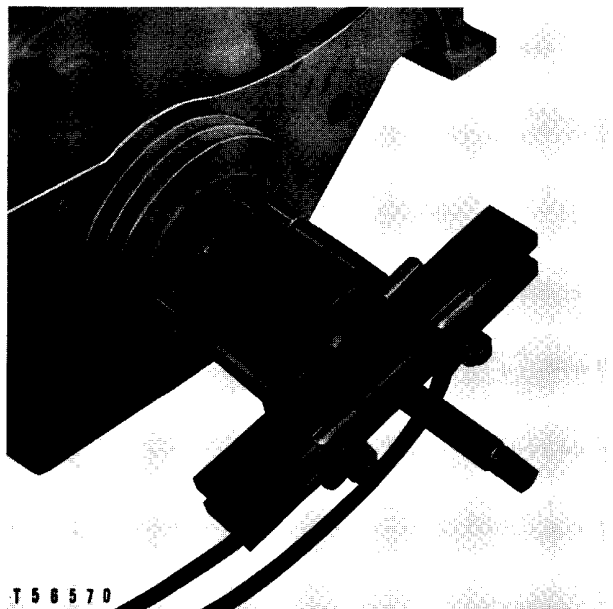
TIMING MARKS

2. Remove the flywheel pointer cover on the flywheel housing and turn the flywheel in the direction of engine rotation until No. 1 piston is at top center on the compression stroke. See the topic, FUEL INJECTION PUMP LIFTER SETTING.

3. The timing marks on the gears should be matched as follows:
 - (a.) C on crankshaft drive gear with C on camshaft gear.
 - (b.) F on camshaft gear with F on accessory drive idler gear.
 - (c.) AB on accessory drive gear with B on accessory drive idler gear.
4. If the timing marks are not properly aligned, see the covering topics for the removal and installation of the particular gears.

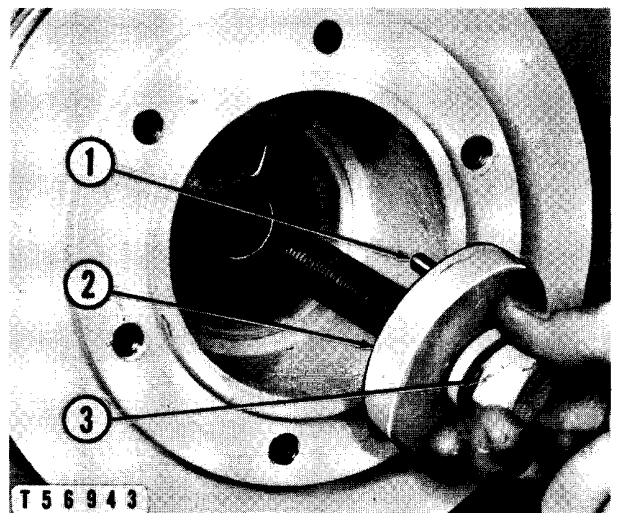
CRANKSHAFT PULLEY AND FRONT SUPPORT REMOVAL AND INSTALLATION

1. Loosen the bolt (3) about five turns.



T 56570

REMOVING CRANKSHAFT PULLEY



T 56943

REMOVING RETAINING PLATE

1-Dowel. 2-Plate. 3-Bolt.

ENGINE TIMING GEARS

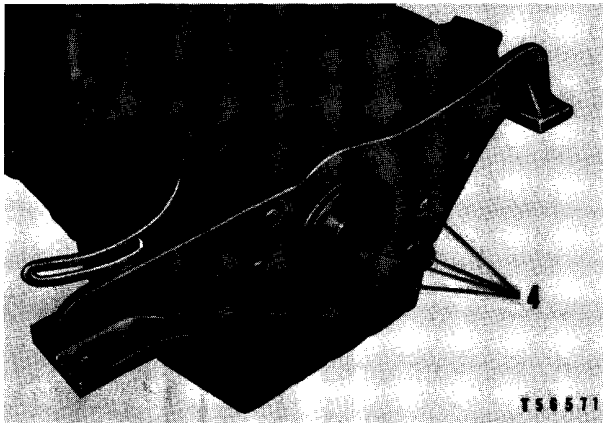
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2. Use an 8B7548 Push Puller, 8B7550 Legs, 8B7557 Adapters, a 5F7353 Washer, a 5F7366 Screw, an 8B7560 Step Plate, a 6F25 Pump Group and a 7F9540 Hydraulic Puller to loosen the crankshaft pulley as shown.
3. Remove the retaining bolt (3), plate (2) and crankshaft pulley.

CAUTION

Be careful not to damage the dowel (1) when installing the plate (2) and bolt (3).

4. Remove the bolts (4) which secure the front support to the timing gear cover, and remove the front support.

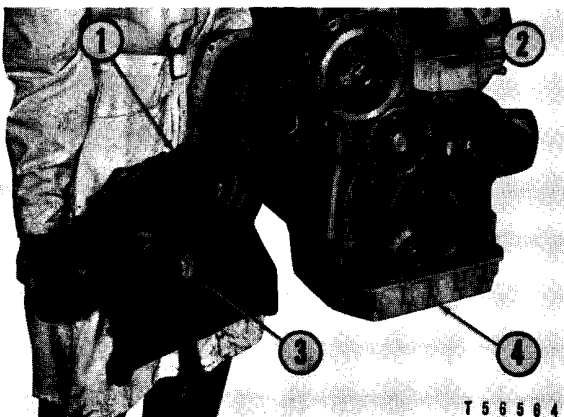


PREPARING TO REMOVE FRONT SUPPORT
4-Bolts.

NOTE

If the front support bushing assembly is to be replaced, see the topic, FRONT SUPPORT.

TIMING GEAR COVER REMOVAL AND INSTALLATION



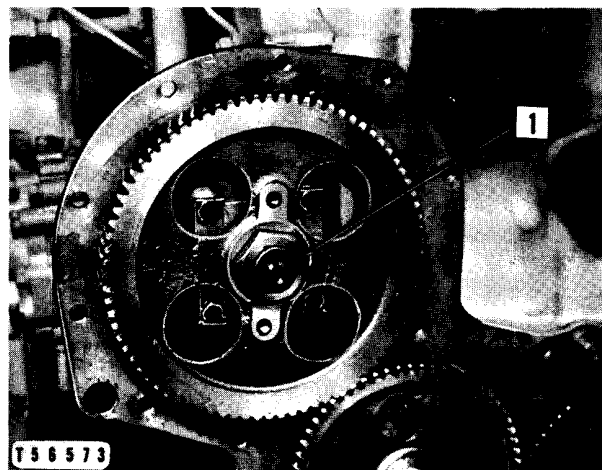
TIMING GEAR COVER REMOVAL
1-Timing gear cover. 2-Timing gear plate.
3-Seal. 4-Oil pan.

1. Mount the engine on an engine positioning stand, if available.
2. Remove the crankshaft pulley as previously described.
3. Remove the bolts that secure the oil pan (4) to the timing gear cover (1).
4. Loosen the bolts which secure the oil pan to the cylinder block.
5. Separate the oil pan from the timing gear cover.
6. Using a putty knife, carefully separate the gasket from the cover.
7. Remove the bolts which secure the timing gear cover to the engine block and plate (2), and remove the timing gear cover.
8. Inspect the seal (3) in the front cover and replace it if it is damaged.
9. If the crankshaft seal has been removed, do not replace the seal until the cover has been installed. Install the seal with the spring-loaded lip in and the dust seal lip out.

ACCESSORY DRIVE SHAFT AND GEAR

Accessory Drive Gear Removal and Installation

1. Remove the timing gear cover. See the topic, TIMING GEAR COVER REMOVAL.



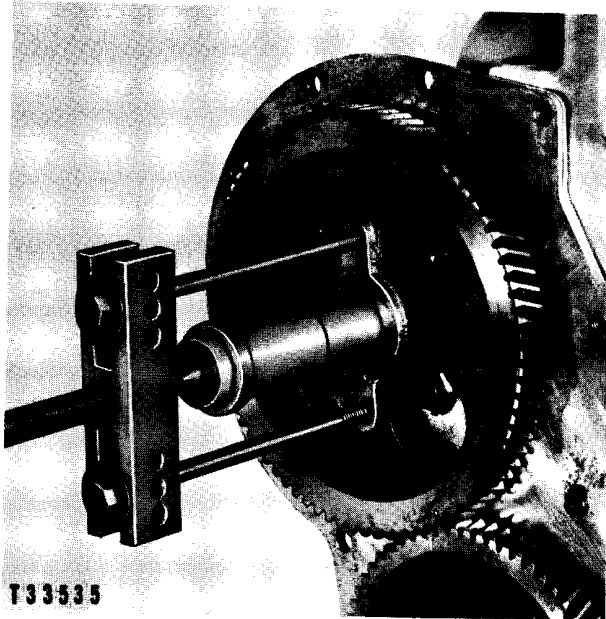
PREPARING TO REMOVE ACCESSORY DRIVE GEAR

☐ Remove

1-Nut.

2. Using a 5F7465 Puller, two $\frac{3}{8}$ " - 16 (NC) bolts $7\frac{1}{2}$ " long, two $\frac{3}{8}$ " washers, 8B7561 Step Plate and 8F3668 Sleeve, pull the accessory drive gear as shown.

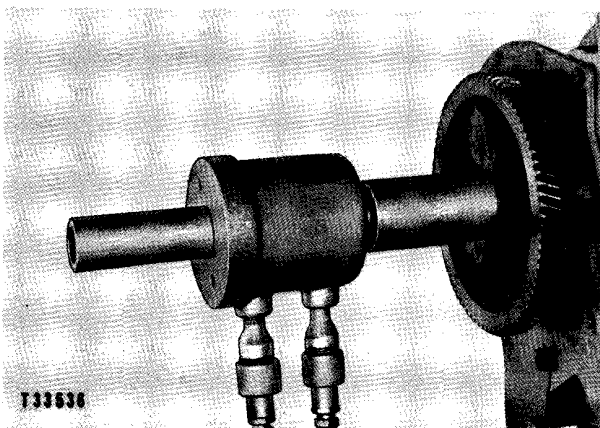
3. Using the 7F9540 Hydraulic Puller Group, 8F3670 Stud, 8F3667 Sleeve, 8F3668 Sleeve and 8F3669 Sleeve, install the accessory drive gear on the shaft as shown.



ACCESSORY DRIVE GEAR REMOVAL

NOTE

When installing the gear on the shaft, check to see that the gear is properly sliding over the key in the shaft and that the timing mark on the gear is aligned with the timing mark on the idler gear. See the topic, TIMING GEARS AND TIMING MARKS.



ACCESSORY DRIVE GEAR INSTALLATION

4. Install the lock and nut.

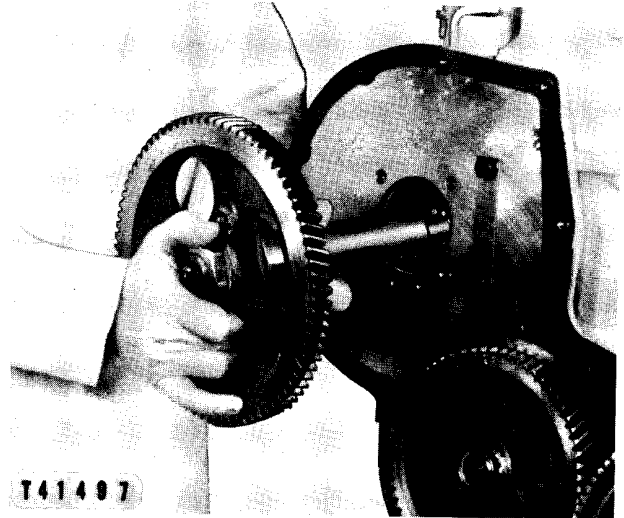
NOTE

Accessory drive gear backlash is listed in the topic, SPECIFICATIONS.

Removal and Installation as a Unit

The accessory drive gear and shaft can be removed as a unit if desired.

With the timing gear cover removed, and the timing marks aligned, remove the bolts which secure the bearing retaining plate to the timing gear plate. See the topic, ACCESSORY DRIVE HOUSING REMOVAL AND INSTALLATION.



REMOVING ACCESSORY DRIVE SHAFT AND GEAR

Remove the shaft and gear as a unit.

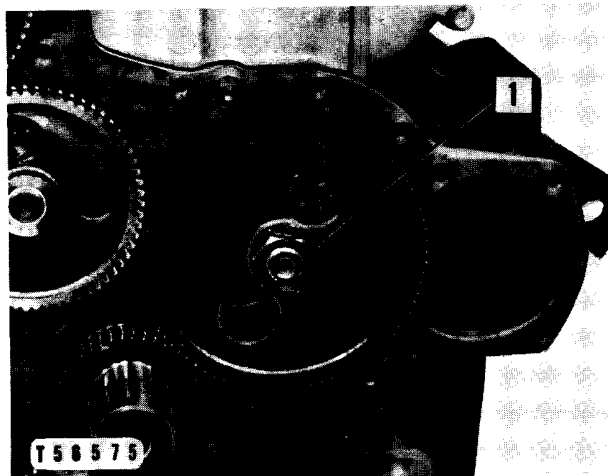
When installing the shaft, be sure to turn the shaft until the off-center tang on the shaft engages with the slot in the fuel injection pump camshaft, and the timing marks on the gear align with the timing marks on the idler gear.

CAMSHAFT

The camshaft is located on the upper left side of the cylinder block and is driven by the camshaft gear. The camshaft is supported in the block by four bearings. The camshaft is positioned by a thrust washer secured to the cylinder block. The cams which actuate the inlet and exhaust valves are forged integrally with the shaft. The camshaft and camshaft gear can be removed as a unit if desired after the timing gear cover has been removed.

Camshaft Gear Removal and Installation

1. Using the identical puller set-up as used to remove the accessory drive gear, pull the gear from the camshaft. See the topic, ACCESSORY DRIVE GEAR REMOVAL AND INSTALLATION.
2. Heat the gear, preferably in oil. Align the keyway on the gear with the key on the camshaft and install the gear. Be sure to align the timing marks.



PREPARING TO REMOVE CAMSHAFT GEAR

☐ Remove

1-Nut.

NOTE

If no suitable means of heating the gear is available, install the camshaft gear as instructed in the topic, ACCESSORY DRIVE GEAR REMOVAL AND INSTALLATION.

3. Install the lock and nut.
4. Check the gear backlash. See the covering topic.

Camshaft Removal and Installation

1. Remove the pushrods and lift the valve lifters clear of the camshaft.
2. Remove the timing gear cover as described in the covering topic.

NOTE

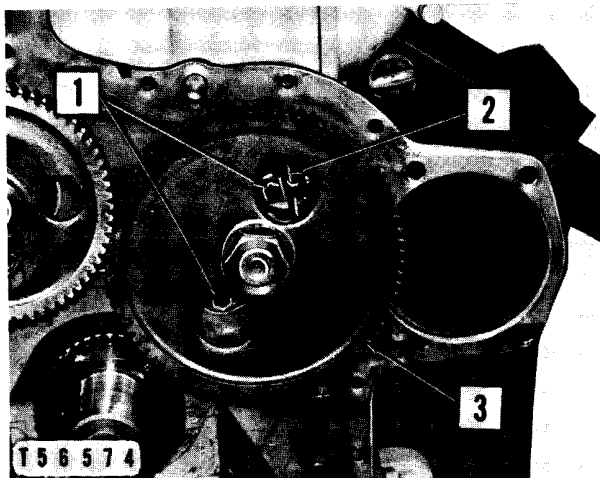
Care should be taken when removing the camshaft and gear not to damage the bearings as the cam lobes pass through them.

3. Check the bearing clearance. For the permissible bearing clearance, see the topic, SPECIFICATIONS.
4. Install the camshaft and gear.

NOTE

If the camshaft bearings are to be removed, see the topic, CAMSHAFT BEARING REMOVAL AND INSTALLATION.

5. Check the camshaft end clearance as described in the topic, CAMSHAFT END CLEARANCE.



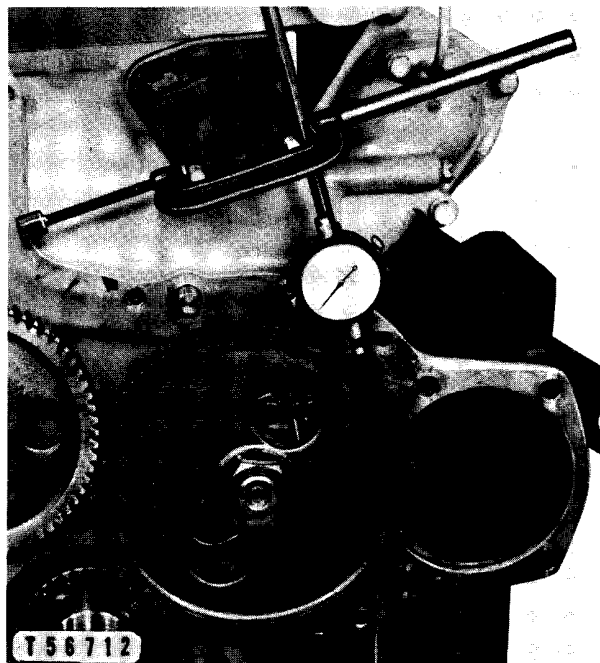
PREPARING TO REMOVE CAMSHAFT

☐ Remove

1-Bolts. 2-Thrust plate. 3-Camshaft and gear.

Checking Camshaft Gear Backlash

The backlash between the camshaft gear and the crankshaft gear can be checked by installing a dial indicator as illustrated. The backlash between the camshaft gear and the crankshaft gear is listed in the topic, SPECIFICATIONS.



CHECKING CAMSHAFT GEAR BACKLASH

When a dial indicator reading of a value greater than the permissible backlash is shown, a further check must be made to determine the cause. Excessive backlash indicates that either the timing gears, the main bearings or the camshaft bearings are badly worn.

ENGINE TIMING GEARS

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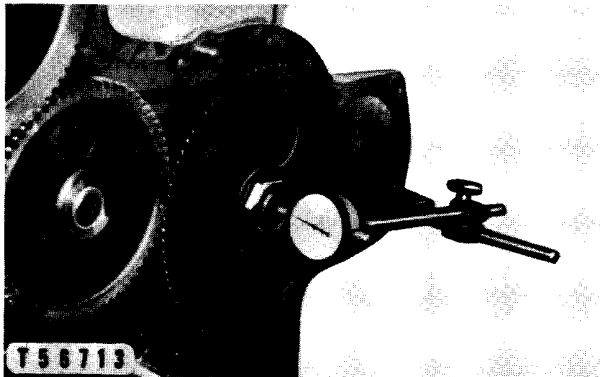
Timing gear wear can be compensated for by adjusting the fuel injection pump lifters. See the topic, FUEL PUMP LIFTER SETTING.

If either the main bearings or the camshaft bearings are badly worn, they should be replaced with new ones.

If a reading of less than the minimum backlash is shown, it is an indication of incorrect assembly, or a burr or rough spot on one of the gears. In this case, take readings every 90° around the camshaft gear to determine the cause. A burr can be removed from a gear tooth, by using a gear file or fine stone, without removing the gear from the camshaft. When removing a burr, cover the remaining exposed parts to keep them clean.

Camshaft End Clearance

The end thrust of the camshaft is taken by the thrust plate.



CHECKING CAMSHAFT END CLEARANCE

The correct end clearance is listed in the topic, SPECIFICATIONS. The thrust plate should be replaced if the end clearance exceeds the permissible end clearance.

The end clearance can be checked by installing a dial indicator as illustrated. After checking the end clearance, if it is necessary to replace the thrust plate, it can be removed as described in the topic, CAMSHAFT REMOVAL AND INSTALLATION.

If the camshaft and camshaft gear were removed as a unit, the thrust plate should be inspected for wear. The end clearance can be checked in the following manner. Push the thrust plate against the gear. Using a thickness gauge, measure the clearance between the thrust plate and the end of the camshaft bearing journal.

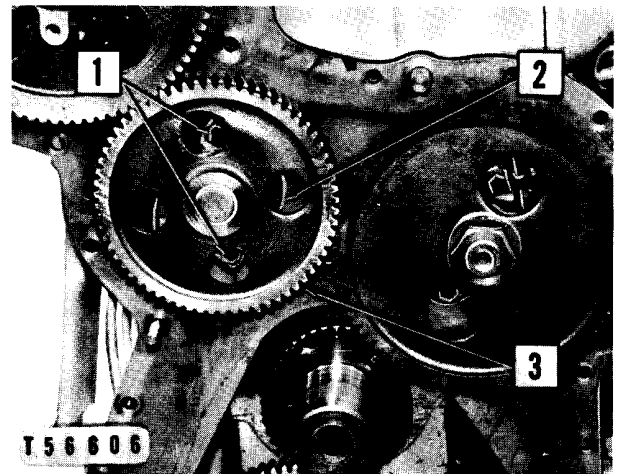
Camshaft Bearing Removal and Installation

1. Place the engine on an engine stand.

2. Remove the camshaft, oil pan, oil pump, oil pan plate, and flywheel housing. See the covering topics.
3. Press the bearings out of the cylinder block.
4. When installing the camshaft front bearing, be sure that the oil hole in the bearing is aligned with the oil hole in the cylinder block.
5. Install the camshaft front and rear bearings so there is $\frac{1}{8}$ " clearance from the face of the cylinder block to the bearing edge.

ACCESSORY DRIVE IDLER GEAR REMOVAL AND INSTALLATION

Inspect the idler gear bearing and the thrust plate and replace if necessary.



ACCESSORY DRIVE IDLER GEAR REMOVAL

☐ Remove

1-Bolts. 2-Thrust plate. 3-Gear.

NOTE

Accessory drive idler gear end clearance is listed in the topic, SPECIFICATIONS.

ENGINE

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PISTONS, RINGS AND CONNECTING RODS

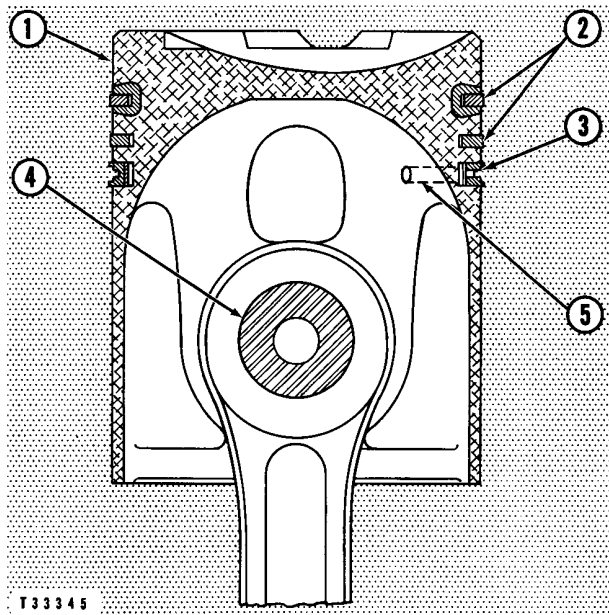
Pistons, Rings and Connecting Rods

PISTONS AND RINGS

The first noticeable symptoms of worn piston rings and cylinder liners are increased oil consumption and excessive vapor from the crankcase breather. Extreme wear will result in poor compression, loss of power and hard starting. See the topic, CYLINDER LINERS.

The cam-shaped aluminum alloy piston (1) has three rings; two compression ring (2) and one oil ring (3). All rings are located above the piston pin bore. The top compression ring seats in an integral cast iron band. The intermediate compression ring has a taper face to provide quick break-in. Holes (5) in the groove for the oil ring above the pin boss provide for the return of oil to the crankcase.

The full floating piston pin (4) is retained by two snap rings which fit in grooves in the pin bore.



PISTON AND RINGS

1-Piston. 2-Compression rings. 3-Oil ring.
4-Piston pin. 5-Oil hole.

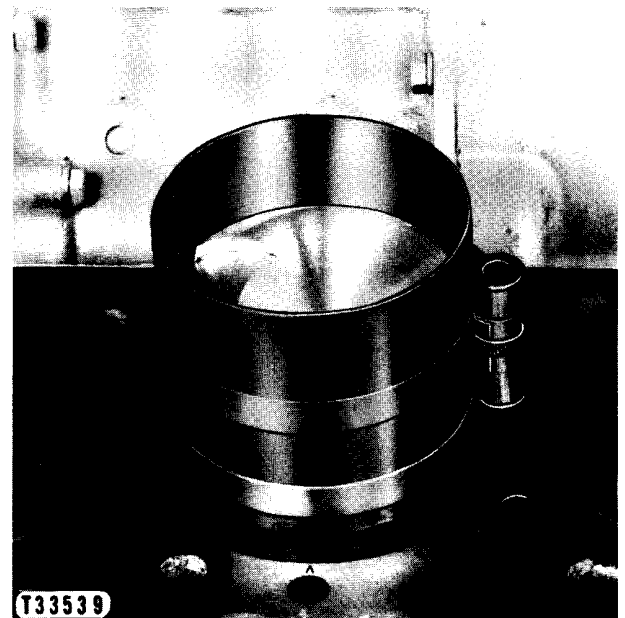
PISTONS

Removal and Installation

1. Drain the lubricating oil and the engine coolant.
2. Remove the cylinder head as described in the topic, CYLINDER HEAD REMOVAL.
3. Remove the carbon from the inside top surface of the cylinder liner.
4. Remove the cylinder block inspection covers or the oil pan and remove the connecting rod bear-

ing caps as described in the topic, CONNECTING ROD BEARING REMOVAL.

5. Rotate the crankshaft until the piston to be removed is at top dead center. Carefully push the connecting rod upward until the piston rings are out of the cylinder.
6. Lift out the piston and connecting rod assembly.
7. Before installing the piston, be sure the liners and pistons are clean and free of carbon.
8. Place the piston ring compressor tool over the cylinder into which the piston is to be installed.
9. Oil the piston and rings and place the piston and connecting rod assembly into the cylinder liner until the ring rests on the compressor tool. Position the V-mark on top of the piston in alignment with the V-mark on top of the cylinder block. This will place the recess for the valves in the piston in the correct relationship to the valves and precombustion chamber in the cylinder head. Push the piston through the compressor and into the cylinder liner.



PISTON INSTALLATION

Cleaning and Inspecting Piston

Pistons which are not worn excessively or scored badly should be cleaned and used again. The ring grooves should be square and smooth. The side clearance between a new ring and the top ring groove should not exceed the value listed in the topic, SPECIFICATIONS.

There are a number of good carbon solvents available for cleaning pistons. A carbon softener for use on diesel engine pistons is fresh, cold water.

ENGINE**PISTONS, RINGS AND CONNECTING RODS****ISSUED 6-60**

Soak pistons in cold water overnight and let them dry (preferably in the sunlight), and most of the carbon including that in the ring grooves can be easily removed.

The use of broken rings or carbon scrapers on ring grooves should be avoided, as grooves can be cut. A stick of hard wood does a good job and will not scratch the piston.

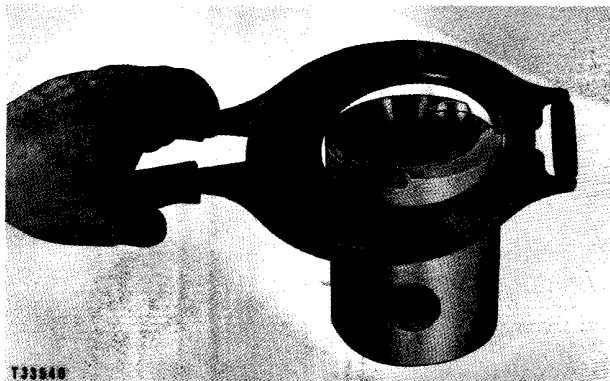
The bottom of each ring groove must be clean and the oil return holes in the oil ring grooves must be open before the installation of new rings.

The area above the top ring can be filed smooth, but pistons badly scored below the top ring groove should be replaced.

RINGS

Piston rings seal compression and control the amount of oil on the cylinder walls. If oil consumption is not excessive and compression is satisfactory, pistons should not be removed nor new rings installed when an engine is dismantled for some other reason.

To avoid damage to the piston, remove the rings with a 5F9059 Piston Ring Expander.

**RING EXPANDER**

The rings should be placed on the piston in the proper groove. Use the 5F9059 Piston Ring Expander when placing the rings in the grooves. The ring expander is not only a time saver but will also prevent breaking or distorting rings and damaging ring grooves.

CAUTION

The intermediate compression ring has a $\frac{1}{2}^\circ$ taper face and has TOP marked. Be sure to install this ring correctly.

When new piston rings are to be used in worn cylinder liners, the ridge at the top of the liner should be raised.

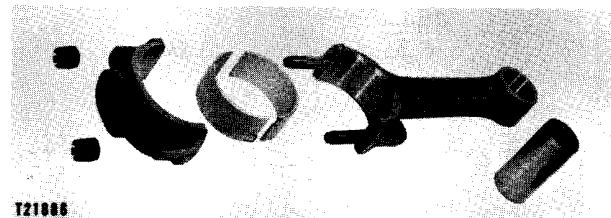
Whenever new rings, piston assemblies or piston and liner groups are installed, be sure to run-in the engine on a conditioning schedule before operating at normal load and speed. Do not run the engine idle for a long period after installing new rings or liners. Rings will not seat during idle operation. See the topic, **RUNNING-IN SCHEDULE**.

RUNNING-IN SCHEDULE

For the correct running-in procedures, follow the information that accompanies piston rings, piston assemblies and piston and liner groups sent from the Caterpillar Parts Department.

CONNECTING RODS

Connecting rod bearings are located in the crankshaft end of the connecting rod. The bearing caps and rods are numbered consecutively, 1-up, from the front of the engine on the left side so the numbers can be seen through the inspection opening on the left side of the cylinder block. The rods should be reassembled with the numbers in this position.

**CONNECTING ROD AND PISTON PIN**

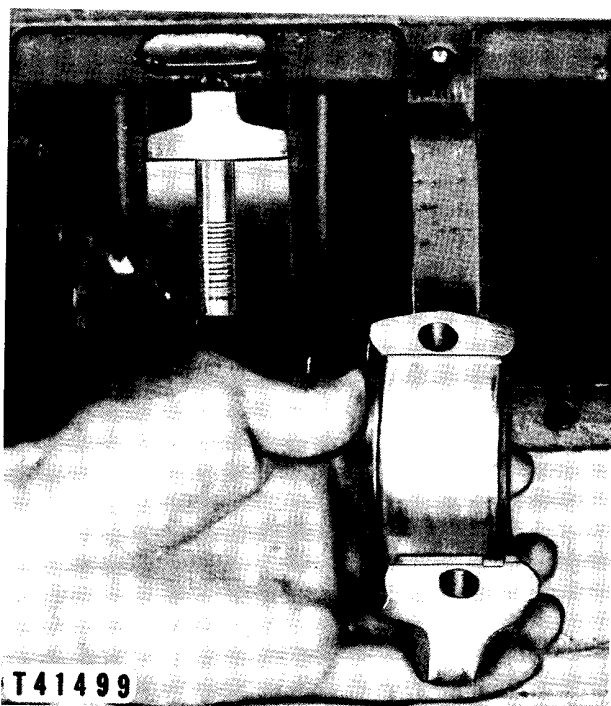
See the topic, **PISTONS**, for information on connecting rod and piston removal.

Connecting Rod Bearing Removal

Connecting rod bearings are the precision type and are to be installed without fitting or scraping. The bearing halves can be removed, inspected and replaced through the inspection opening without removing the rod from the engine.

1. Drain the diesel engine crankcase and cooling system.
2. Remove all items which interfere with the removal of the inspection cover.
3. Remove the inspection cover.
4. With the inspection cover removed, turn the crankshaft until the bearing to be removed is accessible.
5. Remove the cotter pins and nuts from the connecting rod bolts and take out the cap and bearing lower half.

6. To remove the bearing upper half, turn the crankshaft or push the rod up slightly.



REMOVING CONNECTING ROD BEARING CAP

Connecting Rod Bearing Inspection

Connecting rod bearings are steel-backed, aluminum-lined. Larger particles of dirt and abrasives in the oil do not tend to embed in aluminum bearings. Such particles roll around between the bearing and crankshaft journal causing scratches in the aluminum bearing without actually becoming embedded in the aluminum. Such scratches are not necessarily harmful and do not indicate that the bearings should be replaced.

If there is any question about the surface of a bearing, wash it with cleaning solvent to remove the oil. If the surface feels rough and abrasive, install a new bearing. Another indication of dirt in the bearing is excessive crankshaft wear.

New standard connecting rod bearing size and clearances are listed in the topic, SPECIFICATIONS. Bearings .030" undersize are serviced for reground crankshafts.

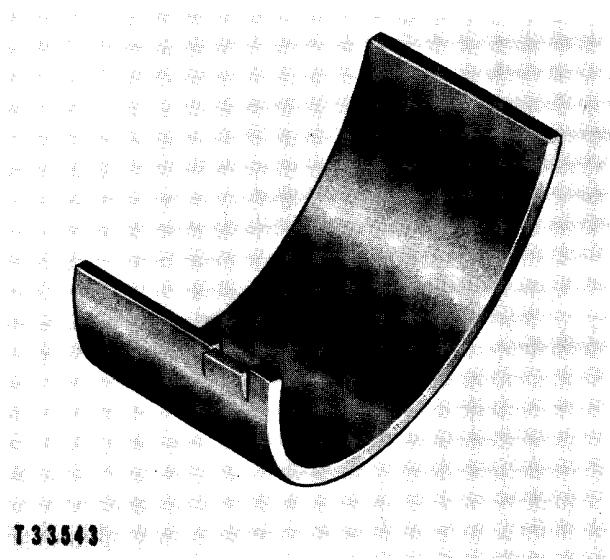
Bent rods should be discarded. Do not attempt to align the connecting rods by bending them.

Connecting Rod Bearing Installation

When replacing the bearings, see that the protruding tab on the back of each bearing half lines up with its corresponding recess in the connecting

rod and bearing cap. This locks the bearing in place and keeps it from rotating.

Tighten the nuts to the torque value listed in the specifications. It is permissible to exceed this value by the amount necessary to turn the nut to the next slot for alignment with cotter pin hole.



BEARING HALF

Piston Pin Bearing

It is not always necessary to replace piston pin bearings at each engine reconditioning. They may last many thousands of hours if the oil has been kept clean.

After the oil has been cleaned from the pin and bearing, it is possible to feel the clearance between them. This normal oil clearance must not be mistaken for wear.

A new bearing, pin, or both, should be installed only when the clearance between the bearing and pin exceeds the value listed in the topic, SPECIFICATIONS.

New connecting rods have the piston pin bearing bored in a special machine which maintains the proper center-to-center distance and parallelism of the connecting rod bearing. Reconditioned rods should be machined in the same manner. A new connecting rod makes a good templet for center-to-center distance which is listed in the specifications.

After pressing a new bearing into place, it should be machined accurately to the inside diameter listed in the topic, SPECIFICATIONS. A 2H6782 Rod Boring Machine is available for this operation.

Cylinder Liners

Cylinder liner surfaces are machined, hardened, ground, honed, and chemically treated to assure proper break-in. The resultant surface is so hard that ordinary boring tools will not machine it. Liners, pistons and rings are available from the factory in standard sizes only and require no fitting when they are installed.

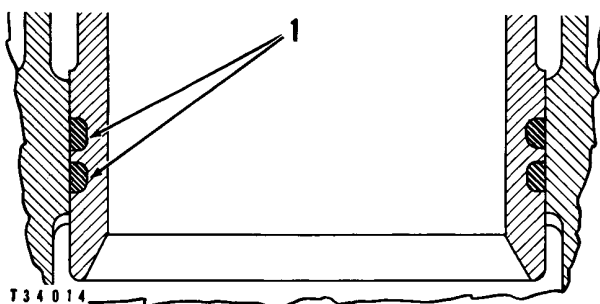


CHECKING LINER WEAR

Cylinder liners should be replaced when they are either worn at the top of the ring travel to a value greater than that listed in the topic, SPECIFICATIONS, or, if they are scratched or scored.

Liner wear should be checked with an inside micrometer as shown.

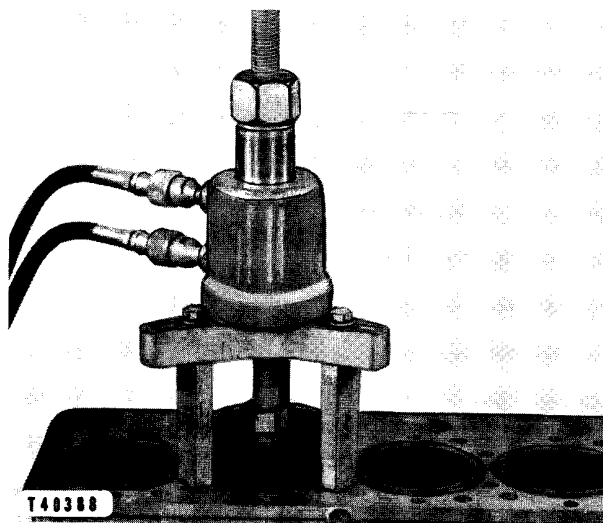
Two O-ring seals (1) fit in grooves on the lower end of each liner as shown to prevent leakage of coolant into the diesel engine lubricating system.



LINER SEALS
1-O-ring seals.

REMOVING AND INSTALLING CYLINDER LINERS

1. Drain the cooling system.
2. Remove the cylinder head, the connecting rods and the pistons as outlined in their respective topics.
3. Place a piece of cardboard or heavy gasket material through the inspection opening to protect the inside of the engine.
4. Install the 7F1857 Puller, the 5F7362 Adapter Plate and the 7F9540 Hydraulic Puller.
5. Remove the cylinder liner and clean the water jacket sediment from the cylinder block.
6. When installing the cylinder liner, always use new rubber seals. Coat the rubber seals with liquid soap to ease installation.



PULLING CYLINDER LINER

7. Lower the cylinder liner carefully into the block. The liners can be driven into place by using a suitable driver, or by placing the puller adapter on the top of the liner. A block of hard wood, to be used as a driving block, is then placed on the puller adapter.
8. Drive the liner into the cylinder block until it bottoms. Then hit the block of wood several light taps, to assure that the liner is in. If the last blow bottoms the liner too hard, the liner may bounce back slightly.

NOTE

Properly installed liners should extend slightly above the face of the cylinder block. This insures proper holding and sealing of the cylinder liner against the cylinder head gasket when the

ENGINE CYLINDER LINERS

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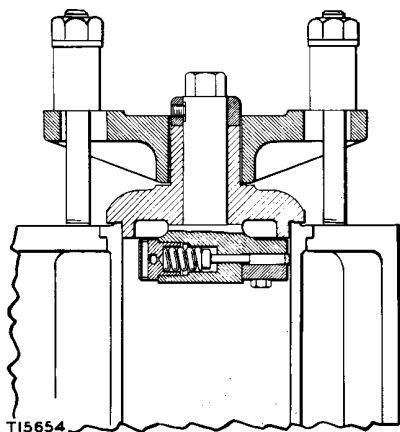
cylinder head is drawn down. Some liners may feel slightly loose in the cylinder block, yet serve satisfactorily without water or antifreeze leaking past the rubber seals.

9. Remove the cardboard and assemble the parts.

RECONDITIONING CYLINDER LINERS

Removing Ridge from Worn Cylinder Liners

When new piston rings are to be used in worn cylinder liners, the ridge in the liner at the top of the ring travel should be raised to provide clearance for the new top ring. Since the liners are too hard for ordinary tools, a liner ridge boring tool should be used. The illustration shows how the tool is installed. Information regarding the tool illustrated is available upon request.



LINER RIDGE BORING TOOL

Use the head bolts or studs and suitable washers as illustrated, to secure the liner ridge boring tool to the cylinder block.

The tungsten carbide tool bit is spring-loaded and specially ground so it will follow the worn contour of the liner and will not cut deeper than the worn surface. Instructions for its use accompany the tool.

CAUTION

Be careful not to rotate the tool counterclockwise when the tool bit is against the liner wall. Doing so will break the cutting edge.

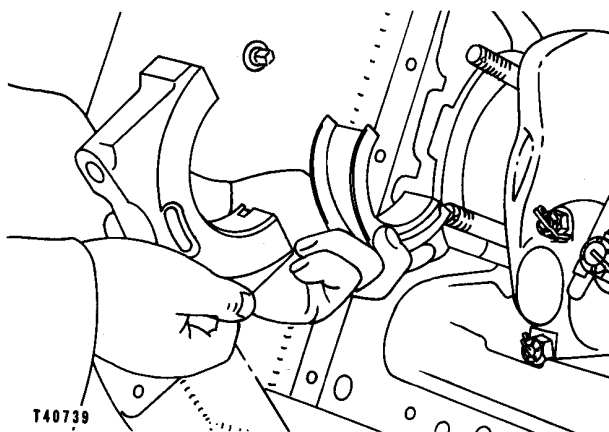
If the engine does not break in properly, follow the information that accompanies piston rings, piston assemblies and piston and liner groups sent from the Caterpillar Parts Department.

Main Bearings

The main bearings and caps support the crankshaft in place in the cylinder block.

The main bearings are of the steel-backed aluminum-lined precision type. Tabs are punched outward on the bearing half at the parting line, and fit into recesses in the cylinder block and bearing cap, securing the bearing and preventing it from rotating. This provides an uninterrupted bearing surface in the most highly-loaded area, improving the loading conditions and giving maximum resistance to fatigue failures.

The crankshaft thrust is taken on the flange of the lower half of the rear main bearing as illustrated. The upper half of the rear main bearing does not include a flange.



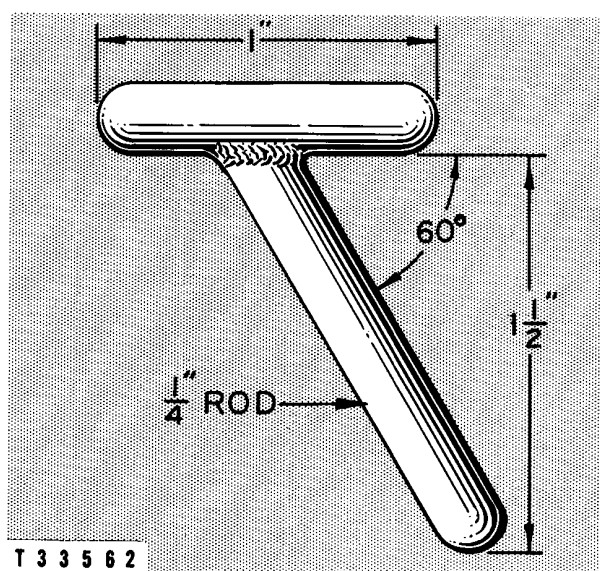
REAR MAIN BEARING AND CAP

REMOVING AND INSTALLING MAIN BEARING

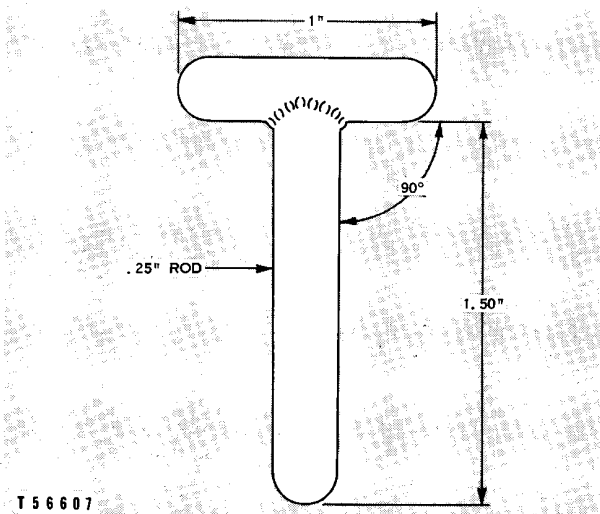
The main bearing caps are identified by numbers stamped on the sides of the caps and the cylinder block. Always return the bearing caps to their proper locations in the block with part number toward front face of block.

All the main bearing caps are held in place by two studs. The rear main bearing also has a dowel to help locate the bearing cap.

1. To remove the main bearings, remove the oil pan, oil pump and plate assembly. See the covering topics.
2. Remove the nuts that secure the bearing caps to the block.
3. Grasp the bearing cap (4) at the recessed portion and pull outward, tapping the cap gently with a soft hammer.
4. The main bearing upper halves can be removed by use of the tool shown in the illustration.



TOOL FOR REMOVING AND INSTALLING
MAIN BEARINGS
(Earlier Engines)



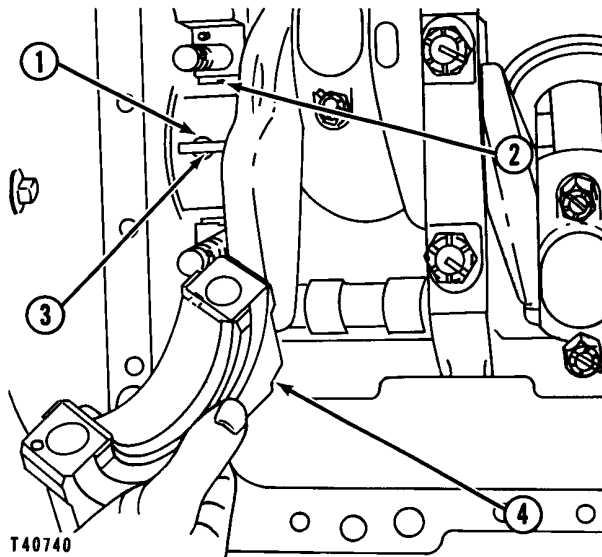
TOOL FOR REMOVING AND INSTALLING
MAIN BEARINGS
(Later Engines)

5. Place the tool (3) in the drilled oil hole (1) as shown, and roll the bearing half (2) out by rotating the crankshaft.

Before installing a bearing, wash it thoroughly and wipe the outer surface dry. Rotate the upper half into position in the same manner in which it was removed. Make certain the bearing tab is aligned with the recess in the block. Remove the tool from the oil hole in the crankshaft. Since only the lower half of the rear main bearing takes the thrust, full contact between the bearing and crank thrust surfaces is assured. Place the lower half in the cap and install the cap. Refer to the topic, SPECIFICATIONS, for the main bearing stud nut torque. It is

ENGINE MAIN BEARINGS

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REMOVING MAIN BEARING

1-Oil hole. 2-Upper half of main bearing. 3-Bearing tool.
4-Bearing cap.

permissible to exceed the specified nut torque by the amount necessary to turn to the next slot for alignment with cotter pin hole. Complete the assembly.

MAIN BEARING INSPECTION

Abrasive materials may roll around between the bearing and crankshaft journal causing scratches in the bearing without actually becoming embedded in the aluminum. Such scratches are not necessarily harmful and do not indicate that the bearings should be replaced.

Make certain that the cylinder block and bearing caps are perfectly clean and free from burrs and high spots. Handle the bearings carefully to avoid marring them. Leave them dry except for the grease holding the lead wire when checking clearances, and lubricate them generously for final installation.

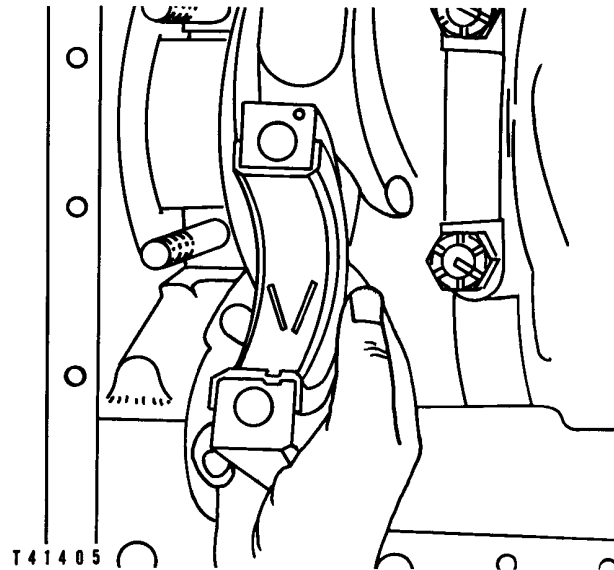
If there is any question about the surface of a bearing, wash it with cleaning solvent to remove oil. If the surface feels rough and abrasive, install a new bearing. Another indication of dirt in the bearing is excessive crankshaft wear.

CHECKING MAIN BEARING CLEARANCE

Refer to the topic, SPECIFICATIONS for the proper main bearing clearance and crankshaft wear limits.

The clearances of the main bearings can be measured without removing the crankshaft if the engine is in an upright position. However, the crankshaft must be held against the upper halves of the main bearings; otherwise the weight of the

crankshaft will compress the lead wire slightly and indicate a lesser clearance than really exists.



CHECKING MAIN BEARING CLEARANCE

The bearing clearance can be checked by placing soft lead wire between the lower bearing half and the crankshaft. Coat two one-inch lengths of the wire with soft grease and place them diagonally on the bearing. The soft grease will keep the wires in position while installing and tightening the cap. Turn the crankshaft one complete revolution. Remove the cap and measure the thickness of the compressed wire with a 0" to 1" micrometer to determine the bearing clearance. Lead wire for the purpose can be obtained from the Parts Department by ordering part number 5B1161.

REPLACEMENT MAIN BEARINGS

Precision main bearing halves, machined to provide proper clearance, are obtainable in complete sets and are to be installed without further machining or hand fitting. Single replacement bearings (both upper and lower half) can also be obtained and installed without special fitting. As a rule, however, it is good practice to replace the complete set if any one of the bearings needs to be replaced. If only one new bearing is installed and the other bearings are worn to any extent, the new bearing will carry more than its normal share of the load and might be damaged as a result.

Precision main bearings are also obtainable for use with crankshafts reground to .030" undersize.

Crankshaft

The crankshaft main bearing journal and the connecting rod bearing journal diameters are given in the topic, SPECIFICATIONS. Steel-backed, aluminum-lined bearings are used. The lower half of the rear main bearing has flanges which take the end thrust of the crankshaft. The crankshaft timing gear is pressed on the front of the crankshaft.

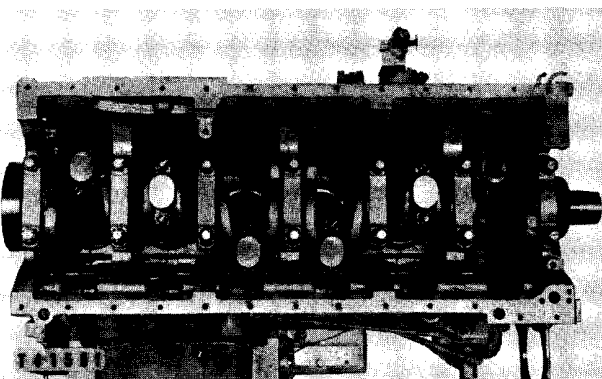
An oil seal pressed into the timing gear housing cover prevents leakage at the front. An oil seal pressed into the flywheel housing and bearing against the crankshaft flange prevents leakage into the flywheel housing. Counter-weights are forged integrally with the shaft and are drilled for balance.

The crankshaft should be reground or replaced if either the wear on the main bearing journals or the wear and out-of-round on the connecting rod journals is greater than the values listed in the topic, SPECIFICATIONS.

Undersize main and connecting rod bearings are available for use with crankshafts reground to .030" undersize.

CRANKSHAFT REMOVAL AND INSTALLATION

1. Remove the radiator. See the topic, RADIATOR REMOVAL AND INSTALLATION.
2. Mount the engine on a 9F3829 Engine Positioning Stand, if available.
3. Remove the water pump, timing gear housing cover and flywheel housing. See the covering topics.



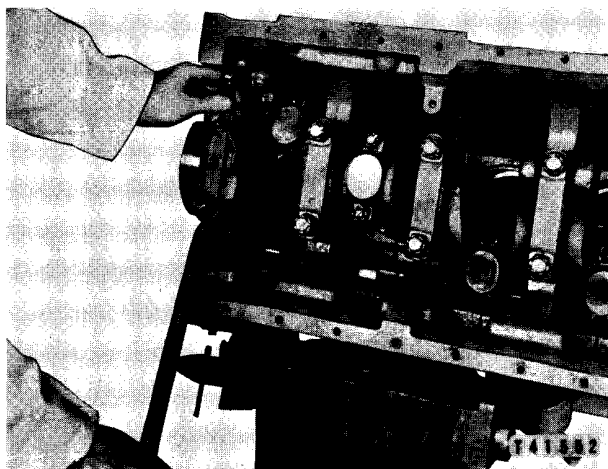
PREPARING TO REMOVE CRANKSHAFT

4. Position the engine on its side as shown and remove the oil pan, plate assembly and oil pump.
5. Remove the connecting rod bearings and caps and push the piston and rod assemblies upward into the cylinders to clear the crankshaft.

6. Wrap the bearing surface on the crankshaft for number two and number five connecting rods with a clean heavy cloth.
7. Place a cable around the cloth and attach the cable to a suitable hoist. Tighten the cable just enough to prevent the crankshaft from moving when the main bearing caps are removed.
8. Cover the bearing cap studs with pieces of rubber hose or something similar to prevent nicking or scratching the crankshaft during removal and installation.
9. Remove the main bearing caps and lift the crankshaft from the engine.
10. Lubricate and wrap the bearing surfaces.
11. Prior to installing the crankshaft, see the topic, TIMING GEARS AND TIMING MARKS, for proper timing of the crankshaft gear.
12. Lubricate the bearings, install the crankshaft and check the bearing clearances. See the covering topic.
13. See the topic, SPECIFICATIONS, for the proper torque on the bearing cap stud nuts.

CRANKSHAFT END CLEARANCE

The crankshaft end thrust is taken on the flanges of the lower half of the rear main bearing described in the topic, MAIN BEARINGS.



CHECKING CRANKSHAFT END CLEARANCE

End clearance can be checked by pushing the crankshaft as far as it will go to one end of the cylinder block and using a thickness gauge to measure the clearance between the machined face of the crankshaft flange and the flange of the lower half of the rear main bearing.

**ENGINE
CRANKSHAFT**

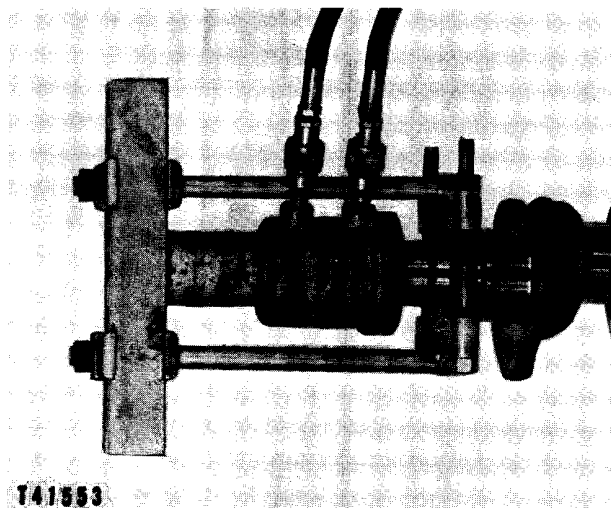
ISSUED 6-60

See the topic, SPECIFICATIONS, for the proper end clearance.

**CRANKSHAFT GEAR REMOVAL AND
INSTALLATION**

Remove the crankshaft. See the covering topic.

Use an 8B7548 Puller, 8B7551 Attachment, 6F25 Pump, 7F9540 Hydraulic Puller and a suitable spacer with the approximate dimensions $2\frac{7}{8}$ inch by $3\frac{1}{2}$ inch to pull the crankshaft gear as shown.

**CRANKSHAFT GEAR REMOVAL****NOTE**

To install the crankshaft gear, heat it in oil to a temperature of 600°F. and push it onto the crankshaft. Make certain the keyway in the gear is aligned with the key in the crankshaft. Be sure the timing marks are to the outside of the gear.

ENGINE

ISSUED 9-62

FLYWHEEL AND FLYWHEEL HOUSING

Flywheel and Flywheel Housing

FLYWHEEL AND RING GEAR

Removal and Installation

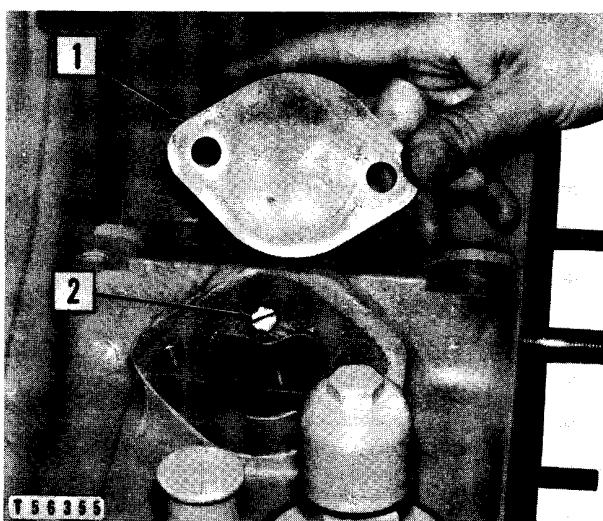
The flywheel and ring gear can be removed with the diesel engine in the machine. For the purpose of clarity, the removal is shown with the engine mounted on a 9F3829 Positioning Stand.

If the engine is removed, the engine should be blocked securely or placed on an engine positioning stand.

Prior to removing the flywheel, make certain the alignment marks on the flywheel and crankshaft

are visible for easier alignment at the time of installation.

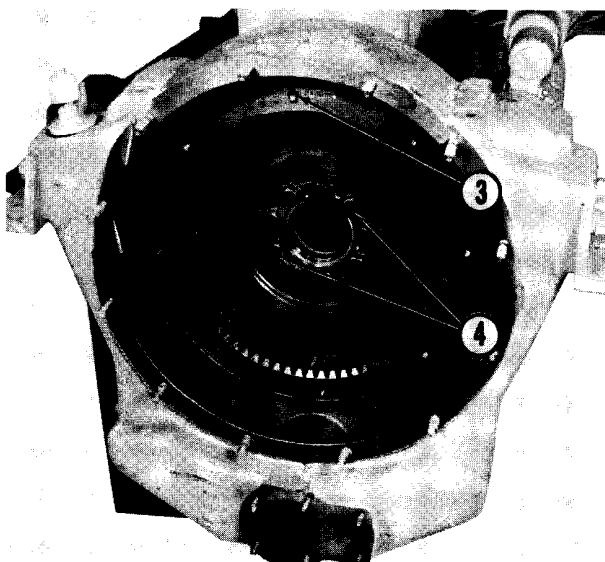
1. Remove the flywheel clutch. See the covering topic.
2. Remove two diagonally opposed flywheel retaining bolts (4) and install two $\frac{5}{8}$ " - 18 NF guide pins.
3. Fabricate a lifting pin of the dimensions shown and install it in the top clutch mounting hole (3).



POINTER REMOVAL

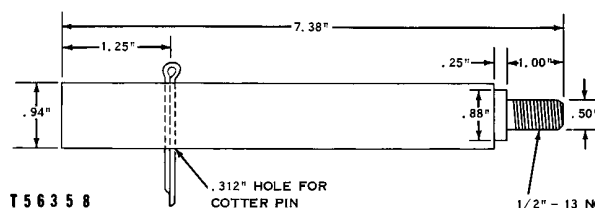
☐ Remove

1-Cover. 2-Pointer.



T56435

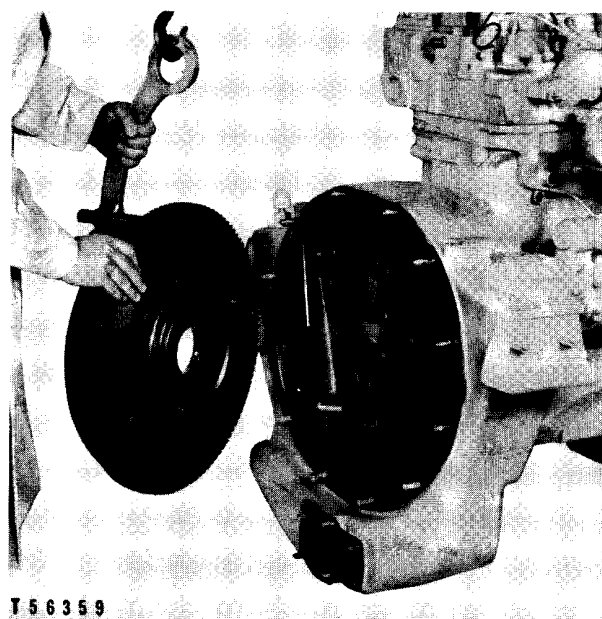
PREPARING TO REMOVE FLYWHEEL
3-Clutch mounting hole. 4-Two diagonally opposed bolts.



T56358

LIFTING PIN

4. Use an 8F4423 Lifting Bracket and a suitable hoist to support the weight of the flywheel. The flywheel weighs approximately 140 pounds.
5. Remove the remaining flywheel retaining bolts and remove the flywheel as shown.



T56359

REMOVING THE FLYWHEEL

6. Press the ring gear off the flywheel.
7. Prior to installing the ring gear, clean both the ring gear and flywheel and remove all burrs.
8. Heat the ring gear in oil, to a temperature not to exceed 600° F., and install it on the flywheel.

ENGINE**FLYWHEEL AND FLYWHEEL HOUSING**

ISSUED 9-62

Make certain the chamfered portion of the teeth on the ring gear are toward the cylinder block when installed.

NOTE

When installing the flywheel, use guide pins as previously described.

9. Align the marks on the flywheel and crankshaft and move the flywheel into position on the crankshaft.
10. Tighten the flywheel bolts evenly and diametrically to the torque value given in the topic, SPECIFICATIONS.

FLYWHEEL HOUSING

The diesel engine must be removed from the machine in order to remove the flywheel housing.

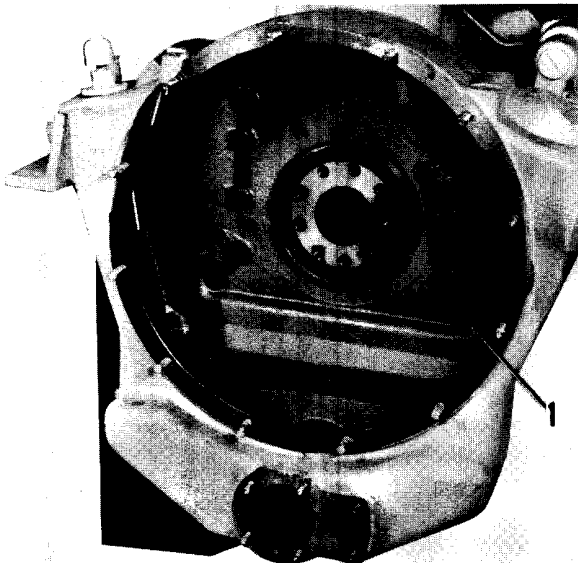
Removal and Installation

1. Remove the diesel engine electric starting motor, if so equipped.

NOTE

It is not necessary to remove the gasoline starting engine to remove the flywheel housing.

2. Remove the flywheel. See the topic, FLYWHEEL AND RING GEAR.

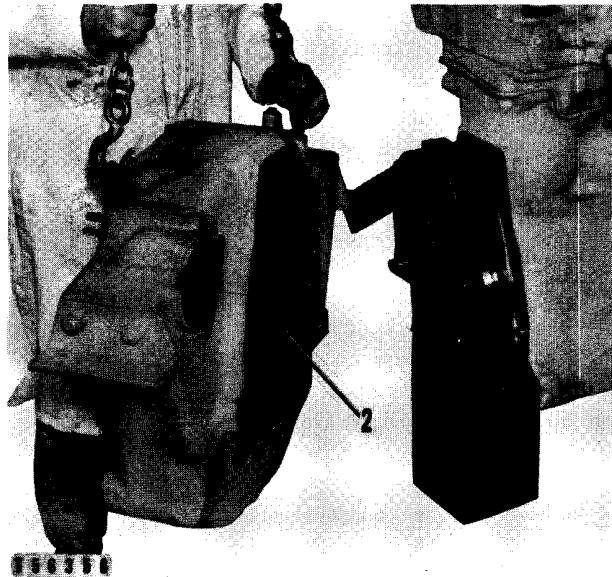


T 56380

PREPARING TO REMOVE FLYWHEEL HOUSING
1-Flywheel housing retaining bolts (thirteen).

3. Remove the bolts which secure the oil pan plate to the flywheel housing.

4. Loosen the bolts which secure the oil pan and oil pan plate to the cylinder block.
5. If the engine is being supported by the oil pan and oil pan plate, raise the cylinder block and insert two 1/4" shims between the cylinder block and oil pan plate at the rear of the diesel engine. Be careful not to damage the oil pan plate gasket.
6. Attach a suitable hoist to support the weight of the flywheel housing, and remove the flywheel housing retaining bolts (1). The flywheel housing weighs approximately 250 pounds.
7. Remove the flywheel housing as shown.



REMOVING THE FLYWHEEL HOUSING
2-Seal.

8. Remove and clean the oil strainer screen in the bottom of the flywheel housing. Replace the screen if it is damaged.
9. Prior to installation of the strainer screen, clean the oil return passage.
10. Inspect the gasket between the flywheel housing and cylinder block and replace it if it is damaged.
11. Prior to installation of the flywheel housing, replace any damaged gaskets, and check the oil seal (2). Replace if necessary. The seal is installed with the lip facing the crankshaft flange.
12. Secure the flywheel housing in place. See the topic, SPECIFICATIONS, for correct retaining bolt torque value.

ENGINE

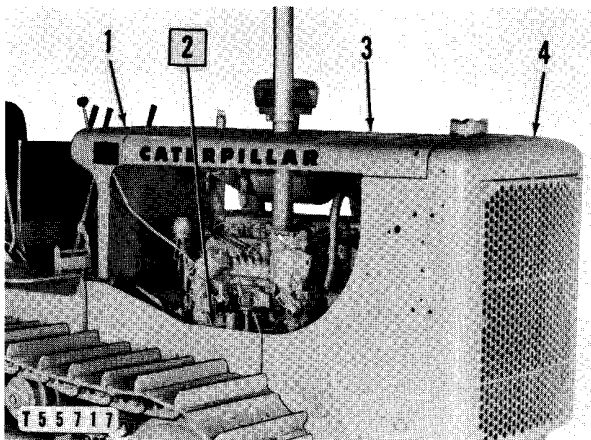
ISSUED 9-62

DIESEL ENGINE REMOVAL AND INSTALLATION

Diesel Engine Removal and Installation

The diesel engine, together with the starting engine and the flywheel clutch, weighs approximately 3000 pounds.

1. Drain the coolant from the radiator.
2. Close the fuel shut-off at the tank.

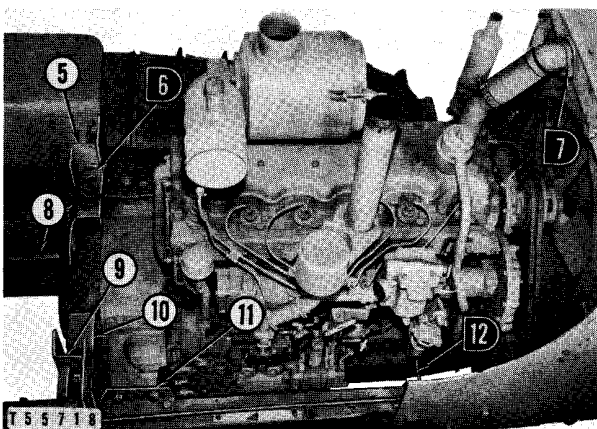


PREPARING TO REMOVE DIESEL ENGINE

☐ Remove

1-Dash. 2-Side plate. 3-Hood. 4-Radiator guard.

3. Remove the hood (3) and the dash (1). See the covering topics.



PREPARING TO REMOVE DIESEL ENGINE

☐ Disconnect

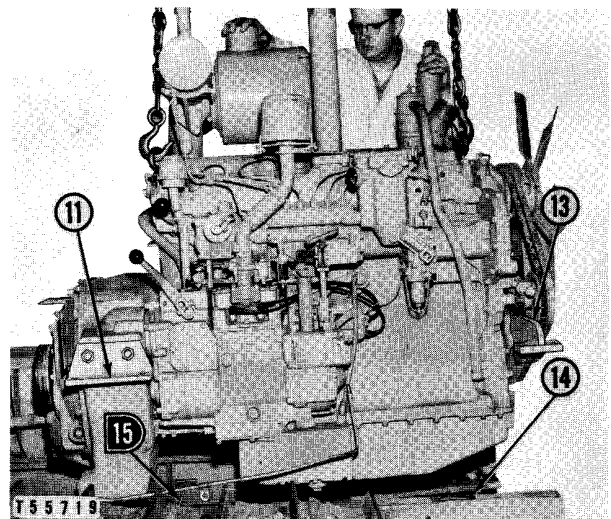
5-Fender support assembly. 6-Plate. 7-Radiator inlet elbow. 8-Flywheel clutch yoke arm. 9-Fender support assembly. 10-Plate. 11-Engine rear support. 12-Fuel line.

4. Remove the universal joint. See the covering topic.

CAUTION

Seal the end of the fuel line and the fuel filter inlet to prevent the entry of any foreign material.

5. Disconnect the plates (6) and (10) from the fender support assemblies (5) and (9).
6. Disconnect the radiator outlet elbow and inlet elbow (7) at the radiator.
7. Remove the radiator guard and radiator. See the covering topic.



REMOVING DIESEL ENGINE

☐ Disconnect

11-Engine rear support. 13-Engine front support. 14-Shims. 15-Equalizer spring pivot pin lubrication tube.

8. Disconnect the equalizer spring pivot pin lubrication tube (15) from the diesel engine crankcase.
9. Remove the bolts that secure the engine supports (11) and (13) to the frame.
10. Install two $\frac{3}{4}$ " - 10 (NC) eyebolts in the cylinder head.
11. Attach a lifting beam and suitable hoist to the eyebolts.
12. Raise the engine slightly and check for any points of interference.
13. The shims which are under the engine supports should be wired together in individual groups and replaced in their original positions when installing the engine.
14. After the engine has been placed in position, its alignment should be checked as recommended in the topic, ALIGNING ENGINE.

ENGINE**DIESEL ENGINE REMOVAL AND INSTALLATION**ISSUED 9-62

ALIGNING ENGINE

Place the amount of shims needed between the engine supports and the tractor frame in order to align the universal joint as closely as possible.

After installation of the engine and universal joint is complete, a visual check of the alignment can be made while rotating the universal joint. If the universal joint wobbles perceptibly closer alignment is needed.

Misalignment

Normally misalignment can be corrected by adding or removing shims as necessary between the frame and the engine supports.

If it is necessary to shift the engine from one side to the other in the frame, loosen the hold-down bolts and shift the engine accordingly.

If the holes for the hold-down bolts are enlarged, dowels should be installed to hold the engine in the proper location after it is bolted down.

Extreme misalignment is probably the result of bent main frame channels, in which case they should be straightened. Extreme wear in the engine front support will also cause misalignment.

Electrical System

Always disconnect the battery from the system prior to working on any of the electrical components.

GENERATOR REMOVAL AND INSTALLATION

The generator is located on the front right side of the engine and is belt driven from the crankshaft pulley.

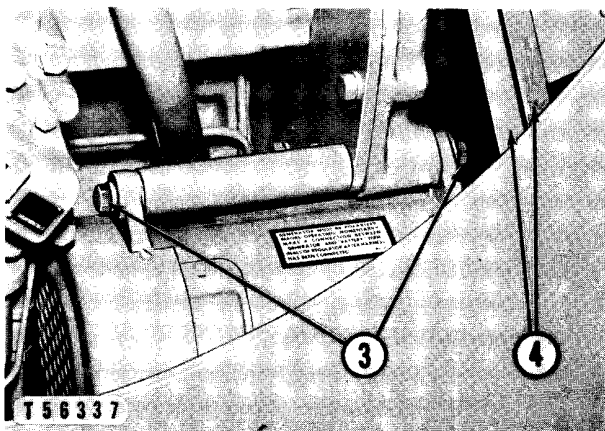


PREPARING TO REMOVE GENERATOR
1-Strap. 2-Bolt.

1. Remove the bolt (2) from the belt tension adjusting strap (1).

NOTE

There is a spacer behind the strap (1) which will fall out when the bolt (2) is removed. Be sure that the spacer is installed upon assembly.



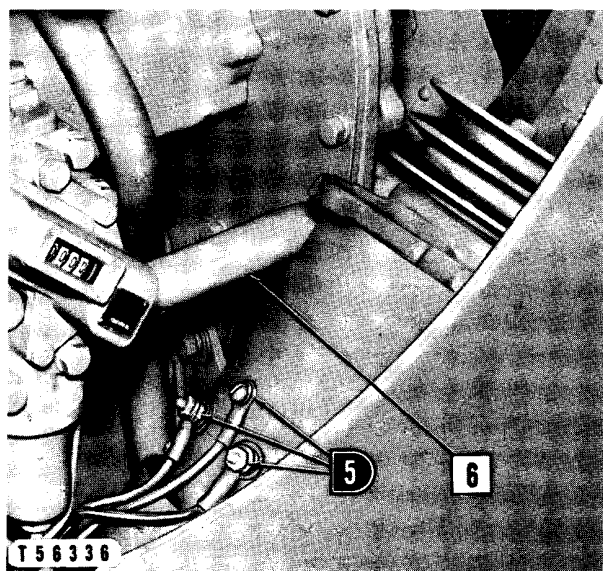
PREPARING TO REMOVE GENERATOR
3-Bolts. 4-Fan belts.

2. Remove the fan belts (4). See the covering topic.

3. Support the generator in some manner to keep it from falling and remove bolts (3).

NOTE

Mark the terminals and wires so they can be correctly connected when the generator is installed.



PREPARING TO REMOVE GENERATOR

☐ Remove ☒ Disconnect

5-Wires. 6-Spacer.



GENERATOR REMOVAL

4. Remove the generator as shown.
5. Install in reverse order of removal.

ENGINE ELECTRICAL SYSTEM

ISSUED 9-62

Polarization

Whenever a DC generator is installed, or reconnected to the generator regulator, it must be polarized before starting the engine. To prevent damage to the generator or generator regulator, the following procedure for polarizing should be used when installing the generator.

Polarizing 24V and 12V generators:

Connect a jumper momentarily between the regulator terminals marked BAT and GEN.

GENERATOR REGULATOR REMOVAL AND INSTALLATION

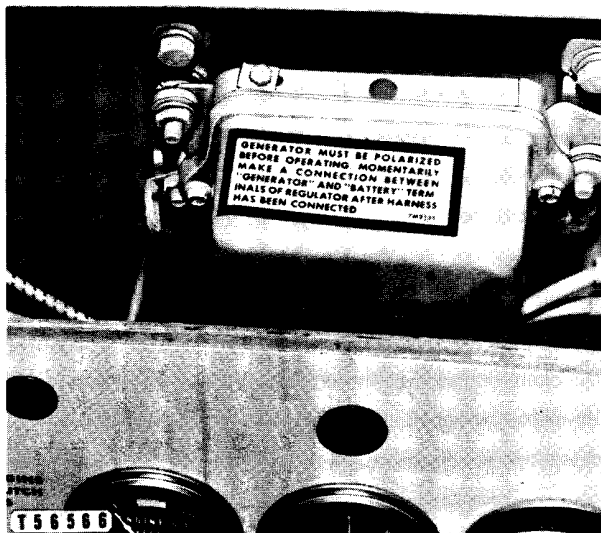
The generator regulator is mounted inside the dash assembly.

Remove the cover from the dash assembly.

Disconnect all wiring leads to the regulator.

NOTE

Mark the terminals and wires so they can be correctly connected when the regulator is installed.



PREPARING TO REMOVE REGULATOR

Install in reverse order of removal.

ELECTRIC STARTING MOTOR REMOVAL AND INSTALLATION

A 24 volt starting motor is used to start the engine if electric starting is used. The starting motor contains an overrunning clutch drive mechanism. It functions in a similar manner as other starting drive mechanisms. The clutch thus serves to protect the pinion and ring gear and also the starting motor when the diesel engine starts.

A heavy duty magnetic switch is used on the starting motor.

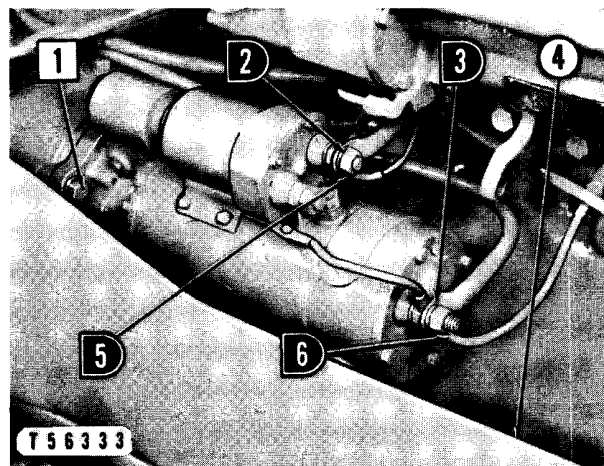
For the wiring diagrams, see the Operation and Maintenance Instructions.

WARNING

Extremely high current flows through the circuit when the starting motor is operating. Therefore, care must be taken to prevent grounding the terminals accidentally because serious damage and injury could result.

NOTE

Mark the cables, wires and terminals so they can be correctly connected when installing the starting motor.



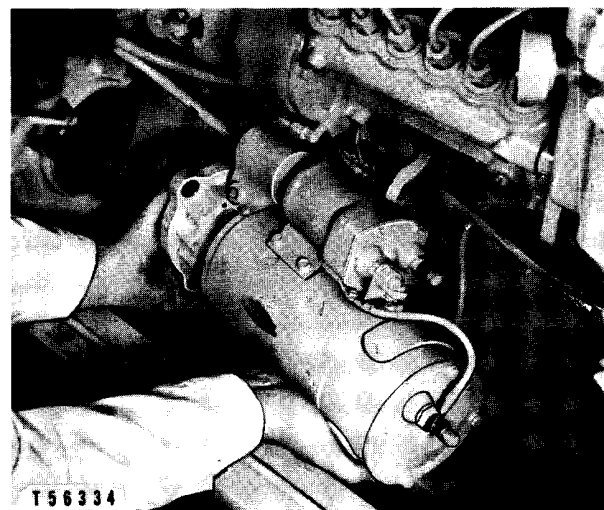
PREPARING TO REMOVE ELECTRIC STARTING MOTOR

☐ Remove ☒ Disconnect

1-Bolts (three). 2-Cable. 3-Cable. 4-Plate assembly.
5-Wire. 6-Wire.

NOTE

The approximate weight of the starting motor is 72 pounds. To facilitate in the removal of the starting motor, the plate assembly (4) can be removed.



STARTING MOTOR REMOVAL

STARTING ENGINE

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STARTING ENGINE

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STARTING ENGINE

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SPECIFICATIONS

Bore and Stroke	2 ³ / ₈ in. x 2 ³ / ₈ in.
Brake horsepower	12 - 15 @ 4500 RPM
High idle speed	5050 RPM
Low idle speed	2200 RPM

Camshaft

Journal diameter	1.498 - 1.499 in.
Bearing bore	1.500 - 1.503 in.
Bearing clearance	.001 - .005 in.
Permissible clearance	.006 in.
End clearance	.001 - .005 in.
Permissible end clearance	.010 in.

Check Valve Spring (Common lubrication system)

Free length	1/2 in.
Lbs. force	2.19
When compressed to	5/16 in.

Clutch

Clutch springs:	
Free length	2 11/64 in.
Lbs. force	69 - 81
When compressed to	13/4 in.
Adjustment:	
Clearance between spherical head of adjusting screw in release lever and the clutch spring retaining plate	.030 - .040 in.
Clearance between brake backup plate and rollers	.030 in.
Free travel in linkage assembly	.030 in.

Connecting Rod

Connecting rod bearing clearance (Earlier)	.0017 - .0039 in.
Connecting rod bearing clearance (Later—with lead-tin overlay)	.0009 - .0031 in.
Permissible bearing clearance	.005 in.
Bolt nut torque	180 - 220 lb. ft.
Center-to-center distance	3.839 - 3.841 in.
Bore in piston pin bearing	.5409 - .5414 in.

Crankshaft

Main journal diameter	1.7497 - 1.7503 in.
Permissible main journal wear	.0065 in.
Main bearing clearance	
Front (earlier)	.001 - .0055 in.
Front (later with lead-tin overlay)	.0005 - .005 in.
Rear (earlier—clearance to be measured vertically)	.0027 - .0043 in.
Rear (later with lead-tin overlay—clearance to be measured vertically)	.0022 - .0035 in.
Permissible bearing clearance (front)	.007 in.
Permissible bearing clearance (rear)	.008 in.
End clearance	.002 - .013 in.
Permissible end clearance	.019 in.
Main bearing bolt torque (rear)	28 - 30 lb. ft.
Connecting rod journal diameter	1.3747 - 1.3753 in.
Permissible rod journal wear	.0045 in.
Permissible out-of-round	.0025 in.

Gear and sprocket retaining bolt torque	118 - 142 lb. ft.
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Cylinder Bore

Diameter	2.376 - 2.377 in.
Cylinder bore wear limit (out-of-round)	.003 in.
Cylinder bore wear limit (taper)	.005 in.
Cylinder diameter for .020" oversize piston	2.396 - 2.397 in.
Cylinder diameter for .040" oversize piston	2.416 - 2.417 in.

Cylinder Head

Bolt torque	31 - 39 lb. ft.
-------------	-----------------

Magneto

Point gap	.015 in.
-----------	----------

Oil Pump

Clearance between gears and end covers (total)	.001 - .003 in.
Shaft diameter	.4919 - .4925 in.
Bearing bore	.4930 - .4936 in.
Bearing clearance	.0005 - .0017 in.
Permissible bearing clearance	.006 in.
Pressure relief spring:	
Free length	1.045 in.
Lbs. force	8.29 - 9.73
When compressed to	.750 in.
Oil pressure, minimum PSI at high idle	40

Pinion Mechanism

Gear backlash between pinion drive gear and clutch gear (through idler)	.004 - .021 in.
Idle gear bearing diameter (ID)	1.1255 - 1.1275 in.
Bearing clearance	.0225 - .0255 in.
Permissible bearing clearance	.030 in.
Overrunning clutch gear bearing diameter (ID)	1.9488 - 1.9494 in.
Bearing clearance	.0005 - .004 in.
Permissible clearance	.006 in.
Plunger spring (Earlier):	
Free length	1.38 in.
Lbs. force	14.2 - 15.8
When compressed to	.97 in.
Plunger spring (Later—plunger guided in cover):	
Free length	1.235 in.
Lbs. force	28.71 - 30.29
When compressed to	.739 in.
Pinion shaft bearing bore (inner)	.8755 - .8775 in.
Bearing clearance	.0015 - .0045 in.
Permissible clearance	.012 in.
Pinion return springs (Earlier):	
Outer spring free length	3.69 in.
Lbs. force	5.00 - 5.52
When compressed to	2.35 in.
Inner spring free length	3.23 in.
Lbs. force	3.87 - 4.27
When compressed to	2.41 in.
Pinion return spring (Later—conical):	
Free length	2.375 in.
Lbs. force	60 - 70
When compressed to	.560 in.

STARTING ENGINE SPECIFICATIONS

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Piston Pins

Clearance in rod bearings	.0007 - .0017 in.
Permissible clearance in rod	.0030 in.
Clearance in piston	.0004 - .0012 in.
Permissible clearance	.0025 in.

Piston Rings

Compression ring gap	.007 - .017 in.
Oil ring gap	.007 - .017 in.
Compression ring side clearance	.0020 - .0035 in.
Oil ring side clearance	.0015 - .0031 in.
Permissible side clearance	.005 in.

Rocker Arm

Clearance between shaft and bearing	.0005 - .0029 in.
Permissible clearance	.005 in.

Spark Plug

Gap	.025 in.
Torque	30 lb. ft.

Starting Motor (12 V—Delco-Remy 1107741, 1107213 and 1107788)

Brush spring tension (minimum)	35 oz.
Rotation (viewing drive end)	Counterclockwise
No load test:	
Volts	10.3
Maximum amps	75
Minimum RPM	6900
Lock test:	
Volts	5.8
Maximum amps	435
Minimum torque	10.5 lb. ft.

Valves

Exhaust valves:	
Clearance (cold)	.010 in.
Stem clearance in head	.0015 - .0035 in.

Permissible clearance with new

valve	.0065 in.
Valve face angle	44 $\frac{1}{4}$ °
Valve seat angle	45°
Inlet valves:	
Clearance (cold)	.010 in.
Stem clearance in head	.0015 - .0035 in.
Permissible clearance with new valve	.0050 in.
Valve face angle	44 $\frac{1}{4}$ °
Valve seat angle	45°

Valve Lifters

Lifter diameter	.3729 - .3735 in.
Bore diameter	.375 - .376 in.
Clearance	.0015 - .0031 in.
Permissible clearance	.0075 in.

Valve Spring

Free length (Earlier—small diameter)	1.5 in.
Lbs. force	25.6 - 28.4
When compressed to	1 $\frac{11}{32}$ in.
Free length (Parts service for earlier machines—small diameter, higher rate)	1.500 in.
Lbs. force	26.4 - 29.2
When compressed to	1.351 in.
Free length (Later—large diameter, higher rate)	1.528 in.
Lbs. force	25.6 - 28.4
When compressed to	1.351 in.

Valve Timing

With valve clearances set correctly cold, dial indicator mounted above valve stem, readings taken with valve .075" off its seat:	
Exhaust opening (before bottom center)	31° 38'
Exhaust closing (before top center)	26° 30'
Inlet opening (after top center)	1° 22'
Inlet closing (after bottom center)	6° 30'

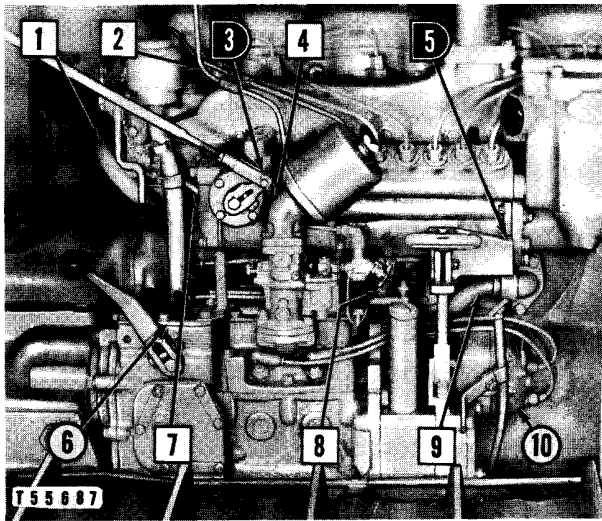
Removal and Installation

1. Drain the diesel engine cooling system.

NOTE

Earlier models incorporated a common lubrication system between the starting engine and the diesel engine. Later models contain a separate lubrication system for each engine. On engines with common lubrication, the diesel engine oil system must also be drained.

2. Shut off the diesel fuel supply and the starting engine gasoline supply at their respective tanks.
3. Drain the fuel filter housing. See the topic, DRAINING FUEL FILTER HOUSING, in the Operation and Maintenance Instructions.
4. Remove the hood. See the topic, HOOD REMOVAL.
5. Remove the right side plate.
6. Remove the fuel injection pump housing and governor as a unit. See the topic, FUEL INJECTION PUMP HOUSING REMOVAL. Cap all connections to prevent dirt from entering the system.



PREPARING TO REMOVE STARTING ENGINE

☐ Remove ☒ Disconnect

- | | |
|--|--|
| 1-Starting engine exhaust pipe. | 6-Cover assembly. |
| 2-Starting engine fuel supply line. | 7-Starting engine coolant outlet pipe. |
| 3-Governor control linkage. | 8-Starting engine mounting bracket. |
| 4-Starting engine air cleaner. | 9-Starting engine coolant inlet pipe. |
| 5-Recoil starter cable mounting bracket. | 10-Magneto. |

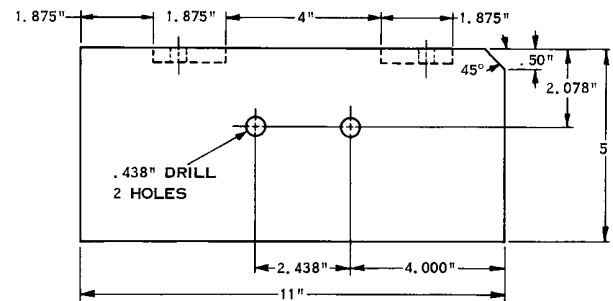
7. Remove the cover assembly (6) and cover the opening in the block to prevent the entry of dirt or other foreign material.

8. Remove the magneto (10). See the topic, MAGNETO REMOVAL AND INSTALLATION.

NOTE

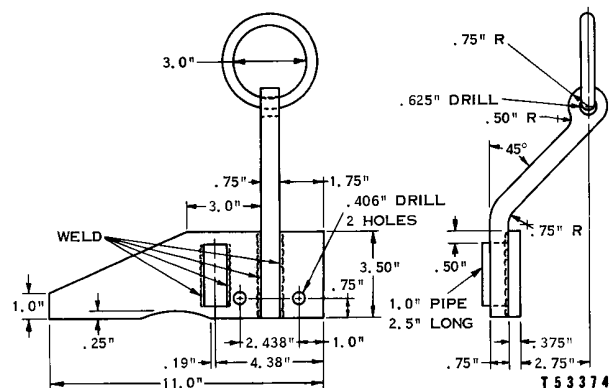
On engines with common lubrication, remove the lubricant elbow from the diesel engine cylinder block to the bottom of the starting engine cylinder block.

9. Attach the starting engine lifting bracket adapter of the dimensions shown to the starting engine.



STARTING ENGINE LIFTING BRACKET ADAPTER

10. Attach the lifting bracket of the dimensions shown to the lifting bracket adapter.

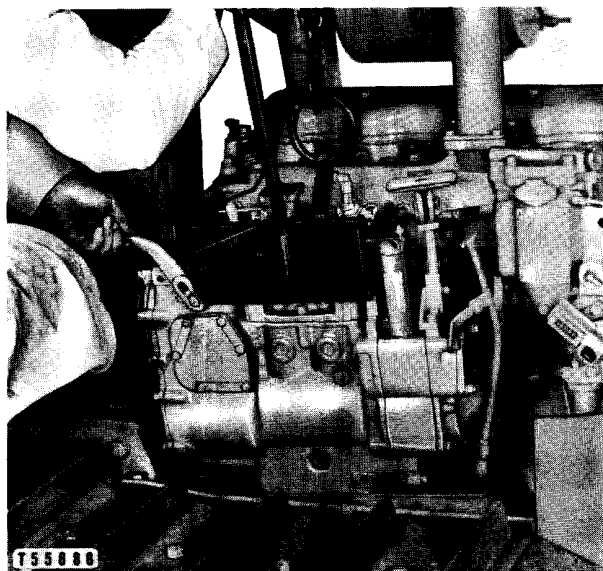


LIFTING BRACKET

11. Attach a suitable hoist to the lifting bracket and support the weight of the engine.
12. Remove the bolts securing the starting engine to the diesel engine.
13. Move the starting engine forward until the pinion housing clears the diesel engine flywheel housing.

STARTING ENGINE REMOVAL AND INSTALLATION

ISSUED 9-62

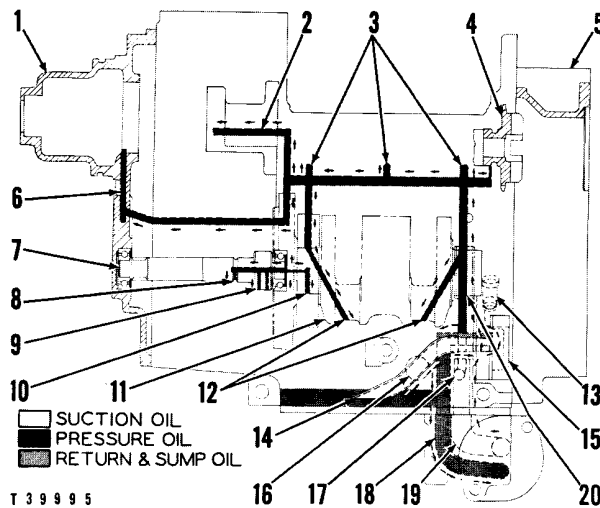


REMOVING STARTING ENGINE

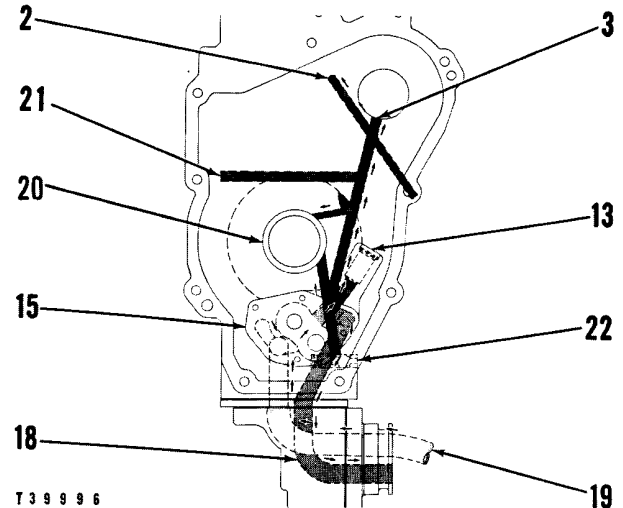
14. Prior to installation of the starting engine, replace all damaged gaskets or seals.
15. Start the starting engine pinion housing into the flywheel housing.
16. Position the pinion housing in the flywheel housing and secure the starting engine to the cylinder block.
17. Complete the installation in the reverse order of removal.

Lubrication System

FLOW OF LUBRICATING OIL



(Right Side View)



(Front View)

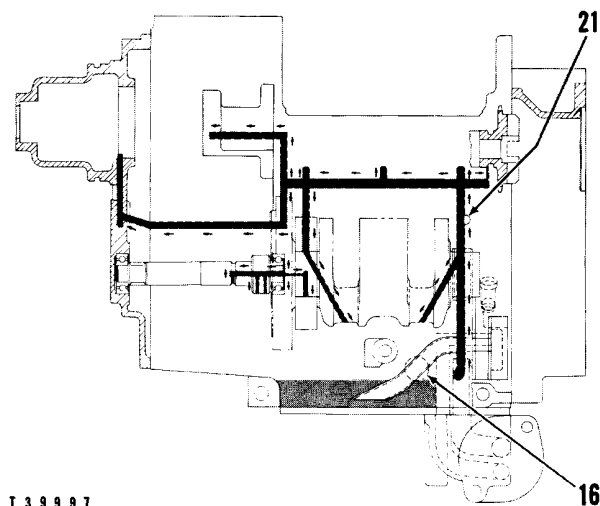
COMMON LUBRICATION SYSTEM (STARTING ENGINE
OPERATING - DIESEL ENGINE INOPERATIVE)

1-Pinion housing. 2-Oil passage to pinion shaft support bearing. 3-Oil passages to camshaft bearings. 4-Camshaft sprocket. 5-Timing chain housing. 6-Oil passage to idler gear stub shaft. 7-Clutch shaft. 8-Oil passage to thrust bearing. 9-Oil passage to clutch plates. 10-Rear main bearing. 11-Crankshaft. 12-Passages in crankshaft to connecting rod bearings. 13-Oil pump pressure relief valve. 14-Oil pump scavenge tube. 15-Oil pump. 16-Starting engine sump drain. 17-Lubricating inlet passage from diesel engine. 18-Oil return line. 19-Oil supply line from diesel engine. 20-Front main bearing. 21-Oil pressure check passage. 22-Check valve.

The starting engine oil pump (15) is a two stage pump. When the starting engine is operating, the first stage draws oil from the diesel engine through the oil supply line (19). The oil is then pumped under pressure to the lubricant passages in the starting engine. A pressure relief valve (13) prevents the pump from building up excessive pressure.

The second stage of the oil pump acts as a scavenger pump for the starting engine sump. Oil enters the pump through the tube (14) and is returned to the diesel engine sump through line (18).

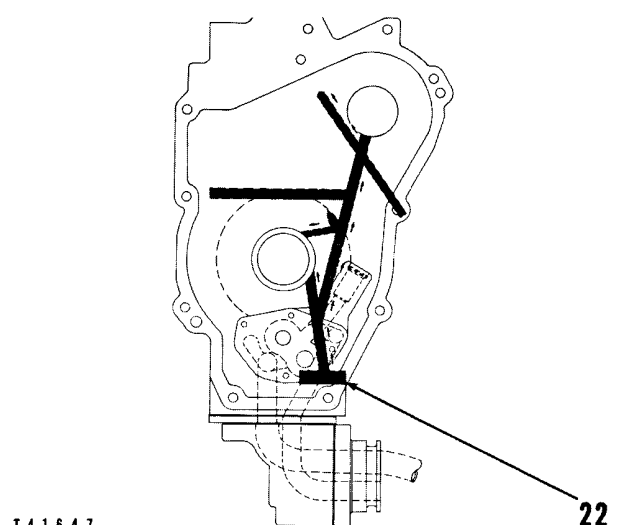
The cylinder walls, piston pins, rocker arms and valve mechanism are lubricated by oil vapors and splash oil.



T 39997

COMMON LUBRICATION SYSTEM (STARTING ENGINE
INOPERATIVE - DIESEL ENGINE OPERATING)

16-Starting engine sump drain. 21-Oil pressure check passage.



T 41647

COMMON LUBRICATION SYSTEM (STARTING ENGINE
INOPERATIVE - DIESEL ENGINE OPERATING)

22-Check valve.

STARTING ENGINE LUBRICATION SYSTEM

ISSUED 6-62

When the diesel engine is operating with the starting engine inoperative, pressure oil from the diesel engine enters the starting engine through the check valve (22). This valve prevents oil from entering the diesel engine when the starting engine is operating. The passage from the check valve by-passes the oil pump. The starting engine is pressure-lubricated through the same passages while it is inoperative. This provides an oil cushion for the various components of the starting engine while the diesel engine is operating.

When the oil in the starting engine sump reaches the level of the drain (16), the oil drains into the diesel engine.

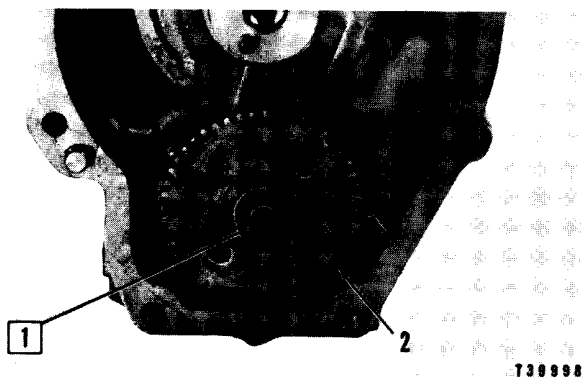
The presence of oil or starting engine oil pressure can be checked by removing the plug in the end of the oil pressure check passage (21). The minimum pressure should be 40 PSI at high idle speed.

OIL PUMP

The oil pump is a two-stage gear-type pump that is gear driven from the crankshaft at the front of the engine. Four bolts secure the oil pump to the cylinder block in the timing chain compartment.

Removal and Installation

Remove the recoil starter housing, timing chain housing and crankshaft gear. See the covering topics.



REMOVING OIL PUMP DRIVE GEAR

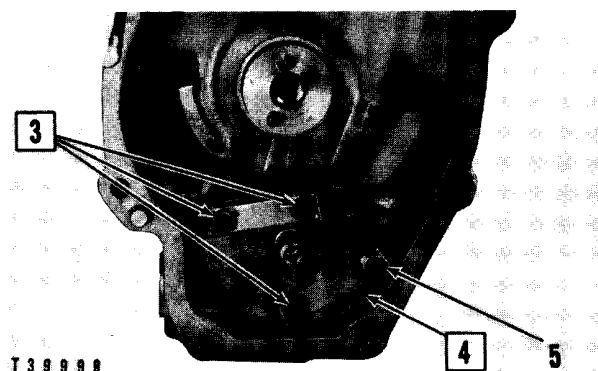
☐ Remove

1-Bolt. 2-Gear.

Pull the gear (2) by using two 3/8" - 24 (NF) bolts 3" long and a 5F7465 No. 5 Puller.

NOTE

Leave bolt (5) in the oil pump to act as an aligning pin.

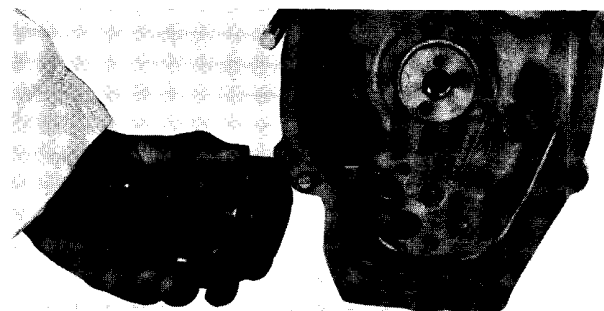


PREPARING TO REMOVE OIL PUMP

☐ Remove

3-Bolts. 4-Oil pump. 5-Bolt.

When installing the oil pump, make certain the gasket between the cylinder block and the oil pump is properly aligned.



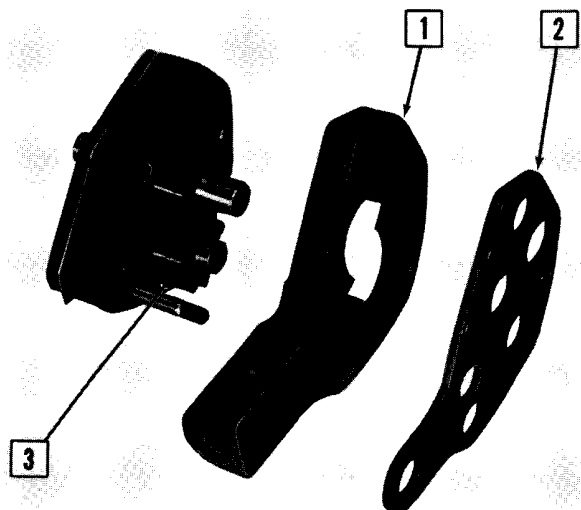
T40000

OIL PUMP REMOVAL

Disassembly and Assembly

1. Remove the plate (7) and shaft assembly (6) by moving the shaft out of the second stage pump housing (8).
2. Remove the housing (8) from the plate (4).
3. Inspect all gears, bearings and shafts and replace if necessary.
4. The gear (9) can be removed by sliding it off the shaft (6). When replacing the gear, make certain the key is in place in the shaft (6).
5. The gear (10) can be removed from the shaft (6) by removing the spring pin that is driven through the gear into the shaft. The gear can then be slid off the shaft. If a new gear is to be installed on a new or used shaft, place the gear in the correct position on the shaft with the key in place. Drill a .125-.129 inch hole .69 inch

deep through the gear at the root diameter into the shaft. Be sure to locate the hole so the drill will not hit the key.

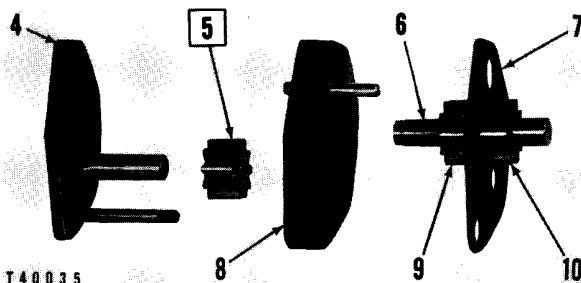


T40034

OIL PUMP DISASSEMBLY

☐ Remove

1-First stage pump housing. 2-Plate. 3-Driven gear.



T40035

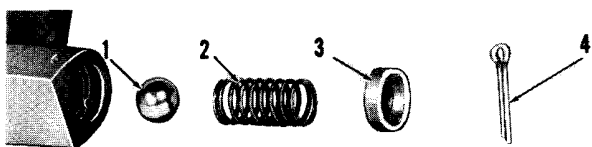
OIL PUMP DISASSEMBLY

☐ Remove

4-Plate. 5-Driven gear. 6-Drive shaft. 7-Plate. 8-Second stage pump housing. 9-Driving gear. 10-Driving gear.

PRESSURE REGULATING VALVE

1. The pressure regulating valve consists of a ball (1), spring (2), core plug (3) and cotter pin (4).



T40036

PRESSURE REGULATING VALVE

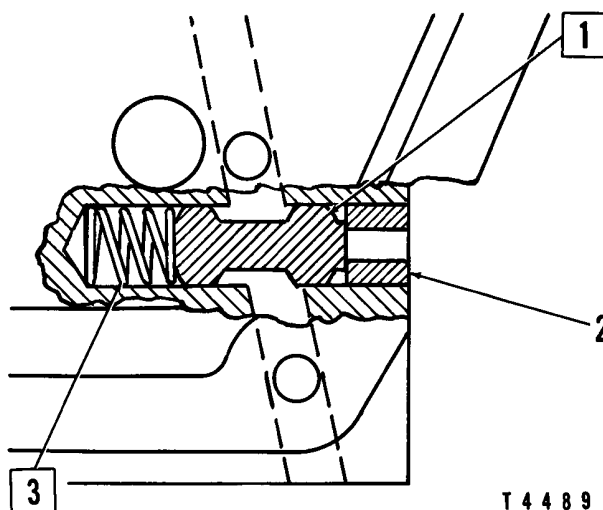
1-Ball. 2-Spring. 3-Core plug. 4-Cotter pin.

2. To remove the valve assembly, remove the cotter pin (4) and core plug (3).
3. Inspect the ball and its seat and replace if necessary.

CHECK VALVE

Removal and Installation

1. Insert a hooked instrument through the adapter (2) and pull it out of the starting engine cylinder block.



T44891

CHECK VALVE

☐ Remove

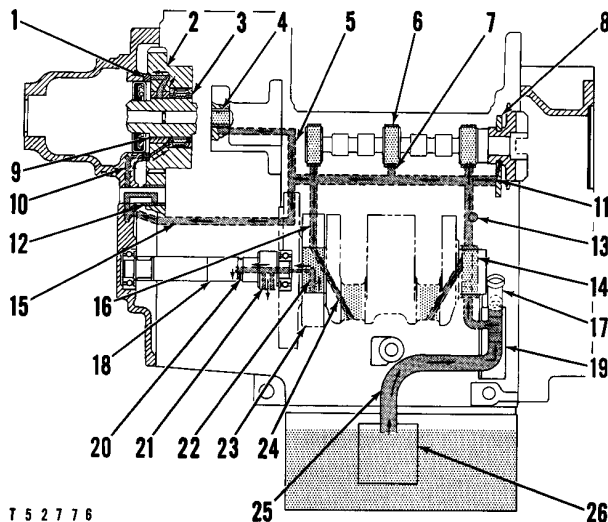
1-Check valve. 2-Adapter. 3-Spring.

2. Inspect the check valve (1) for any sign of wear or damage. Replace if necessary.
3. Inspect the seating surface of the adapter (2). Replace if necessary.
4. Inspect the spring (3). See the topic, SPECIFICATIONS.
5. Install the spring (3), check valve (1) and adapter (2) into the starting engine cylinder block.

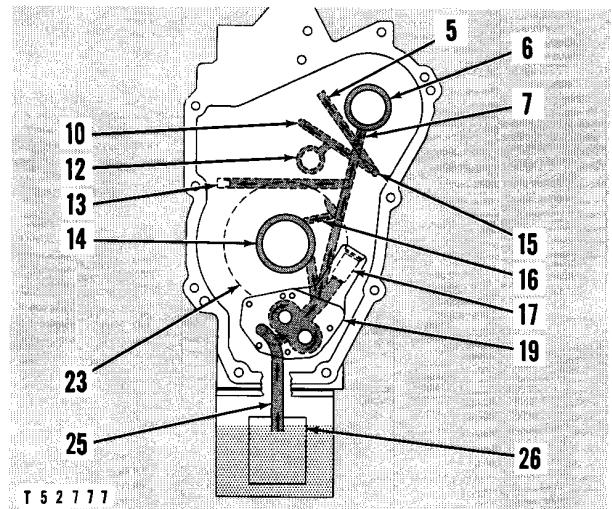
NOTE

Install the adapter with the chamfered edge toward the plunger. When the adapter is properly installed, the outside edge should be flush with the starting engine cylinder block.

Lubrication System FLOW OF LUBRICATING OIL



(Right Side View)



(Front View)

STARTING ENGINE LUBRICATION SYSTEM

- 1-Thrust washer.
- 2-Pinion drive gear.
- 3-Overrunning clutch.
- 4-Support bearing.
- 5-Oil passage to support bearing.
- 6-Camshaft bearing.
- 7-Oil passage to camshaft bearing.
- 8-Camshaft thrust plate.

- 9-Pinion drive gear bearing.
- 10-Oil passage to starter mechanism.
- 11-Oil passage in camshaft thrust washer.
- 12-Idler gear bearing.
- 13-Oil pressure check passage.
- 14-Crankshaft front main bearing.

- 15-Oil passage to pinion housing.
- 16-Oil passage in crankshaft rear main bearing.
- 17-Oil pump pressure regulating valve.
- 18-Clutch shaft.
- 19-Oil pump.
- 20-Oil passage to clutch thrust bearing.

- 21-Oil passage to clutch plates (two).
- 22-Oil passage in crankshaft to clutch shaft.
- 23-Crankshaft rear main bearing.
- 24-Oil passage in crankshaft to connecting rod bearings.
- 25-Oil supply tube.
- 26-Suction bell.

The starting engine oil pump (19) is a single section pump. When the starting engine is operating, oil is drawn through the supply tube (25), through the oil pump (19) and into the lubricant passages within the starting engine. A pressure regulating valve (17) prevents the pump from creating too much pressure.

Pressure oil is supplied to the starter pinion mechanism housing through the passage (15). The idler gear bearing (12) is pressure lubricated. Two passages are drilled in the pinion drive gear (2). When each passage, respectively, is aligned with the passage in the starter pinion housing and thrust washer (1), lubricant is supplied to the overrunning clutch (3) and the pinion drive gear bearing (9) respectively.

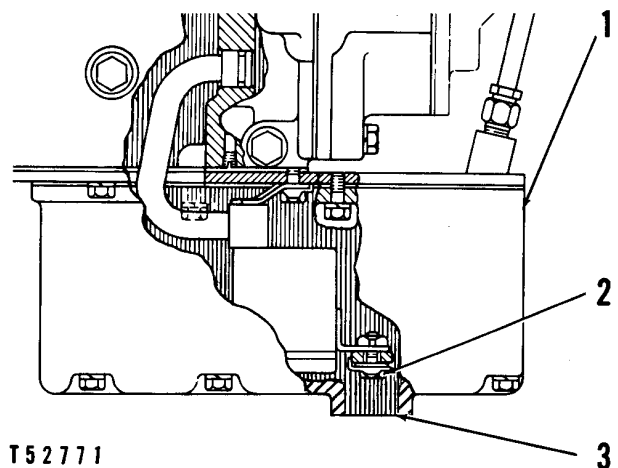
The cylinder walls, piston pins, rocker arms and valve mechanism are lubricated by oil vapors and splash oil.

Starting engine oil pressure can be checked by removing the plug in the end of the oil pressure check passage (13). Refer to the topic, SPECIFICATIONS, for the minimum oil pressure at high idle RPM.

OIL PAN, OIL PAN PLATE AND SUCTION BELL

Removal and Installation

1. Drain the oil from the oil pan.



T 5 2 7 7 1

PREPARING TO REMOVE OIL PAN (D4 Illustrated)

- 1-Oil pan. 2-Bolt and lock. 3-Opening.

2. Remove the bolt and lock (2) by working through the opening (3).

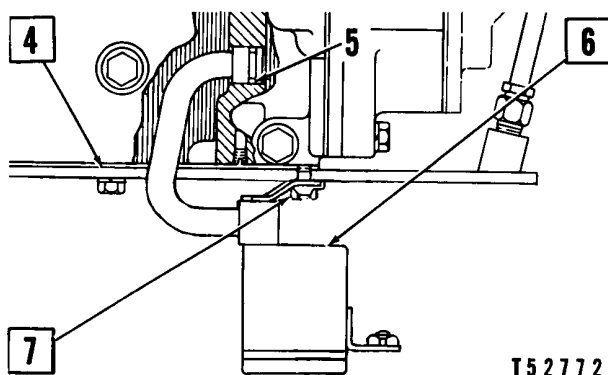
STARTING ENGINE LUBRICATION SYSTEM

ISSUED 5-62

3. Remove the remaining bolts securing the oil pan (1) to the plate assembly and remove the oil pan.

NOTE

On 944, 955 and 966 models, there are no bolt and lock (2). On D330 and D333 industrial, electrical and marine applications, the opening (3) has a core plug installed in it and there is an O-ring seal connection between the tube assembly and the suction bell. The bolt and lock (2) need be removed only if the suction bell is to be removed from the oil pan. Caution should be used to avoid damaging the O-ring seal when removing and installing the suction bell. The D6, 561, No. 12, No. 14, No. 112, and 977 are similar in construction to the D4.



PREPARING TO REMOVE OIL PAN PLATE
AND SUCTION BELL
(D4 Illustrated)

☐ Remove

4-Oil pan plate assembly. 5-O-ring seal. 6-Suction bell assembly. 7-Bolt and lock.

4. Inspect the seal (5) and replace if necessary.

NOTE

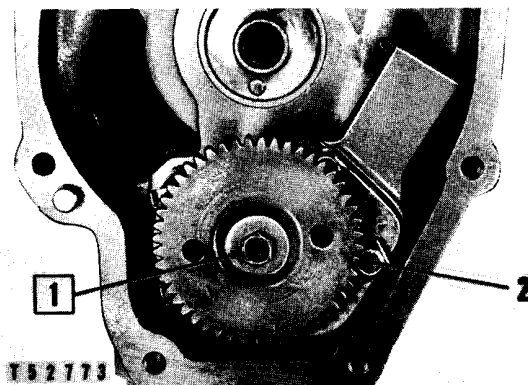
When installing the suction bell assembly (6) into the block, use caution to avoid damaging the seal (5).

OIL PUMP

The oil pump is a single element gear-type pump that is gear driven from the crankshaft gear at the front of the engine. Four bolts secure the oil pump to the cylinder block in the timing chain compartment.

Removal and Installation

Remove the recoil starter housing, timing chain housing and crankshaft gear. See the covering topics.

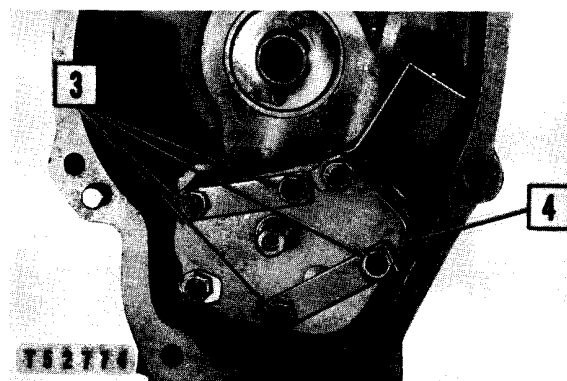


REMOVING OIL PUMP DRIVE GEAR

☐ Remove

1-Bolt. 2-Gear.

Pull the gear (2) by using two $\frac{3}{8}$ " - 24 (NF) bolts 3" long and a 5F7465 No. 5 Puller.

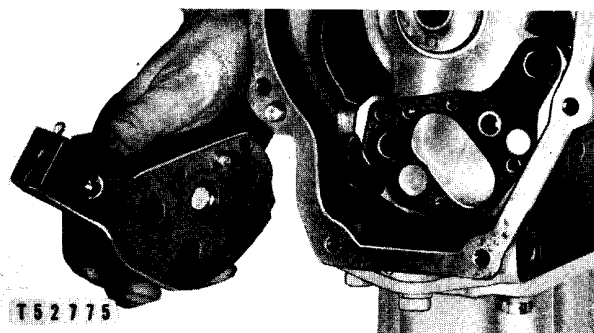


PREPARING TO REMOVE OIL PUMP

☐ Remove

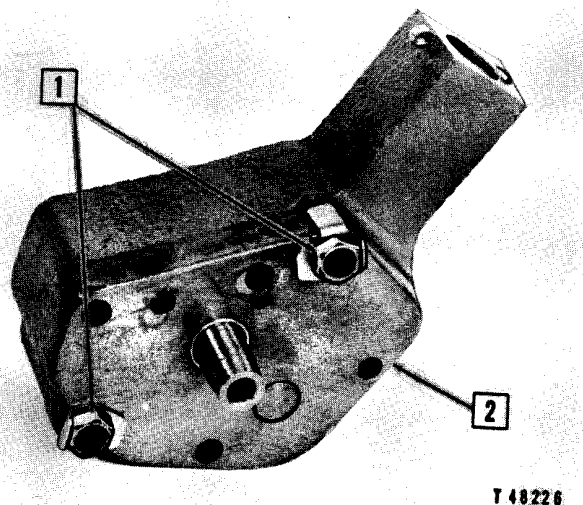
3-Bolts and locks. 4-Oil pump.

When installing the oil pump, make certain the gasket between the cylinder block and the oil pump is properly aligned.



OIL PUMP REMOVAL

Disassembly and Assembly

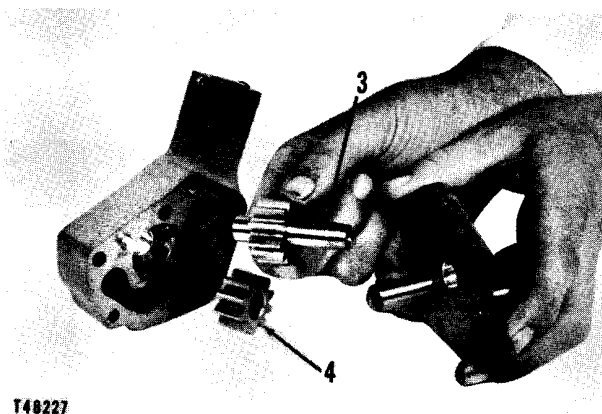


PREPARING TO DISASSEMBLE OIL PUMP

☐ Remove

1-Bolts. 2-Cover.

Remove the oil pump drive shaft assembly (3) and idler gear (4) from the cover as illustrated.



DRIVE SHAFT AND IDLER GEAR REMOVAL

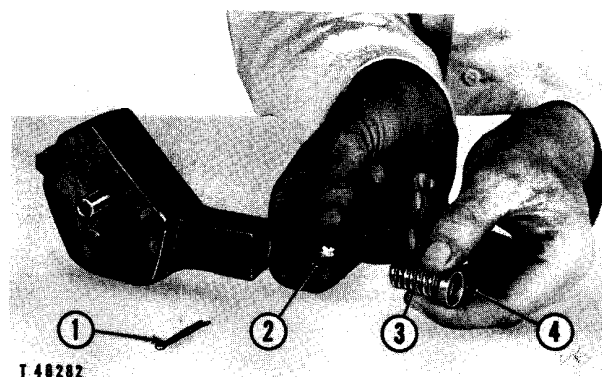
3-Shaft assembly. 4-Gear.

Check the clearance between the gears and cover. See the topic, SPECIFICATIONS, for the correct clearance.

Inspect the gear, shaft, shaft assembly, and bearings for wear or damage and replace if necessary. See the topic, SPECIFICATIONS, for the proper clearances and dimensions.

PRESSURE REGULATING VALVE

1. The pressure regulating valve consists of a ball (2), spring (3), core plug (4) and cotter pin (1).



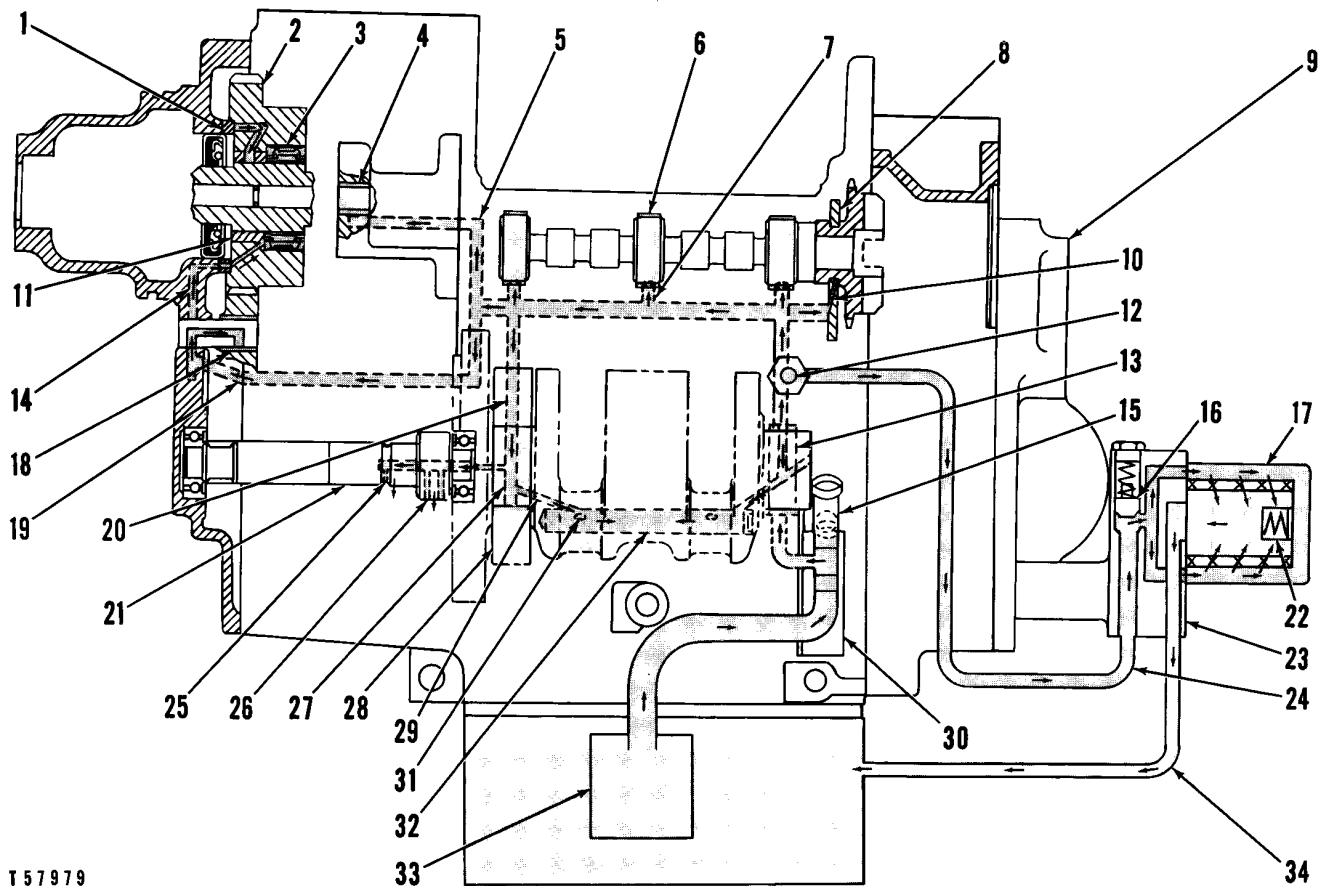
PRESSURE REGULATING VALVE

1-Cotter pin. 2-Ball. 3-Spring. 4-Core plug.

2. To remove the valve assembly, remove the cotter pin (1) and core plug (4).
3. Inspect the ball for signs of grooving. Replace if necessary.
4. Inspect the spring. See the topic, SPECIFICATIONS.

NOTE

When installing a new core plug, a hole must be drilled in it for the cotter pin. Install the plug flush with the outer surface of the housing and, using the existing holes in the pump body as a guide, drill the necessary hole.

Lubrication System
FLOW OF LUBRICATING OIL

T 57979

STARTING ENGINE LUBRICATION SYSTEM
(Right Side View)

- | | | | |
|------------------------------------|--|--|---|
| 1-Thrust washer. | 11-Pinion drive gear bearing. | 20-Oil passage in rear main bearing. | 29-Oil supply passage. |
| 2-Pinion drive gear. | 12-Oil pressure check point plug. | 21-Clutch shaft. | 30-Oil pump. |
| 3-Overrunning clutch. | 13-Front main bearing. | 22-Bypass valve. | 31-Connecting rod bearing oil supply passage. |
| 4-Support bearing. | 14-Oil passage to starter mechanism. | 23-Oil filter base. | 32-Sludge pocket. |
| 5-Oil passage to support bearing. | 15-Oil pump pressure regulating valve. | 24-Oil filter supply line. | 33-Suction bell and tube assembly. |
| 6-Camshaft bearing. | 16-Valve. | 25-Oil passage to clutch thrust bearing. | 34-Oil filter outlet line. |
| 7-Oil passage to camshaft bearing. | 17-Oil filter. | 26-Oil passage to clutch plates (two). | |
| 8-Camshaft thrust plate. | 18-Idler gear bearing. | 27-Oil passage in main bearing journal. | |
| 9-Recoil starter housing. | 19-Oil passage to pinion housing. | 28-Rear main bearing. | |

The starting engine oil pump (30) is a single section pump. When the starting engine is operating, oil is drawn through the suction bell and tube assembly (33), through the oil pump (30) and into the lubricant passages within the starting engine. A pressure regulating valve (15) limits the maximum pressure in the oil system.

Pressure oil is supplied to the starter pinion mechanism housing through the passage (19). The idler gear bearing (18) is pressure lubricated. Four passages are drilled in the pinion drive gear (2). When each passage is aligned with the passage in

the starter pinion housing and thrust washer (1), lubricant is supplied to the overrunning clutch (3) and pinion drive gear bearing (11), respectively.

The passages (25) and (26) provide lubrication to the clutch plates and thrust bearing.

The connecting rod bearings are lubricated from drilled passages (31) in the connecting rod bearing journals. This oil is supplied through the passage (29) from the oil passage (27) in the rear main bearing journal and from similar passages in the front main bearing. The centrifugal force of the rotating

STARTING ENGINE LUBRICATION SYSTEM

ISSUED 12-62

crankshaft keeps foreign material in the oil from going through the passages to the connecting rod bearings. These particles are collected in the sludge pocket (32) in the crankshaft bearing journals. A plug in the end of the journals can be removed and the pocket cleaned. See the topic, CRANKSHAFT AND MAIN BEARINGS.

The cylinder walls, piston pins, rocker arms and valve mechanism are lubricated by oil vapors and splash oil.

The starting engine contains a bypass type lubricating oil filter. A portion of the pressure oil in the camshaft oil manifold is diverted through the oil filter supply line (24) to the starting engine oil filter base (23). The oil filter base is mounted on the front of the recoil starter housing (9). A valve (16) in the base prevents oil from going through the filter (17) until the system pressure is high enough to open the valve. Filtered oil is passed through the oil filter outlet line (34) to the starting engine oil pan.

If the oil filter (17) becomes plugged, a filter bypass valve (22) in the filter element opens and allows the oil to bypass the filter element and flow directly to the oil pan.

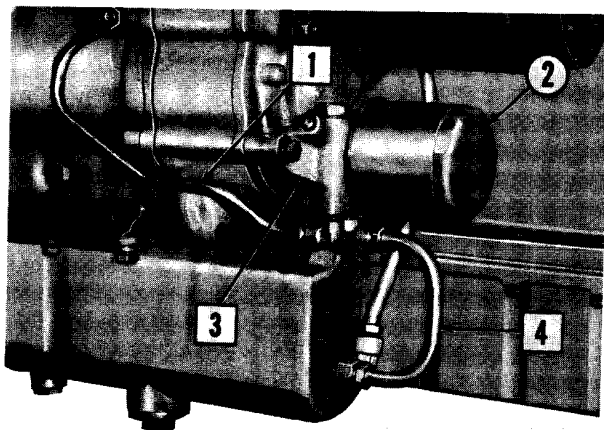
Starting engine oil pressure can be checked by removing the plug (12) from the head of the bolt which secures the oil filter supply line to the cylinder block.

OIL FILTER

Removal and Installation

NOTE

A 9M9852 Wrench is available, if needed, to aid in removing and installing the oil filter.



T 57 976

PREPARING TO REMOVE OIL FILTER

☐ Remove

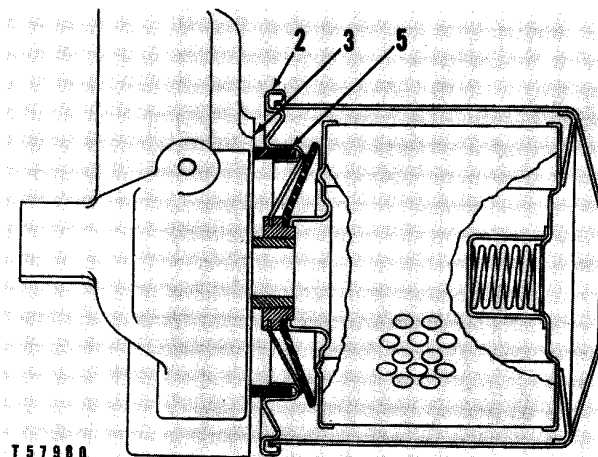
1-Oil filter inlet tube. 2-Oil filter. 3-Oil filter base.
4-Oil filter outlet tube.

1. Drain the oil from the oil pan.
2. Remove the filter (2) by turning counterclockwise. The oil filter (2) is the throw-away type and should be discarded if the gasket (5) is damaged or at the suggested service interval. See the Operation and Maintenance Instructions.

NOTE

On the 977 Traxcavator the oil filter outlet tube (4) enters the oil pan on the side, but is removed in the same manner.

3. Inspect all parts for damage and replace if necessary.
4. Install in the reverse procedure of removal.
5. Seat the gasket properly in the end of the filter and apply a film of clean light oil to the end of the gasket (5) which contacts the filter base.
6. Install the filter (2) by turning it clockwise.



INSTALLING OIL FILTER

2-Oil filter. 3-Oil filter base. 5-Gasket.

7. Turn the filter until the gasket (5) just contacts the base (3). Tighten the filter one additional turn from this point.

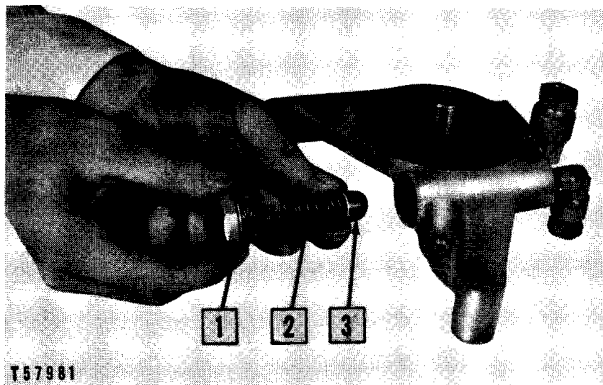
CAUTION

Do not overtighten the filter. Damage may result to the gasket which could result in oil leakage.

8. Fill the crankcase with oil to the specified level. See the covering topic in the Operation and Maintenance Instructions.

Disassembly and Assembly

Inspect all parts for wear or damage and replace if necessary.



REMOVING VALVE FROM FILTER BASE

☐ Remove

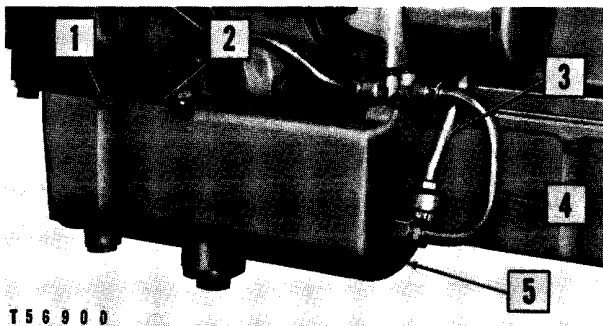
1-Plug. 2-Spring. 3-Plunger.

Inspect the spring. See the topic, SPECIFICATIONS.

Assemble in the reverse order of disassembly.

OIL PAN, OIL PAN PLATE AND SUCTION BELL**Removal and Installation**

1. Drain the oil from the starting engine.



PREPARING TO REMOVE OIL PAN

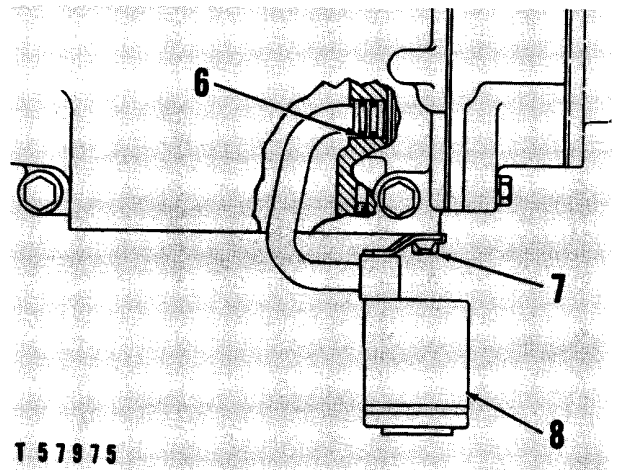
☐ Remove

1-Bolts. 2-Oil pan plate. 3-Oil level gauge tube assembly. 4-Oil filter outlet tube. 5-Oil pan.

2. Remove the bolt (7) which secures the suction bell assembly to the cylinder block.
3. Remove the suction bell assembly (8).
4. Inspect all parts for wear or damage and replace if necessary.
5. Inspect the O-ring seals (6) and replace if necessary.

NOTE

Use caution when installing the suction bell assembly (8) into the cylinder block to avoid damaging O-ring seals (6).

REMOVING SUCTION BELL ASSEMBLY
6-O-ring seals (two). 7-Bolt. 8-Suction bell assembly.

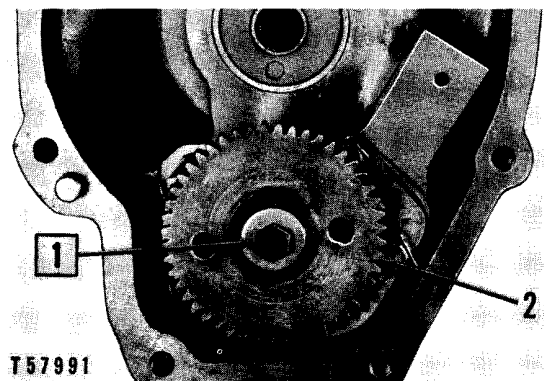
6. Install in the reverse order of removal.

OIL PUMP

The oil pump is a single element gear-type pump that is gear driven from the crankshaft gear at the front of the engine. Four bolts secure the oil pump to the cylinder block in the timing chain compartment.

Removal and Installation

Remove the recoil starter housing, timing chain housing and crankshaft gear. See the covering topics.



REMOVING OIL PUMP DRIVE GEAR

☐ Remove

1-Bolt. 2-Gear.

Pull the gear (2) by using two $\frac{3}{8}$ " - 24 (NF) bolts 3" long and a 5F7465 No. 5 Puller.

When installing the oil pump, make certain the gasket between the cylinder block and the oil pump is properly aligned.